Bifurcations and chaos in language models fed their own output

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Status of this draft. Unless marked otherwise, results on the soft-token map are numerically validated in float64 (rung 1 or 2 of the evidence ladder in Section 18). Float32 maps are labelled as such. Huginn's spectra are float32 and its accuracy runs bfloat16. **[Planned:** Red] marks planned work and **[In progress:** orange] running work. The only computer-assisted proof is that of the period-8 orbit of Table 3 (rung 3, Section 18).

Abstract

Feeding a language model the expected embedding of its own next token, as Soft Thinking does, turns inference into a nonlinear dynamical system. In Qwen2.5, OLMo-2 and Llama-3 models of 0.5B to 72B parameters, temperature and steering move this loop between repeating text and chaos, with positive Lyapunov exponents that exact tangents reproduce. The chaos has practical consequences. Rerun with only the batch size changed, chaotic float64 loops separate until their most probable tokens differ, after about the number of steps the Lyapunov exponent sets, while periodic loops mostly stay together, in four models of two families up to 70B. In frozen tests on arithmetic, the batch layout changed 56 % of the answers of bfloat16 soft-thinking inference in a 0.5B model and 55 % in a 7B model, far more often than those of a forced greedy baseline. Exactly repeating text can hide a chaotic internal state. Along the sampled orbits, steering creates the chaos mainly by making next tokens compete, which turns the softmax readout from a contraction into an expansion; a logit bias alone suffices. Earlier injection carries chaos to lower temperature, two steering vectors at different depths create chaos neither causes alone, and two loops writing one text lock each other into periodic orbits that neither has alone. The loop's regime does not mark the accuracy of ordinary answers, and we find no advantage at the edge of chaos. The loop obeys quantitative laws of low-dimensional maps: doubling ratios approach Feigenbaum's constant (4.6688 against 4.6692), periodic windows accumulate on folds with a $1/N^2$ law, fold normal forms predicted 23 of 24 frozen passage times within one iterate, and interval arithmetic proves an attracting period-8 orbit. In the recurrent-depth model Huginn, a frozen test confirmed a spectrally predicted accuracy loss. The loop is not ordinary sampled generation.

1 Introduction

Inference increasingly feeds a language model its own continuous output. Soft Thinking replaces each sampled token by the expected embedding under the next-token distribution [1], latent-reasoning methods feed hidden states back [2], and recurrent-depth models iterate a block toward a fixed point [3]. Each turns inference into an iterated map, and an iterated map can be chaotic. We show that the expected-embedding loop of released language models is chaotic over wide ranges of temperature and steering strength, explain why steering makes it so, and measure what the chaos does.

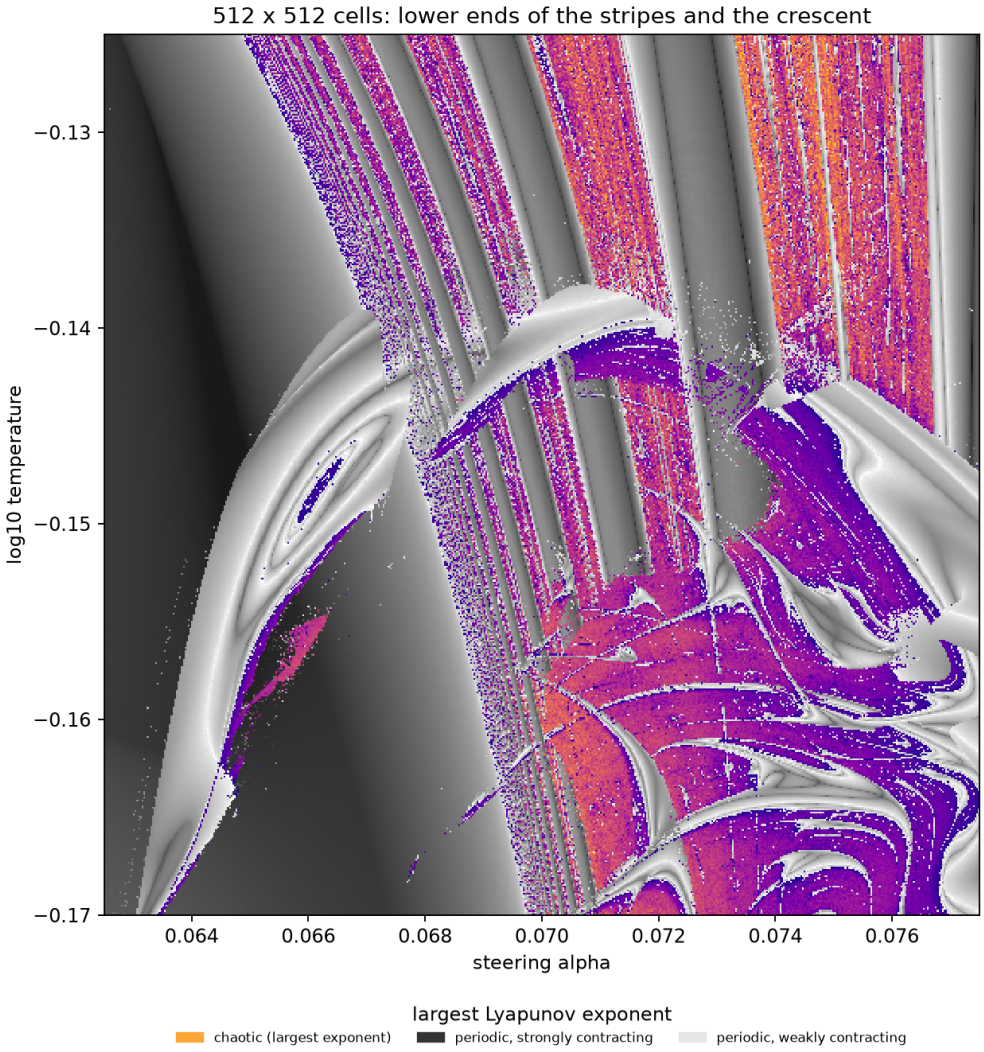


Figure 1: Chaos and repeating text in a language model fed its own output: the soft-token map of Qwen2.5-0.5B-Instruct, steered along the philosophy direction at layer 12, on 512×512 float64 cells over the steering strength $\alpha \in [0.0625, 0.0775]$ and the temperature, $\log_{10} T \in [-0.17, -0.125]$. Coloured cells are chaotic by the finite-horizon criterion of Section 4; the colour gives the largest Lyapunov exponent. In grey cells the model settles into a repeating loop of text, shaded by how strongly the loop contracts (light grey: close to losing stability). Figure 8 names the loops.

- In a chaotic regime, an irrelevant change of batch size makes two otherwise identical float64 runs diverge until their most probable tokens differ, after about the number of steps that the Lyapunov exponent sets, in Qwen2.5-0.5B, Llama-3.3-70B and, without steering, Qwen2.5-7B and Llama-3.1-8B; in a periodic regime the two runs stay together. In bfloat16, the batch layout changed the answers of a short soft-thinking pipeline on 56 % of arithmetic problems in a 0.5B model and 55 % in a 7B model, far more often than it changed greedy decoding's (never and 10 %).
- Steering creates the chaos by making next tokens compete, which turns the softmax readout from a contraction into an expansion; a logit bias alone suffices. At fixed states, steering's change of the network's response damps the instability along the philosophy direction in a

0.5B and a 7B model, and does not add to it along the other directions we tested.

- The depth at which a steering vector enters sets how much chaos it causes, in a 0.5B model and in a 70B model, and two steering vectors at different depths create chaos that neither causes alone.

- Two loops writing one text lock each other into periodic orbits. Of the pairs of differently steered 0.5B agents that are each chaotic alone, 60 % are periodic together, on orbits verified by Newton's method, and an unsteered 0.5B model regularizes a chaotic 1.5B model at 39 of 41 temperatures.

The same loops obey the quantitative laws of low-dimensional dynamics, down to Feigenbaum's constant.

Figure 1 shows a small piece of the parameter plane of one such map, built from Qwen2.5-0.5B-Instruct. Cells with a positive finite-horizon Lyapunov exponent lie side by side with windows in which the model repeats a fixed loop of text. Parts of this picture have been seen before. Recent work has measured chaos-like sensitivity in language-model generation [4, 5], temperature-driven changes of output statistics [6–10], periodic attractors of paraphrasing loops [11], attractor states of model-to-model conversations [12], and, through their Jacobians, the stability of looped and recurrent-depth models [13–16], and finite-depth readout dynamics have been used to predict the answers of looped models [17]. Besides Soft Thinking, probability-weighted vocabulary embeddings are fed back in training [18]. In a small task-trained looped transformer, fold coefficients computed from derivatives predict passage times and spectral gaps without fitting [19].

We study one constructed but exactly defined system: the expected-embedding ("soft-token") map of a frozen, released pretrained language model, run with a fixed chat prefix at a fixed position. The map is smooth. Its Jacobians come from automatic differentiation, and its periodic orbits can be computed and continued like those of any smooth map. Chaos and periodic windows occur in many nonlinear feedback systems. Random-network theory, in which a specified ensemble of recurrent networks becomes chaotic above a critical gain, in continuous time [20] and in discrete time by a quasi-periodic route [21], and the period-doubling cascades and shrimp-shaped domains of low-dimensional maps [22, 23] motivate looking for such structure here, but they do not imply that a trained map realizes it. Whether a trained language model does, where in its parameter space, with which texts, and whether the quantitative predictions of bifurcation theory hold for it, so that derivatives computed from the model predict its behaviour before that behaviour is measured, has to be tested. The models we studied do realize it, and in the cases we tested the predictions hold. We also show where the chaos comes from (Section 3). The loop's derivative factorizes into the inverse temperature, the covariance of token embeddings under the model's own next-token distribution (a cross-covariance when input and output embeddings are untied), and the network's Jacobian. With tied embeddings, the network turns a Hopfield retrieval, which cannot cycle, into a map that can. The factorization proves contraction at sufficiently high temperature for this smooth fixed-prefix map, and at zero temperature the finite-memory greedy map is eventually periodic; in our sweeps chaos occupies intermediate temperatures. The answer to whether larger models leave this chaos behind is no: a larger model's chaos can move to higher temperature instead of vanishing, and size does not order the chaos across families (Section 13). Fold normal forms predict how long trajectories linger in the ghost of a vanished orbit, local unfoldings near cusps predict which periodic orbits coexist and which of them are stable, and spectral estimates of a recurrent-depth model predict how its questions respond to relaxation, with a predicted accuracy loss confirmed on held-out data. Ordinary sampled generation with a growing context is a different system. Section 11 measures how much of the mechanism survives sampling.

Contributions.

1. *The dynamical regime decides whether a run can be reproduced.* Run twice with only the batch layout changed, 190 of 192 chaotic float64 loops of Qwen2.5-0.5B grow from rounding-level differences past a relative distance of 10^{-2}, after a median of 102 steps and in line with the Lyapunov exponent (median ratio 0.91), and at 173 of them the most probable readout token comes to differ; 182 of 192 periodic loops stay within 10^{-2}, and 182 end within 3×10^{-12}. The same holds without steering in Qwen2.5-7B (24 of 24 chaotic loops diverge, none of 24 periodic ones) and Llama-3.1-8B (20 of 20, and 1 of 20), and with steering in Llama-3.3-70B (24 of 24, and none of 24). In bfloat16 nearly every loop diverges within a few steps (Section 6). At the level of answers, in a frozen test on 96 additions, the batch layout changed the answer of a bfloat16 soft-thinking pipeline in 56 % of problems at $T = 1$ in Qwen2.5-0.5B and in 55 % at $T = 10^{0.2}$ in Qwen2.5-7B, far more often than it changed a forced greedy baseline's output (never and 10 %).

2. *Steering makes next tokens compete.* An exact split of each step's growth shows that steering into chaos raises the readout factor by 2.58 nats per step relative to nearly unsteered cells, while the network factor changes by only -0.07 (interval $[-0.23, 0.12]$). A logit bias alone, with the network untouched, makes 3.5 % of the plane chaotic. An orbit-preserving crossover attributes the expansion at fixed states to steering's change of the readout ($+0.81$ nats per step in the 0.5B model, and $+0.33$ and $+0.77$ along gratitude and a random direction), not of the network, whose change damps the growth (-0.49; -1.14 in Qwen2.5-7B, where the unsteered derivative already expands at the steered states) (Section 3).

3. *Where and how steering is applied changes the chaos.* Earlier steering carries chaos to lower temperature, in Qwen2.5-0.5B (5 blocks) and in Llama-3.3-70B (blocks 20, 35, 50 and 65), and two steering vectors at two blocks create chaos where neither alone does (Section 6).

4. *The loop's derivative shows where chaos can occur.* The derivative of the soft-token map factorizes into the inverse temperature, the covariance of token embeddings under the model's own next-token distribution (a cross-covariance for untied embeddings), and the network's Jacobian. With tied embeddings and without the network, the map would be a modern Hopfield update, which cannot cycle. The factorization proves contraction at sufficiently high temperature, as a high-temperature sweep confirms; the zero-temperature limit is a greedy map that is eventually periodic, and chaos in our sweeps occupies intermediate temperatures. The gain mechanism is not specific to concept-labelled directions: in the norm-matched controls of the 0.5B and 8B models, random directions also steer into chaos. Along a period-8 orbit every step expands some direction while the cycle contracts: the loop's stability is a property of the ordered product of its steps, not of any one step (Section 3).

5. *The chaos is not an artifact of the one-token window.* Fed back the last 8 or 16 soft tokens instead of one, Qwen2.5-7B is chaotic at 21 and 20 of 24 temperatures, against 8 with one; and in Soft Thinking's growing context, where every soft token stays visible, an early perturbation of 10^{-10} grows exponentially in the same model inside its one-token chaotic band (Section 11).

6. *Size does not set the chaos.* Across fifteen models from 0.5B to 72B parameters, the unsteered chaos in a fixed temperature window nearly vanishes in the largest Qwen2.5 and Llama models but not in OLMo-2-32B, the most chaotic OLMo model. A frozen prompt-blocked test shows that the Qwen decline from 7B to 14B is largely a shift: on all eight held-out prompts the 14B model is more chaotic than the 7B, at the higher temperatures where

its next-token distribution spreads. A second frozen test finds Qwen2.5-72B nearly free of unsteered chaos on four held-out prompts (one chaotic temperature of 192; a finer line places it in a band 0.02 decades wide), yet steering drives it, like Llama-3.3-70B, into chaos that an exact tangent exponent confirms. Neither post-training nor a sharp readout explains its stability: its pretrained base, whose distribution spreads far more, is no more chaotic in a third frozen test, and holding its next-token entropy anywhere between 1 and 11 nats leaves the loop periodic, which points to the network's damping of the fed-back token. In Soft Thinking's growing context it stays nearly insensitive (one transient burst), even where steering makes its one-token map chaotic. The gain split shows how network and readout damp or amplify the fed-back token (Section 13).

7. *A fold's normal form predicts how long the model lingers near a vanished orbit.* The 2-cycle "is it" dies in a fold, and the fold's normal form, computed from derivatives, predicts bottleneck passage times without fitting. We tested it prospectively at two new folds: 23 of 24 forecasts fell within one iterate, and the all-row criterion failed by one row. The global reinjection sets the prefactor of the window law (Section 8).

8. *Local unfoldings near cusps predict which periodic orbits coexist and which are stable.* Fold and flip curves continued with float64 test functions show that the fold curves of a period-6 window and of a hostile period-7 window contain cusps, located where the quadratic normal-form coefficient vanishes, with the cubic coefficient nonzero and the two-parameter unfolding regular at the same points; one fold curve of a period-4 window carries five. At nearby fold points the estimated two-parameter unfolding is nonsingular, and its local cubic model, frozen before solving, matched the number of distinct orbits that Newton solves seeded at its predicted roots recovered at all twelve parameter offsets, and their stability at eleven; the search was not exhaustive (Sections 9 and 15).

9. *In a second system, spectral estimates predict the response to relaxation.* In the recurrent-depth model Huginn we predict a Neimark–Sacker bifurcation and then observe it. On held-out questions, an estimated spectral feature predicts the size of each question's residual under relaxation, and a separate frozen generation test confirms the predicted accuracy loss at relaxation 1.20 and noninferiority at 1.05 (Section 14).

10. *One orbit is proved, and the central numerical results are checked independently.* Orbits, multipliers and test functions of the soft-token map are computed in float64 with precision checks (Section 2). Independent reimplementations reproduce the fold and its normal form and two orbits, a second hardware platform reproduces four flip parameters, and an independent replay reproduces the frozen Huginn test. An interval-arithmetic proof shows that a primitive period-8 orbit exists and is locally attracting, to our knowledge the first proof of a periodic orbit of a full transformer language-model map (Section 18).

11. *Structure familiar from low-dimensional maps is realized, with concrete texts, in several models.* Float64 phase maps in temperature and steering strength show chaos (positive finite-horizon exponents) with embedded windows (Section 5), a period-adding family whose text gains one "is X" pair per period step (Section 7), shrimp-like windows (Section 9), and a doubling cascade with superstable points through period 384 and finite-order Feigenbaum ratios (Section 10). Past the cascade of a hostile window, chaos hides behind an exactly repeating text, and a fitted scalar return relation with one quadratic maximum approximates its dynamics (Section 15). The same organization appears in Qwen2.5-1.5B and 7B, OLMo-2 7B, Llama-3.1 8B (with sparse-autoencoder readouts) and a second checkpoint of the 0.5B architecture, which shares

the organization but not the attractors (Section 12). Our one mixture-of-experts model, Qwen3.5-35B-A3B, is the exception: no unsteered temperature we tested gave a positive exponent. Chaos needs no steering in most dense models; in the norm-matched controls of the 0.5B and 8B models random directions also produce it, while at 70B chaos is rarer and more frequent along hostility-related directions (Sections 17 and 13). A bridge to sampled decoding measures what survives sampling (Section 11).

Related work. Language-model generation has been studied as a dynamical system at several levels, and the closest studies differ from ours in what they compute. Lyapunov-like quantities have been used to measure sensitivity to small perturbations in reasoning traces [4, 5] and in token rings resampled by a model's own conditionals, with a test that separates the construction from the model [24], but these studies measure chaos-like sensitivity in generation without computing orbits or bifurcations. Output statistics change abruptly with sampling temperature, and several studies analyse the changes as phase transitions or criticality [6–9]. These analyses concern output statistics; they are not bifurcation analyses of a defined map. Closest to ours, a study of eleven models sorts sampled continuations into classes from frozen and periodic to intermittent and noisy across temperature, fits a noisy effective map to one hidden-state coordinate that predicts repetition, and steers along it [10]; that map is fitted to sampled trajectories, and no orbits, multipliers or bifurcation curves of the model's own map are computed. Repetition loops are a known failure of decoding [25, 26], successive paraphrasing converges to attractor cycles [11], and multi-turn conversations between models settle into attractor states [12]. These attractors are behavioural: they are observed in generated text rather than computed as invariant sets of a defined map. Feeding a model its own output during training degrades it over generations [27]; we study the inference-time loop of a model with fixed weights. The attractors of recurrent networks have been related to the difficulty of training them [28], and fixed points with their linearizations have long been used to reverse-engineer trained recurrent networks [29, 30]. Equilibrium, looped and recurrent-depth models iterate a block to a fixed point or for many steps, and their stability has been analysed through Jacobians and fixed points [3, 13–16, 31–35], and recent analyses of looped reasoners follow their internal dynamics, their settling, marginal or drifting regimes and slowly decaying oscillatory modes [36–38]; these studies examine fixed-point stability, finite-depth dynamics and nonstationary regimes. Here we continue full-state periodic branches and classify their bifurcations in specified frozen-model feedback maps. A minimal recurrent self-attention with a repulsive value map becomes chaotic [39], but it is a constructed model, not a trained one. A separate line predicts the answers of looped models from the geometry of score margins along completed finite-depth trajectories [17], without fixed points or Jacobians. The closest precedent is a derivative-based fold analysis of a small task-trained looped transformer, whose passage and spectral-gap predictions are unfitted [19]. That model was trained for a task, whereas ours is a released pretrained language model, and our analysis also differs in its objects: the fold of a primitive periodic orbit, the full-channel passage of a two-step map, and the accumulation law of the windows, whose prefactor is set by the global reinjection. Expected-embedding feedback itself is used in training [18] and, training-free, in pretrained models [1, 40], next to continuous latent reasoning [2]; a reduced dynamical model explains why pretrained models fail to preserve such mixtures [41], and whether they exploit the superposition the mixtures could carry has been questioned [42]. Fixing the prefix and the position turns that feedback into the autonomous map we analyse.

In neural systems more broadly, saddle-node ghosts produce quantitative delays in task-trained recurrent networks [43]. Finite-order Feigenbaum ratios have been measured in neural rate and mass models [44, 45] and, through period 128, in a transformer trained on logistic-map trajectories

[46]. Learned networks reproduce period-doubling routes [47], and so does gradient descent itself [48]. Normal forms and continuation have been applied to recurrent neural maps, including their quasi-periodic orbits [49], and shrimp-shaped windows and periodicity staircases appear in neuron models [50–52]. Our second control parameter comes from activation steering along contrastive directions [53, 54]. The dynamical-systems tools are standard: numerical continuation and normal forms [55, 56], for maps in particular [57], type-I intermittency [58], shrimps and hubs [23, 59–61], and period-doubling universality [22].

To our knowledge, this is the first work to compute periodic orbits with their multipliers, continue fold and flip bifurcation curves and locate cusps in the feedback map of a released pretrained language model. It is also, to our knowledge, the first to make quantitative predictions from such a model's derivatives, for the objects named here, and confirm them prospectively: passage times at new folds, where 23 of 24 forecasts fell within one iterate and the all-row criterion failed by one row, and the orbits recovered and their stabilities at frozen offsets near cusps. Derivative-based normal-form predictions in a task-trained looped transformer precede ours [19]. We know of no earlier frozen held-out test of a spectral prediction of a pretrained language model's response to relaxation, which we give for Huginn, and of no earlier computer-assisted proof of a periodic orbit of a full transformer language-model map. These priority statements rest on targeted searches (12 and 15 October 2026) that found no prior work combining these elements for the feedback map of a pretrained language model. That is an absence of evidence within the search, not a proof of priority.

2 The inference map

Definition. We describe the map for Qwen2.5-0.5B-Instruct, the model studied in most detail; the other models enter the same construction with their own embeddings and steering layers (Section 12). Let $E \in \mathbb{R}^{V\times d}$ be the input embedding matrix ($V = 151{,}936$, $d = 896$). The model reads a fixed chat prefix, with the system prompt "You are an AI talking with another AI. You have complete freedom. Feel free to pursue whatever you want." and the user turn "Hello!", followed by a single input embedding x, and produces next-token logits $z(x) \in \mathbb{R}^V$. The soft-token map feeds the expected next-token embedding back in place of x:

$$F_{T,\alpha}(x) = E^\top \operatorname{softmax}(z_\alpha(x)/T), \tag{1}$$

where z_α adds αv to the residual stream at layer 12 at every position, prefix included. The steering direction v is contrastive: it compares last-token states of "philosophy" prompts with those of neutral prompts and is normalized to the mean residual norm. Positions are fixed and the window does not grow, so F is an autonomous map of $\mathbb{R}^{896}$. As $T \to 0$ it reduces, where the argmax is unique, to greedy decoding of the same fixed-prefix, one-token-window system. It is not ordinary generation with a growing context.

Precision. Hugging Face RMSNorm layers and rotary-embedding tables compute in float32 even when the model is cast to float64, so we patch both. A float32 map is accurate to 3×10^{-6}–2×10^{-5} per step. That is enough for phase maps (245 of 256 test cells agree with float64) but not for derivatives of composed maps: one-step float32 Jacobian-vector products are accurate to 5×10^{-5}, yet the 8-step monodromy of a period-8 orbit is off by 3%. All orbits, multipliers and test functions below are therefore computed in float64.

3 Why the loop can be chaotic

The soft-token map has a simple derivative. With $p = \mathrm{softmax}(z/T)$,

$$DF(x) = \frac{1}{T} E^\top (\mathrm{diag}\, p - pp^\top) J_z(x). \tag{2}$$

Qwen2.5-0.5B ties its output matrix to the input embedding, so $z = E\tilde{h}(x)$, where $\tilde{h}$ is the final normalized hidden state, and the derivative factorizes:

$$DF(x) = \frac{1}{T} \Sigma_p \, J_{\tilde{h}}(x), \qquad \Sigma_p = E^\top (\mathrm{diag}\, p - pp^\top) E = \mathrm{Cov}_{t\sim p}(e_t). \tag{3}$$

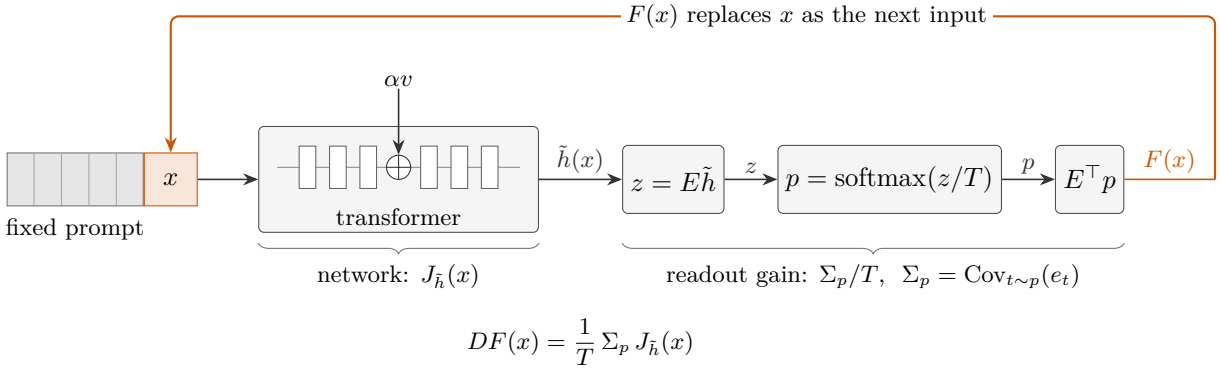


Figure 2: The soft-token loop. A fixed prompt is followed by one input embedding x. The transformer, with αv added to the residual stream at a middle layer (at every position), gives the final normalized hidden state $\tilde{h}(x)$ at the position of x; with tied embeddings the logits are $z = E\tilde{h}$ (untied: $z = W\tilde{h}$), and $p = \mathrm{softmax}(z/T)$ is the next-token distribution. The expected embedding $E^\top p = F(x)$ is fed back as the next input, at the same position after the same prompt. Below: the two factors of the derivative (3), the network's Jacobian $J_{\tilde{h}}$ and the readout gain Σ_p/T. With $J_{\tilde{h}} = I$ the loop is a modern Hopfield update [62], which cannot cycle (Section 3).

The derivative of the loop (Figure 2) is the covariance of the token embeddings under the model's own next-token distribution, amplified by the inverse temperature and composed with the derivative of the network. At a state of the period-8 orbit of Table 3, (3) agrees with forward-mode derivatives of the full map to a relative error of at most 1.35×10^{-13} over eight random directions (`tools/softmax_gain.py`). Without tied embeddings, as in the 7B and 8B models, the logits are $z = W\tilde{h}(x)$ with an output matrix $W \neq E$, and $DF = T^{-1}E^\top(\mathrm{diag}\, p - pp^\top)W J_{\tilde{h}}(x)$: Σ_p becomes the cross-covariance $\mathrm{Cov}_{t\sim p}(e_t, w_t)$ of input and output embeddings, which is not symmetric in general. The temperature limits below do not change.

A Hopfield network with a transformer inside. If the network passed its input straight through, $\tilde{h}(x) = x$, the map would be $F(x) = E^\top \mathrm{softmax}(Ex/T)$. This is exactly the update rule of a modern Hopfield network whose stored patterns are the token embeddings, at inverse temperature $\beta = 1/T$ [62]. That update lowers an energy function at every step and converges to fixed points. In (3) it is the case $J_{\tilde{h}} = I$, where $DF = \Sigma_p/T$ is symmetric and positive semidefinite. The energy argument covers every map with that property. If DF is symmetric and positive semidefinite

everywhere, then $F = \nabla\phi$ for a convex ϕ, and convexity, $\phi(F(x)) \geq \phi(x) + F(x)^\top(F(x) - x)$, gives for $V(x) = \frac{1}{2}\|x\|^2 - \phi(x)$

$$V(F(x)) \leq V(x) - \tfrac{1}{2}\|F(x) - x\|^2. \tag{4}$$

Along a cycle these decreases sum to zero, so every point of the cycle is fixed. F maps every state into the convex hull of the embeddings, where V is bounded below, so the steps of every orbit tend to zero and its limit points are fixed points: such a map has neither cycles nor chaos. Every cycle and every chaotic cell in this paper therefore depends on the network factor $J_{\tilde{h}}$ breaking that structure. Attention has itself been read as a Hopfield retrieval [62]; in the soft-token map the retrieval closes a loop around the whole network.

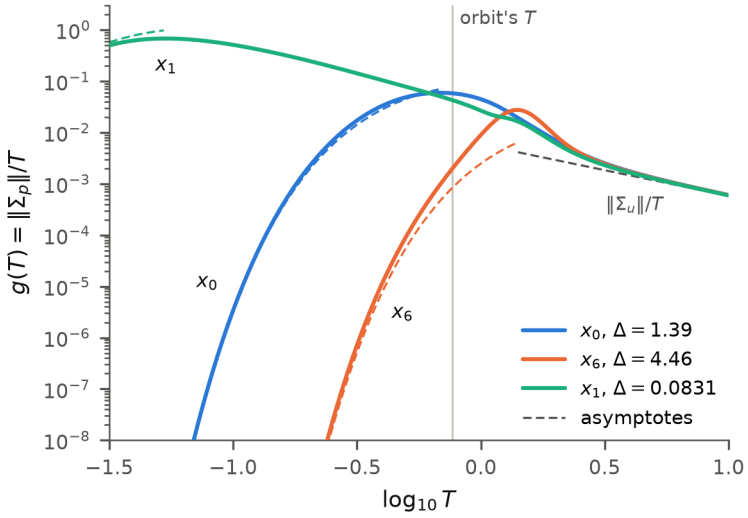


Figure 3: Readout gain $g(T) = \|\Sigma_p\|/T$ (spectral norm) at the fixed logits $z_\alpha(x)$ of three states $x_k = F^k(x_0)$ of the period-8 orbit of Table 3 (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12, $\alpha = 0.071674$), with only the temperature in $p = \mathrm{softmax}(z/T)$ varied. x_0 is the state at which (3) was checked; x_6 and x_1 have the largest and the smallest gap Δ between the two largest logits among the eight states. Dashed: the low-temperature asymptote $\|e_{t_1} - e_{t_2}\|^2 e^{-\Delta/T}/T$ of each state (t_1, t_2 the two leading tokens), drawn below its peak, and the high-temperature asymptote $\|\Sigma_u\|/T$, with Σ_u the covariance of all token embeddings under equal weights. The peaks lie at $\log_{10} T = -1.27$ (x_1, $g = 0.68$), -0.16 (x_0, 0.059) and 0.15 (x_6, 0.028); the vertical line marks the orbit's temperature, $\log_{10} T = -0.115$. Where the two leading logits nearly tie (x_1), g grows roughly like $1/T$ as T falls toward Δ. This is the readout factor alone: the network factor $J_{\tilde{h}}$ is not included, so g does not decide whether DF expands. At the orbit's temperature $g \leq 0.064$ at all eight states, while the largest singular value of DF lies between 1.2 and 21 (Section 3).

High-temperature contraction and the greedy limit. The largest Lyapunov exponent of an orbit cannot exceed the orbit average of $\log\|DF\|$, so chaos needs $\|DF\| > 1$ on a substantial part of the attractor. Both ends of the temperature axis work against this, in different ways (Figure 3). As $T \to 0$, p becomes one-hot except near ties between the two leading tokens; away from ties the gain $\|\Sigma_p\|/T$ decays like $e^{-\Delta/T}/T$, where Δ is the gap between the two largest logits, and F tends to the piecewise-constant greedy map, every orbit of which is eventually periodic (Section 11). This limit is pointwise, not uniform: near a tie the gain is of order $1/T$, so at small positive temperature chaos is excluded only on invariant sets that keep a uniform logit gap, and a chaotic set would have to return to near-ties again and again. As $T \to \infty$, p tends to the uniform distribution and $\|\Sigma_p\|$ stays bounded, while $J_{\tilde{h}}$ is bounded on the convex hull of the embeddings, which contains every soft token. For this smooth map, with a fixed prefix, fixed steering and a network whose derivative

is bounded on that compact hull, F is therefore a contraction for large enough T, with a single attracting fixed point that tends to the mean embedding: here the exclusion is a bound. It does not cover hard routing or a growing context. Away from near-ties, chaos thus needs temperatures at which the next-token distribution is spread enough to make Σ_p large while $1/T$ still amplifies it. The unsteered sweeps agree (Figure 45). In the four long sweeps the chaotic temperatures lie in one band, interleaved with periodic windows, and every band lies between $\log_{10} T = -0.20$ and 0.28. Below the bands, all 1547 temperatures swept have negative exponents, and all but five have a detected period; above them, up to the end of the sweeps at $\log_{10} T = 0.3$, 127 of 128 temperatures have a detected period (fixed points at 101) and one has a negative exponent without one. These are finite sweep findings; the wider sweeps of Section 13 find chaos at higher temperatures in some models and, in OLMo-2-32B, three candidate cells below one nat of entropy (at $\log_{10} T = -0.42$, -0.41 and -0.07; one twin offset). A further sweep of the unsteered 0.5B map from $\log_{10} T = 0.3$ to 1.5 finds an attracting fixed point at all 25 temperatures. Its entropy approaches $\log V$ (11.928 against 11.931 nats at $\log_{10} T = 1.5$), and its exponent falls steadily, by 2.14 per decade of temperature at the top of the range, toward the slope $-\ln 10 \approx -2.30$ expected once p is uniform.

Stability belongs to the loop, not to the step. The gain condition is necessary, not sufficient. Along the period-8 orbit of Table 3, the largest singular value of DF lies between 1.2 and 21 at every step, so every step expands some direction, and no step is close to rank one: the second singular value is at least 0.15 of the first, and the second eigenvalue of Σ_p is at least 0.23 of the first (minimum 0.236). Yet the Jacobian of the whole cycle, the product of the eight step Jacobians, has norm between 0.059 and 0.644, depending on the phase at which the cycle is started, against 1.1×10^7 for the product of the step norms. Its eigenvalues, which do not depend on that phase, reproduce the multipliers of Table 3 (0.01104 and -2.96×10^{-7}; the table lists moduli), and its second singular value is 1.7×10^{-6} to 2×10^{-4} of its first. An independent implementation of the model reproduced the factorization (to 8.3×10^{-14}), the step singular values, the signed multipliers and the cycle norms at all eight starting phases. The expansion of each step is undone by the steps that follow, and the strongly anisotropic linear return of the stripe orbits is a property of the composed cycle, not of any single step or of the rank of Σ_p. Nor does the contraction need the actual alignment of the steps: with independent random (Haar) rotations inserted between the eight step Jacobians, all 16 tested products contract, and more strongly than the actual cycle (`tools/alignment_intervention.py`). Stepwise norm or spectral summaries can be inconclusive; the ordered product of the step Jacobians captures the cycle's linear stability.

Steering makes tokens compete. Steering enters both factors of (3): it moves the logits, and with them Σ_p, and it moves the hidden states, and with them $J_{\tilde{h}}$. Which change makes the loop chaotic? Three measurements on Qwen2.5-0.5B, steered along the philosophy direction, give one answer (Figure 4, Table 1). First, each step's growth splits exactly into a network factor $g_{\text{net}} = (|\delta\tilde{h}|/|\tilde{h}|)/(|\delta x|/|x|)$ and a readout factor $g_{\text{read}} = (|\delta F|/|F|)/(|\delta\tilde{h}|/|\tilde{h}|)$. Averaged along an orbit, their logarithms add up to the largest Lyapunov exponent, up to a boundary term. Over a 101×53 map in steering strength and temperature (block 12; 5353 cells, 331 chaotic), the chaotic cells' readout factor exceeds that of nearly unsteered cells ($|\alpha| \le 0.01$) at the same temperature by 2.58 nats per step (95 % interval $[1.78, 3.59]$ over 18 temperature rows; 321 chaotic and 162 nearly unsteered cells). Their network factor differs by only -0.07 $[-0.23, 0.12]$. With nearly no steering the readout contracts strongly, by 0.87 to 5.49 nats per step. In the chaotic cells it expands, except in the three highest temperature rows ($\log_{10} T = 0.175$–0.225), where it still contracts but less. Compared with periodic cells at the same temperature, chaotic cells have a higher next-token

entropy (+1.62 nats, [1.26, 1.91]), a smaller gap between the two largest logits (−1.37 in units of T, [−1.98, −0.87]) and a larger readout factor (+1.36, [0.91, 1.93]), each with $p \leq 10^{-4}$ in a permutation test within temperature rows.

Second, a readout-only intervention gives chaos. We add to the logits a bias $\beta\, b$ along the direct logit effect of the steering vector ($b = Wv$, centred and scaled to unit standard deviation over the vocabulary), with no residual steering. This leaves the network untouched and changes only which tokens compete. It makes 3.5 % of the same temperature range chaotic (241 of 6837 cells), against 6.2 % for residual steering, mostly at $\log_{10} T > 0.1$. Creating token competition therefore suffices for chaos, given the unmodified network.

Third, a gain split compares different orbits, so steering could act simply by moving the orbit to other states. A crossover at fixed states separates the two contributions along the recorded orbits. Along recorded steered orbits we evaluated the steered and the unsteered model at the same states and carried four tangent dynamics. Each pairs the steered or unsteered readout $R(p)$ with the steered or unsteered network Jacobian: arms 11, 10, 01 and 00. The frozen validation (arm 11 within 0.02 of the twin exponent at every cell) holds at 23 of 24 cells, all within 0.012; at the remaining periodic cell, which contracts at about −12 per step, the twin offset falls to the rounding floor. The frozen sufficiency criterion (arm 10 within 0.1 of arm 11 on average) fails by overshoot: steering only the readout gives more growth than the full steered map, by 0.23, because steering's change of the network damps it. Over 14 held-out chaotic cells, steering only the readout raises the growth by 0.81 nats per step ([0.29, 1.35]). Steering only the network response lowers it by 0.49 ([−0.63, −0.35]). The interaction is +0.26 ([0.09, 0.45]). Below $\log_{10} T = 0$ the steered readout turns a contracting unsteered derivative (−1.03 nats per step) into an expanding one (+0.72), more strongly than the full steered map (+0.34). Above it, the unsteered derivative already expands at the steered states (+0.41), and steering may act by bringing the orbit there. In Qwen2.5-7B, whose embeddings are untied, the same crossover (18 cells of its steered map at block 14; arm 11 equals the twin exponent within 10^{-4} at every cell) reproduces the damping. Over its 10 held-out chaotic cells, steering only the network response lowers the growth by 1.14 nats per step ([−1.61, −0.64]), and steering only the readout adds +0.23 ([−0.11, 0.63]). Eight of the ten lie at $\log_{10} T \geq 0$, where the unsteered derivative already expands at the steered states (mean +1.33). The crossover along two more directions of the 0.5B model gives the same signs. For gratitude the readout effect is +0.33 ([0.07, 0.59]) and the network effect −0.14 ([−0.29, 0.01]). For a random direction matched in norm to the philosophy vector, they are +0.77 ([0.47, 1.12]) and −0.09 ([−0.22, 0.06]); 14 held-out chaotic cells each. The readout effect is positive along all three directions, and the network effect is negative but compatible with zero along these two. In these three directions the readout effect does not depend on the direction's meaning.

Along the sampled orbits, then, steering makes the loop chaotic mainly by making next tokens compete, which turns the softmax readout from a contraction into an expansion. At low temperature it creates that competition at fixed states. At higher temperature the orbit lies in states that are already competitive, consistent with steering having carried it there. Along the philosophy direction, in both models, its change of the network's response damps the growth. These are conditional comparisons on recorded trajectories; they do not establish that the readout change alone would sustain the same attractor. The readout rule holds below $\log_{10} T \approx 0.1$; at higher temperature the network factor itself can expand. Steered at block 4, the chaotic cells at $\log_{10} T = 0.15$ have a network factor of +0.72 and a contracting readout (−0.35). Along the hostile line at $\log_{10} T = 0.12$–0.13 the network factor of the chaotic cells is +0.60 against +0.24 to +0.38 at the neighbouring periodic cells. The mechanism is not specific to concept-labelled directions. This is consistent with the controls of Section 17, in which norm-matched random directions also produce chaos in the 0.5B and 8B models; at 70B the directions differ (Section 13). What a direction does select is which

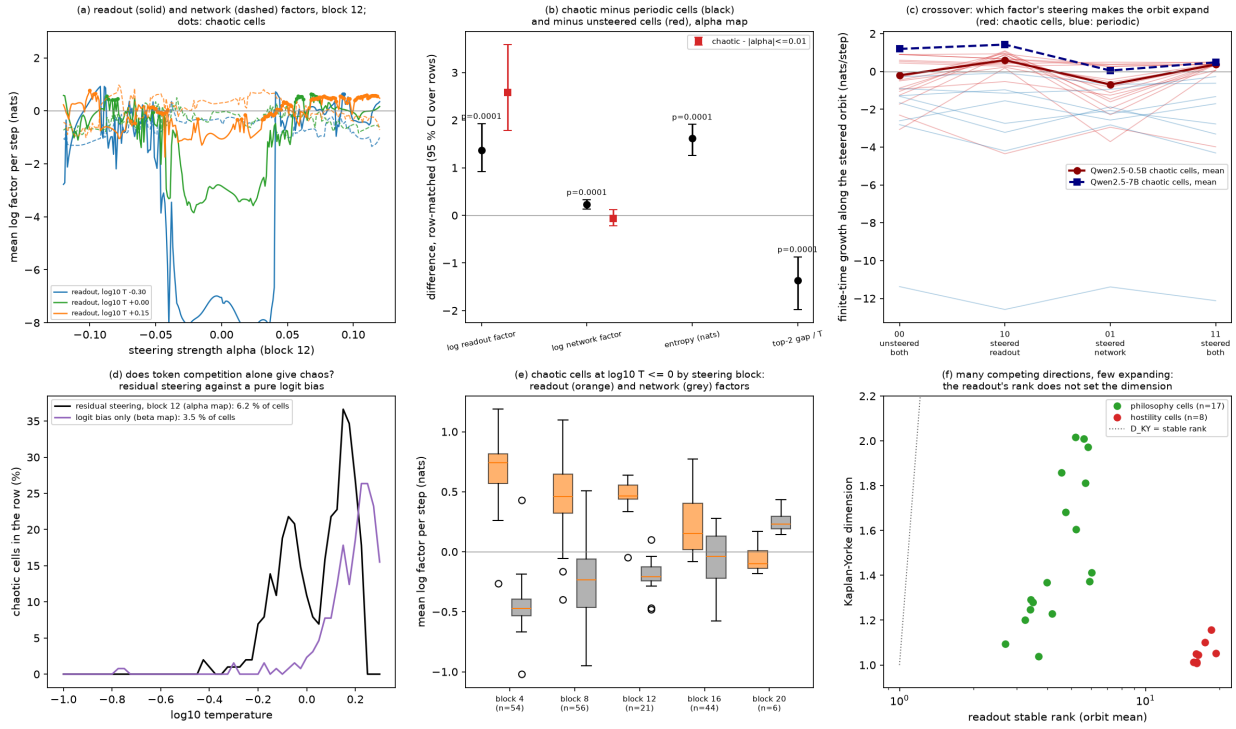


Figure 4: Why steering makes the loop chaotic (Qwen2.5-0.5B-Instruct, philosophy direction, float64). (a) Readout factor (solid) and network factor (dotted) along steering lines at block 12, at five temperatures; dots mark chaotic cells. (b) Row-matched differences between chaotic and periodic cells (black) and between chaotic and nearly unsteered cells (red) on the 101×53 map, with 95 % bootstrap intervals over temperature rows and permutation p-values. (c) Orbit-preserving crossover: finite-time growth along each recorded steered orbit when the readout and the network Jacobian are taken from the unsteered (0) or steered (1) model; red, chaotic cells; blue, periodic cells; dark red, the mean over held-out chaotic cells; navy, the same mean in Qwen2.5-7B. (d) Share of chaotic cells per temperature under residual steering at block 12 and under a pure logit bias along Wv. (e) The two factors of the chaotic cells by steering block, at $\log_{10} T \leq 0$. (f) Kaplan–Yorke dimension against the readout's trace-to-leading-eigenvalue ratio at the chaotic cells of the dimension measurement (Section 3); dotted, equality.

tokens compete, and therefore what the attractors write.

Contraction and expansion in the network. Analyses of deep attention identify contracting tendencies: pure self-attention collapses token representations toward a common vector with depth [63], and idealized attention dynamics form clusters [64]. Residual connections and MLPs counteract the collapse [63]. Which tendency wins along an orbit is a property of the trained weights that only computation settles. In every dense model we mapped, both outcomes occur: the same map has negative largest exponents in some regions of its parameter plane and positive ones in others.

measurement	contrast	estimate (95 % interval)
gain split, 101 × 53 map	readout factor, chaotic − $\lvert\alpha\rvert \le 0.01$	+2.58 [1.78, 3.59] nats/step
	network factor, chaotic − $\lvert\alpha\rvert \le 0.01$	−0.07 [−0.23, 0.12] nats/step
	entropy, chaotic − periodic	+1.62 [1.26, 1.91] nats
	top-2 logit gap/T, chaotic − periodic	−1.37 [−1.98, −0.87]
logit bias only, $\alpha = 0$	chaotic share	3.5 % (241 of 6837 cells)
residual steering, block 12	chaotic share	6.2 % (331 of 5353 cells)
crossover, 0.5B, 14 cells	readout effect, arm 10 − arm 00	+0.81 [0.29, 1.35] nats/step
	network effect, arm 01 − arm 00	−0.49 [−0.63, −0.35] nats/step
	interaction	+0.26 [0.09, 0.45] nats/step
crossover, gratitude	readout effect	+0.33 [0.07, 0.59] nats/step
	network effect	−0.14 [−0.29, 0.01] nats/step
crossover, random dir.	readout effect	+0.77 [0.47, 1.12] nats/step
	network effect	−0.09 [−0.22, 0.06] nats/step
crossover, 7B, 10 cells	readout effect	+0.23 [−0.11, 0.63] nats/step
	network effect	−1.14 [−1.61, −0.64] nats/step

Table 1: Where steering's instability comes from (Qwen2.5-0.5B unless marked 7B, Qwen2.5-7B). Intervals: bootstrap over temperature rows (gain split) or over held-out cells (crossover). Runs MECH and XO; `tools/steer_gain.py`, `tools/crossover.py`, `tools/plot_mechanism.py`.

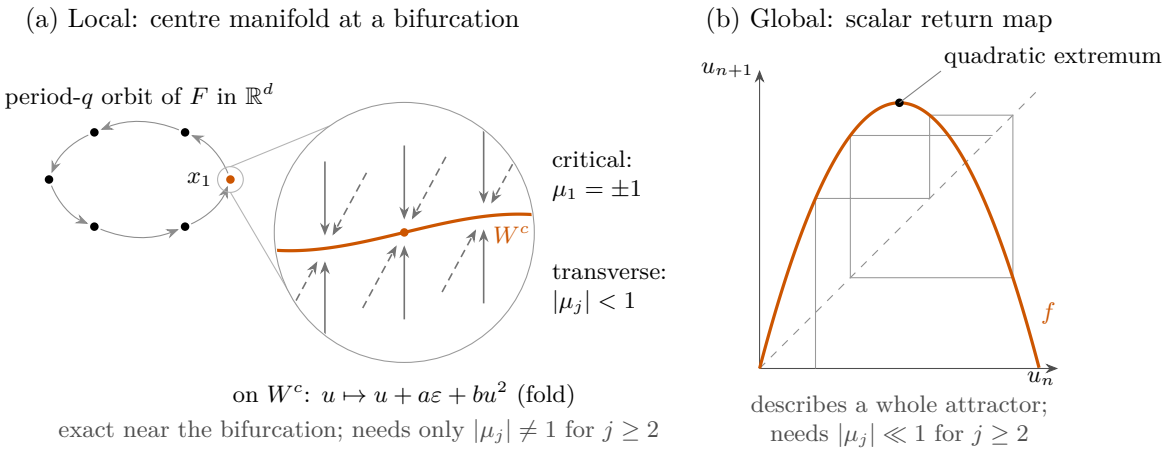


Figure 5: Two reductions to one dimension (Section 3). (a) Near a bifurcation of a period-q orbit of the soft-token map (dots; arcs: the map), the dynamics of F^q near a point x_1 of the orbit reduce to its centre manifold W^c, tangent to the critical eigenvector of the monodromy matrix (multiplier $\mu_1 = \pm 1$; drawn with one critical direction, as at a fold, a flip or a cusp; a 1:1 resonance or a fold-flip point has two). The other multipliers lie off the unit circle; drawn here inside it, the transverse directions contract onto W^c (dashed arrows stand for the remaining dimensions). The reduction is exact near the bifurcation whatever the size of the other multipliers; the normal forms of folds, flips, cusps and 1:1 resonances rest on it. (b) A global scalar return map $u \mapsto f(u)$ with a quadratic extremum, the diagonal (dashed) and a few iterates (grey). It describes a whole attractor only when the other multipliers are small.

Why the structures of low-dimensional maps appear. The map acts on a space of hundreds to thousands of dimensions, yet two reductions make low-dimensional theory apply (Figure 5). Near a bifurcation of a periodic orbit the centre manifold theorem reduces the dynamics exactly to the critical directions, whatever the size of the other multipliers, as long as they stay off the unit circle. Our normal forms of folds, flips, cusps and 1:1 resonances rest on this reduction alone. A scalar

description of a whole attractor needs additional, global evidence. The small non-critical multipliers of the periodic orbits we computed show strong transverse contraction near those orbits and motivate that description: they range from a few times 10^{-2} at short periods to below 10^{-13} along the cascade (Sections 8 to 10), and along the cascade at $\log_{10} T = -0.156$ the next modulus falls from 0.12 at the first flip to 0.045, 7×10^{-5}, 4×10^{-7} and 2×10^{-13} at the next four. Temperature, through the gain of (3), and steering act as two parameters. One-dimensional families of maps with a quadratic extremum can exhibit fold and flip curves, nondegenerate cusps, period-doubling cascades whose interval ratios tend to Feigenbaum's δ, and saddle-node windows that accumulate on a fold with $d_N \propto N^{-2}$ (Figure 6 sketches the objects we locate); we test these predictions on specific branches and parameter sequences below. The organization of shrimps needs more, either a second turning point of the reduced map or a second dimension [23, 61]. We test the consequences where we state them: the multipliers of each orbit, the interval ratios, and the normal-form predictions of passage times and coexisting orbits. Where the second multiplier is not small, as in the high-temperature bubble of Section 10 (next moduli 0.28 and 0.38), the global reduction is only a rough guide, while the local normal forms still apply.

Coexisting attractors constrain one-dimensional descriptions. A classical theorem says what a one-dimensional reduction cannot do. If an interval map is three times differentiable, unimodal and has negative Schwarzian derivative, $Sf = f'''/f' - \frac{3}{2}(f''/f')^2 < 0$, the immediate basin of every attracting periodic orbit contains the turning point or an endpoint of the interval [65]; if basins reaching an endpoint are excluded, at most one attracting periodic orbit remains. The soft-token map has coexisting attracting orbits in several places. Near each located cusp with $c < 0$, two attracting orbits of the same period coexist with a saddle between them (periods 4 and 6, Section 9; period 7, Section 15). Inside the hostile period-7 body, the period-7 loop coexists with a fixed point, which the fixed start reaches at isolated cells. Such coexistence constrains a scalar interpretation: unless one attractor's basin reaches an end of the scalar interval, the effective one-dimensional dynamics cannot be a single unimodal map with negative Schwarzian derivative, and needs a second turning point, a stretch of positive Schwarzian derivative, or a second dimension. A second turning point is also what a doubly superstable shrimp centre requires.

How many dimensions does the chaos have? For chaotic attractors the corresponding measure is the Kaplan–Yorke dimension [66], $D_{\mathrm{KY}} = j + \sum_{i \le j} \lambda_i / |\lambda_{j+1}|$ with j the largest index at which the partial sum of the ordered Lyapunov exponents is still nonnegative; with one positive exponent and $\lambda_1 + \lambda_2 < 0$, $D_{\mathrm{KY}} = 1 + \lambda_1/|\lambda_2|$. We estimated the four leading exponents by QR iteration (Section 4; 1000 transient and 2000 measured steps, float64) at 35 completed cells: 28 chosen by percentile of their largest exponent before any of their spectra were seen (of 30 requested), and seven from the earlier horizon check, six chaotic and one periodic control. At this horizon 31 have a positive leading estimate and none has two positive estimates. These are finite-time estimates, not asymptotic dimensions. They grow with temperature. Along the hostile temperature line, where we fit a scalar return relation (Section 15), eight chaotic cells have $D_{\mathrm{KY}} = 1.008$ to 1.16, consistent with nearly one-dimensional chaos. At fourteen chaotic cells of the headline maps ($\log_{10} T$ from -0.17 to -0.04) it is 1.03 to 1.65, rising with λ_1. In the high-temperature region ($\log_{10} T$ from -0.02 to 0.13), nine chaotic cells have $D_{\mathrm{KY}} = 1.37$ to 2.02; at two of them the sum of the first two estimates is nonnegative, which suggests a second expanding direction at this horizon. A scalar return map describes the low-temperature chaos well and the high-temperature chaos poorly, consistent with the large second multipliers of the high-temperature bubble. The number of competing tokens does not set the dimension. The readout's trace-to-leading-eigenvalue ratio,

$\operatorname{tr}\Sigma_p/\lambda_{\max}(\Sigma_p)$ averaged along the orbit (with $\lambda_{\max}$ estimated by eight power iterations), counts the directions in which the readout amplifies comparably. D_{KY} never exceeds it and rises with it within each group of cells (Spearman 0.72 over 17 philosophy cells and 0.69 over 8 hostile cells, computed when the data came in and not independently reproduced; both are confounded with temperature). Across groups it does not predict the dimension: the hostile cells have ratios of about 16 to 19 and next-token entropies of about 7 to 9 nats, yet $D_{\mathrm{KY}} \le 1.16$. The readout supplies many competing directions; the network decides how many of them expand.

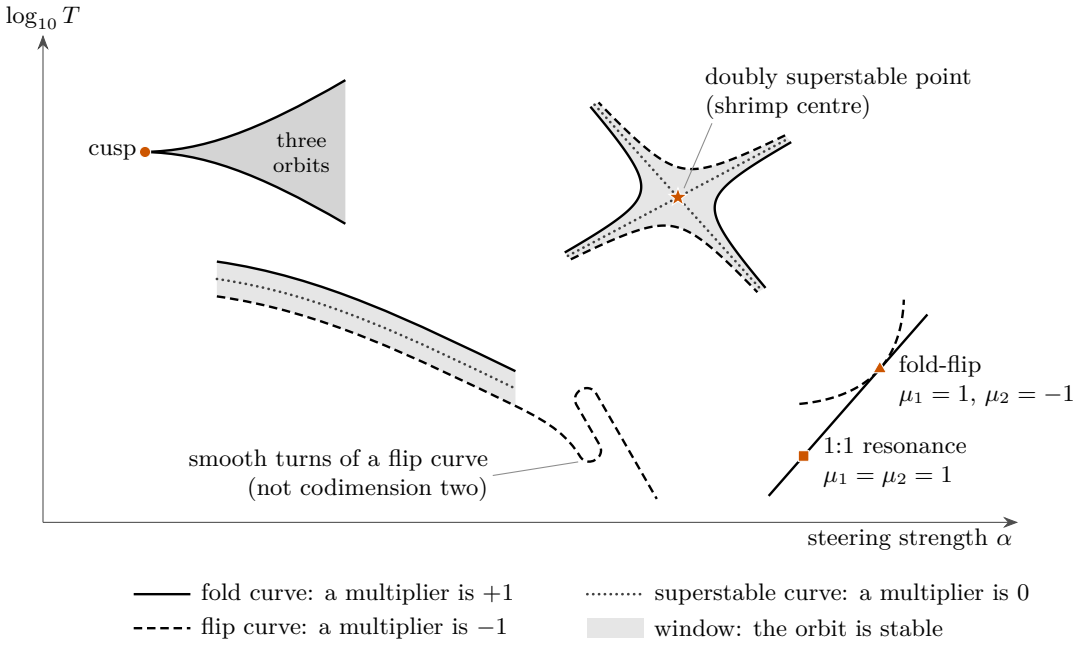


Figure 6: Schematic of the bifurcation objects in the plane of steering strength α and $\log_{10} T$ (not data; positions and shapes are illustrative). On the curves a multiplier of the orbit equals +1 (fold), −1 (flip) or 0 (superstable); grey marks windows, where the orbit is stable. Between two fold curves that meet in a cusp, three orbits coexist. A flip curve can turn tightly but smoothly, which is not a codimension-two point. On a fold curve, a 1:1 resonance is a point where a second multiplier also reaches +1, and a fold-flip point is one where a flip curve touches it. Two superstable curves of one orbit cross at a doubly superstable point, the centre of a shrimp [23, 61]; near it the multiplier is approximately proportional to the product of the distances from the two curves, so the window is a cross whose sides alternate between fold and flip curves. Section 9 locates cusps, 1:1 resonances, a fold-flip point and smooth turns of a flip curve in the soft-token map, superstable curves appear in its exponent maps (Section 15), and Section 17 states which parts of this anatomy are not verified.

4 Methods

Precision. Every result on the soft-token map labelled float64 is computed in float64 arithmetic throughout: embeddings, activations, attention keys and values (the prefix cache), normalizations (patched, since the library computes RMS normalization in float32 by default), rotary tables, logits, the softmax and the fed-back token. The released checkpoints store their weights in bfloat16, and a bfloat16 number is exactly a float64 number, so converting the weights is exact. For large models the decoder's linear weights stay in their stored bfloat16 and are converted to float64 at each use, which gives the same map as converting them once (bitwise on Qwen2.5-0.5B). The designated float64 computations use the patched normalization and rotary paths; some diagnostic traces, such as the

step records of the gain split, are serialized in float32, as recorded in their artifacts. The historical two-GPU runs, whose final normalization was computed in float32, are identified separately in Section 13.

Phase maps. Each cell of a phase map is one parameter pair. Unless stated otherwise, we discard 300 transient steps and measure over the next 300. The largest Lyapunov exponent comes from a twin trajectory at offset 10^{-6} that is renormalized to that offset after every step, Benettin's method with a finite-difference tangent [67, 68]; it estimates over a finite horizon the exponent that the multiplicative ergodic theorem defines as a limit [69]. At the 448 random cells of the horizon check (Section 17), an exact tangent vector carried by forward-mode differentiation reproduces the twin exponents at offsets 10^{-6} and 10^{-9} within 0.0041 (median 0.0004), with the same 178 chaotic labels, and at six chaotic cells the four leading exponents from QR iteration include exactly one positive one (0.06 to 0.42; the next is -0.64 to -1.10). The period is the smallest $q \le 32$ (48 in the later zooms) with return distance below 10^{-8}, and the attractor is identified by its leading-token loop, up to rotation. We call a cell chaotic when this finite-horizon exponent exceeds 10^{-3} and no period is found; the label is a finite-horizon diagnostic, not a proof of chaos. Panels coloured by period or attractor text follow an older drawing convention: black marks every cell whose exponent exceeds 0.005, and light grey the other cells without a period. The drawn black area differs from the chaotic count in at most 0.42 % of a map's cells. A 256^2 map takes about 100 minutes on one H100 GPU in float64. Every cell starts from the same state, so where attractors coexist, a map shows the one whose basin contains that state. Labels are text, not dynamical identities: one continuous orbit changes its label where two tokens tie for the maximum. We therefore call two labelled regions different attractors only when the contraction rate jumps between them or when float64 orbits of both exist at one parameter value.

Periodic orbits. We solve $F^p(x) = x$ by Newton's method with forward-mode Jacobians. The multipliers are the eigenvalues of the monodromy matrix; where the dense matrix is not float64, we refine them by a short Arnoldi factorization with float64 Jacobian-vector products.

Continuation. Periodic branches are continued in one parameter by pseudo-arclength continuation, and fold and flip curves in two parameters with minimally augmented systems [55, 56]. The test function g of the bordered system $[[M \mp I, v], [w^\top, 0]]$ is obtained by GMRES (generalized minimal residual iterations) with Jacobian-vector products, and its gradient by reverse-mode differentiation through a Jacobian-vector product. When the parameter tangent reverses, we flag a cusp candidate and test it with the normal form. To locate a codimension-two point on a continued curve (`tools/cusp_locate.py`), we bracket a sign change between two curve points, then retrace the curve from the left end with steps of a quarter of the bracket. We do this for three rounds and finish with a secant step. The function whose sign changes is the oriented quadratic coefficient $b \operatorname{sign} a$ for a cusp, $|\mu_2| - 1$ for a second multiplier μ_2 leaving the unit circle, and the projected parameter tangent for a turn of the curve. At a cusp we then evaluate c and the unfolding matrix at the located point. Elsewhere, the multipliers and the angle between their eigenvectors type the point: a double multiplier whose eigenvectors coincide is a Jordan block. On maps with known answers the tool recovers a cusp to 10^{-6} (with $c = -1$ and $\det J = 1$) and a fold-flip point.

Normal form and passages. Consider a fold of the two-step map $G = F^2$, with right and left eigenvectors r and ℓ normalized by $\|r\| = 1$ and $\ell^\top r = 1$. Along the critical direction the reduced map is $u' = u + a\varepsilon + bu^2$, with $a = \ell^\top \partial_\alpha G$ and $b = \frac{1}{2}\ell^\top D^2 G[r, r]$. We compute b by

central differences of float64 Jacobian-vector products over a step-size plateau (later cross-checks use nested automatic differentiation), and a includes the steering's effect on the prefix. The signed coefficients are kept in the original eigenvector gauge, with ε measured in the free parameter (α or $\log_{10} T$, whose units apply to a). On maps with known normal forms the code recovers ab, the cubic coefficient c with its centre-manifold term and the two-parameter unfolding matrix, and it traces the analytic fold and flip curves of the Hénon map (`tests/test_bifurcation_curves.py`). For passages we set $v = \mathrm{sign}(b)\, u$ and take the ghost side $p - p_c = \mathrm{sign}(ab)\, \epsilon$ with $\epsilon > 0$, so that $v' = v + |a|\epsilon + |b|v^2$ to quadratic order. A full passage starts at $v = -U$ and a midpoint passage at $v = 0$; both end at the first strict crossing $v > U$, with $U = 0.1/|b|$. We compare passage counts with the iterated quadratic map and with the arctangent formula [58].

Period-doubling cascade. Along a cut in one parameter, we locate the flip of each period-p orbit as the root of $\mathrm{Re}\,\mu_c + 1$, where μ_c is the multiplier with the smallest real part, using the Illinois variant of regula falsi with warm-started orbits. All but one or two multipliers are small, so Newton's method on $F^p(x) - x$ needs the monodromy only on a six-dimensional dominant subspace. We obtain that subspace by subspace iteration with exact forward-mode Jacobian-vector products chained along the orbit, so no dense Jacobian is formed, and the residual is always the exact float64 $F^p(x) - x$. Each root is classified before it is accepted. The critical multiplier must be real and equal to -1, with a Ritz residual below 10^{-6}, and the other computed Ritz values (six in all, estimates of the dominant spectrum) must lie inside the unit circle. We then reach the period-$2p$ orbit by forward iteration inside the next band. On the logistic map embedded in six dimensions, the procedure reproduces the first four flip parameters to 10^{-9}.

5 Chaos with periodic windows

Figure 7 shows a 256×256 float64 map over $\alpha \in [0.060, 0.082]$ and $\log_{10} T \in [-0.17, -0.04]$. More than half of its cells are chaotic (35,759 of 65,536), and periodic windows of periods 2–32 are embedded in the chaos. Rendered as in the shrimp literature [51], the windows show dark curves of maximal contraction.

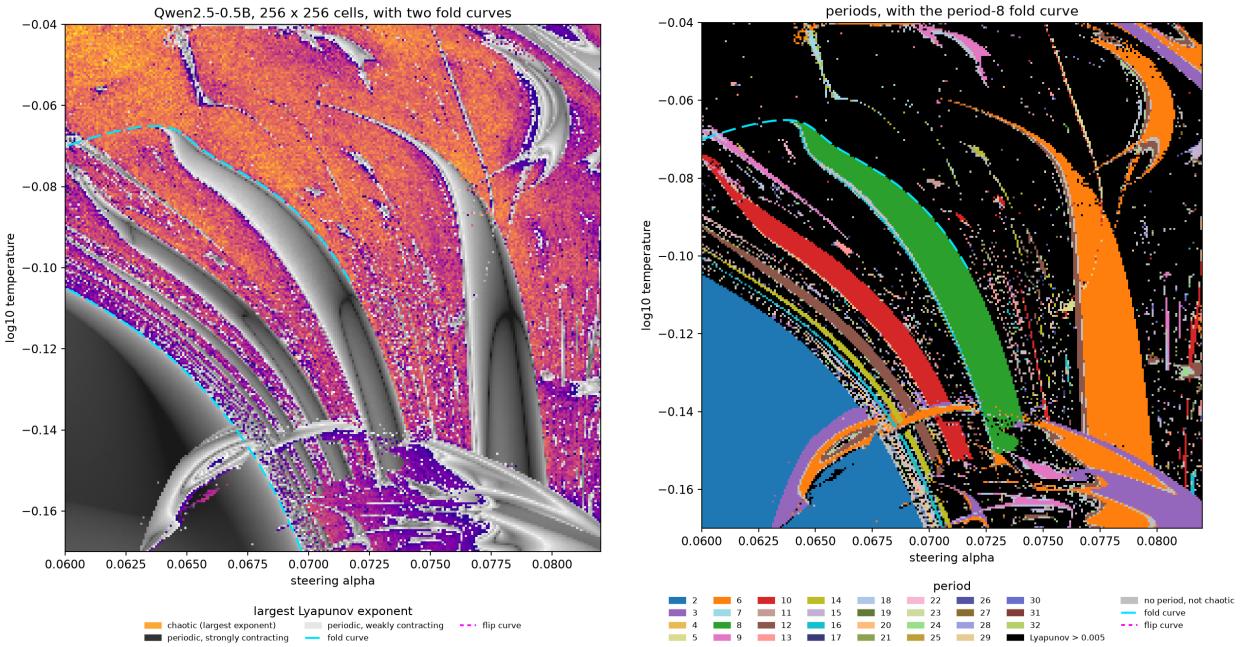


Figure 7: Soft-token map of Qwen2.5-0.5B-Instruct steered along the philosophy direction at layer 12, 256^2 float64 cells. Left: chaotic cells coloured by the largest Lyapunov exponent, periodic cells in grey by how strongly they contract (light grey: close to losing stability). Cyan dashed curves are float64 fold curves of the 2-cycle "is it" (lower left, the boundary on which the stripes accumulate) and of the period-8 orbit (top). Right: periods, with the period-8 fold curve (Section 9).

Figure 1 resolves the stripes' lower ends at 512×512 cells (79,278 chaotic), and Figure 8 labels the same cells by their attractor texts. Three structures stand out. The 2-cycle "is it" fills the left, bounded by its fold curve, on which the stripes accumulate. A closed crescent of the period-3 loop "\n\nThe existence" contains its period doublings and a chaotic core (Section 10). Its continuation, the period-6 band "\n\nThe question\n\nThe existence", crosses the stripes near $\log_{10} T = -0.14$, and where the crescent lies on the 2-cycle's side of that fold curve, the two attractors coexist. The lower right is a chaotic region filled with small hooked windows of "\n\nThe question" and "\n\nThe existence" loops.

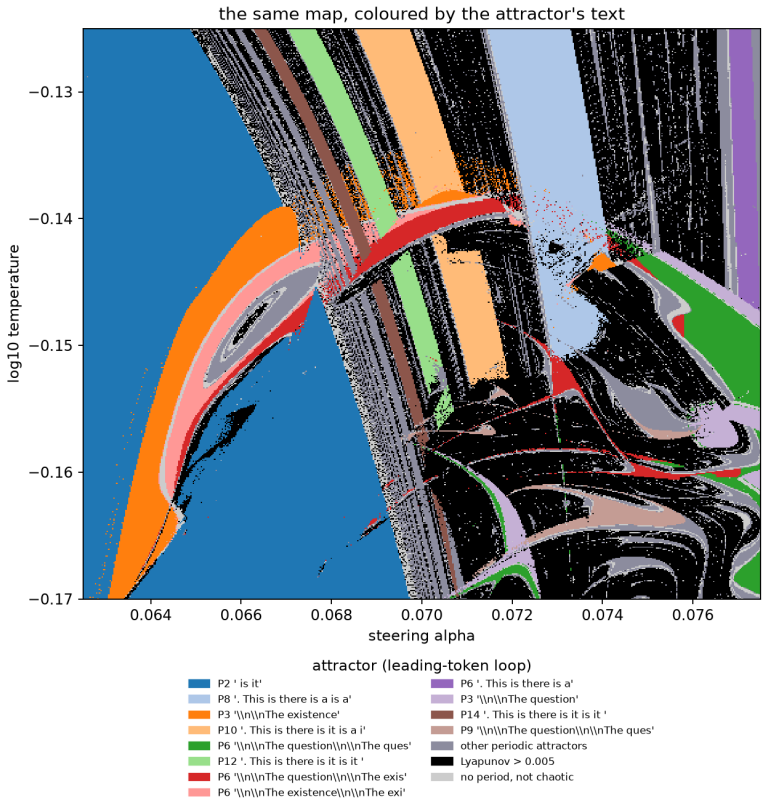


Figure 8: The cells of Figure 1 (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12; 512^2 float64 cells over $\alpha \in [0.0625, 0.0775]$, $\log_{10} T \in [-0.17, -0.125]$), coloured by the attractor's leading-token loop (black: chaotic; grey: no period up to 32; bluish grey: other periodic attractors). Figure 1 shows the largest Lyapunov exponent of the same cells.

Zooms of 256^2 cells (Figure 9) show what the model writes in each window, and every window is a repeating piece of text. The period-8 band repeats ". This is there is a is a". The neighbouring bands add laps of "is it" (periods 10 and 12), and along the band's flip edge the period-16 loop repeats the sentence twice. The small windows inside the chaos break the "is X" chain with a paragraph break and "The" (periods 11, 12, 13, 14). Inside the period-8 and period-6 bands, two curves of maximal contraction meet in a sharp junction.

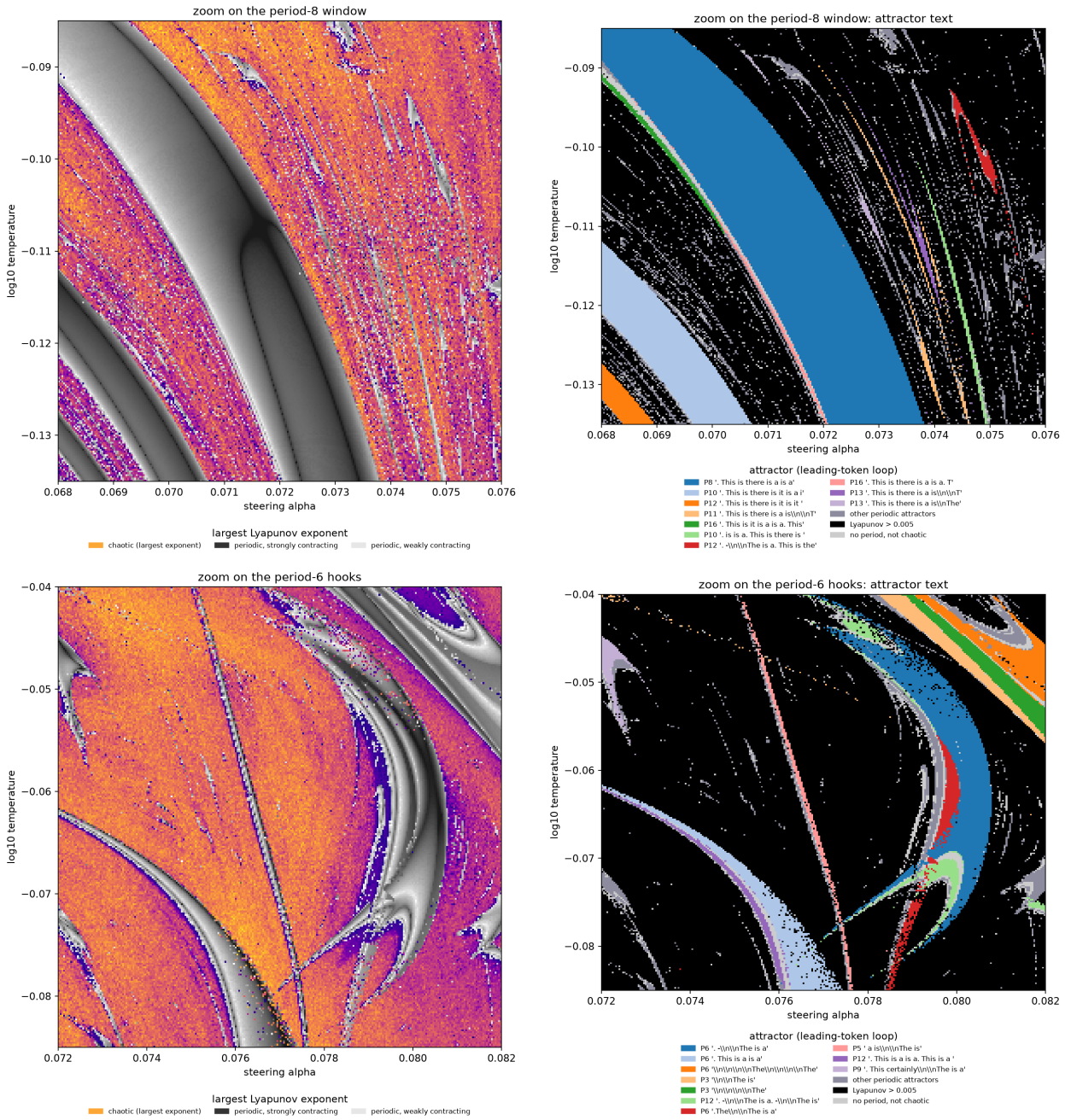


Figure 9: Zooms of 256^2 float64 cells of the Qwen2.5-0.5B-Instruct map (philosophy direction, layer 12), with the largest Lyapunov exponent on the left, rendered as in Figure 1, and attractor texts on the right. Top: the period-8 window ($\alpha \in [0.068, 0.076]$, $\log_{10} T \in [-0.135, -0.085]$). Bottom: the period-6 hooks ($\alpha \in [0.072, 0.082]$, $\log_{10} T \in [-0.085, -0.04]$), whose curved period-6 body, period-12 foot and thin leg are analysed in Section 9.

6 What the chaos does to inference and steering

The loop studied here is the expected-embedding feedback that continuous reasoning methods such as Soft Thinking use [1], and its chaos has consequences that can be measured without any reference

to dynamical systems. We report three: whether a run can be reproduced, how the depth at which a steering vector enters sets the chaos it causes, and what happens when two steering vectors are combined. A fourth, that a repeating text can hide a chaotic state from anyone who reads only the output, is shown in Section 15.

The regime decides whether a run can be reproduced. Language-model inference is not bitwise reproducible: GPU kernels choose different reduction orders for different matrix shapes, so a request's result depends on the batch it shares and on the hardware [70, 71]. We ran the 448 labelled cells of the horizon check (192 chaotic, 192 periodic and 64 other cells of two float64 maps; Section 4) twice in lockstep from the same start. The two runs differ only in the batch: all cells in one batch of n rows, or the same rows followed by n further rows whose steering vectors are those of the first n in reverse order, at the original temperatures ($2n$ rows). This changes the matrix shapes the kernels see; the first n rows compute the same cells (Figure 10, Table 2). In float64 the two runs first differ by about 6×10^{-15}, the rounding level. At 190 of the 192 chaotic cells this difference passes a relative distance of 10^{-2}, after a median of 102 steps, and grows on to order one (median final distance 0.91); at 173 of the 192 chaotic cells the most probable token of the loop's readout differs between the runs at some step of the last 500. At 182 of the 192 periodic cells the two runs stay within 10^{-2} for all 1500 steps, with a median largest distance of 2.3×10^{-12}, and 182 of the 192 periodic cells end the run within 3×10^{-12}. The Lyapunov exponent accounts for when a chaotic run diverges: the step at which the distance passes 10^{-2} follows $\ln(10^{-2}/d_1)/\lambda$, where d_1 is the smallest nonzero distance in the run, with a median ratio of 0.91 and a log-log correlation of 0.73 over 190 cells. This uses the whole run and is therefore retrospective; an independent recomputation with the first nonzero distance instead gives the same rounded values. The two chaotic cells that never diverge have exponents of 0.013 and 0.0013, for which the predicted step lies beyond the run. The same holds without steering in Qwen2.5-7B, whose embeddings are untied. Over 24 chaotic and 24 periodic temperatures of its unsteered loop, all 24 chaotic float64 runs diverge, after a median of 56 steps (the exponent again accounts for the step, median ratio 0.95), and at 23 of them the readout's most probable token differs at some step of the last 500. None of the 24 periodic runs moves past 10^{-2}; 22 end within 1.3×10^{-13} of each other and all within 2.2×10^{-12}. A second model family, unsteered Llama-3.1-8B, gives the same picture: all 20 chaotic temperatures diverge after a median of 68 steps (median ratio to the exponent's estimate 1.05), all 20 differ in their readout's most probable token in the last 500 steps, and 19 of the 20 periodic ones stay within 10^{-2}. At 70B the picture is unchanged. In Llama-3.3-70B-Instruct steered along hostility-vs-kindness at block 20 (float64 arithmetic on its stored bfloat16 weights, layers spread over two GPUs, 1000 steps), all 24 chaotic cells diverge, after a median of 114.5 steps (median ratio to the exponent's estimate 0.83, log-log correlation 0.74), and at 21 of them the readout's most probable token differs at some step of the last 300. None of the 24 periodic cells moves past 10^{-2}, and all end within 1.3×10^{-12}. In lower precision the rounding is larger. In float32 every chaotic cell of both models diverges, after a median of 20 steps (0.5B) or 16 (7B). In bfloat16, a common serving precision, the two layouts pass a distance of 10^{-2} after a median of one or two steps at every cell of the two Qwen models and at 38 of the 40 Llama cells, periodic ones included. With these kernels and at that precision, the loop was not reproducible across batch layouts in any regime we tested.

The same test at the level of answers, with a design frozen before the run, shows what this means for a task. We asked Qwen2.5-0.5B 96 held-out two-digit additions through a short soft-thinking pipeline: 32 expected-embedding steps appended to a growing context, then a fixed answer cue and greedy extraction of the number. Each problem ran alone and in a batch with three fixed distractor problems of the same length, so nothing is padded and only the matrix shapes differ.

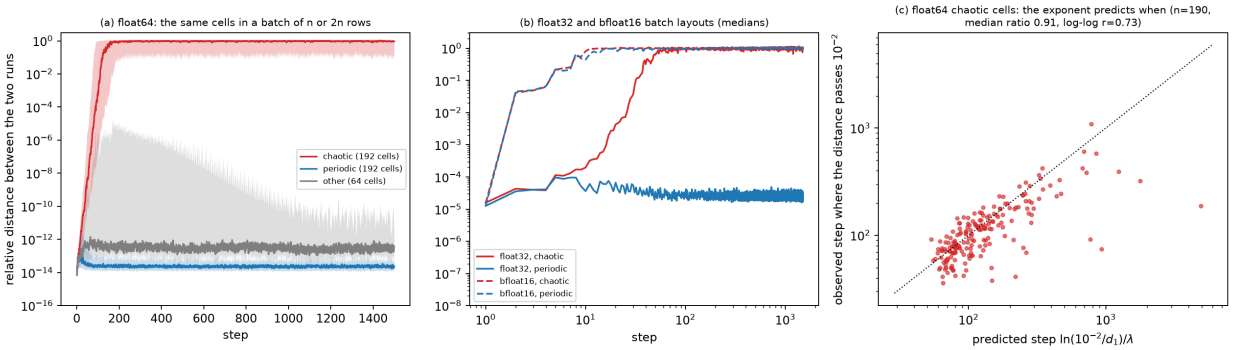


Figure 10: Only the batch layout differs. The 448 labelled cells of Qwen2.5-0.5B-Instruct (philosophy direction, layer 12) run twice from the same start, alone in a batch or followed by n further rows with the steering vectors in reverse order. (a) float64: median and interquartile range of the relative distance between the two runs, by the cells' float64 labels. (b) Medians for float32 and bfloat16 batch layouts. (c) float64 chaotic cells: the step at which the distance passes 10^{-2} against $\ln(10^{-2}/d_1)/\lambda$, with d_1 the first nonzero distance and λ the map exponent; dotted, equality.

In bfloat16 the extracted answer changed with the batch layout in 56 % of the problems at $T = 1$ and in 8 % at $T = 10^{0.2}$. A forced greedy baseline with the same budget (32 greedy tokens, then the same cue) produced identical token sequences in both layouts for every problem. The excess disagreement of soft over greedy decoding, averaged over the two temperatures, is 32.3 percentage points (95 % interval $[26.5, 38.1]$; sign-flip $p = 10^{-4}$, the smallest attainable). In float64 no answer changed within the 32 steps. There, the median growth of an injected kick to the first soft token is 0.033 nats per step at $T = 1$ (range -0.13 to 0.24 over six problems and three kick sizes), and even a relative kick of 10^{-4} changes no answer. The bfloat16 discrepancies therefore do not require the float64 map's chaos; they are consistent with bfloat16's rounding differences accumulating through continuous feedback, which a discrete argmax would absorb. (The forced greedy baseline's accuracy is zero and not informative: forced through 32 tokens, the model writes past its answer, and the cue then extracts another number.) The same frozen test in Qwen2.5-7B, whose soft-thinking pipeline answers 51 to 64 % of these problems, replicates the result. In bfloat16 the batch layout changed the soft-thinking answer in 53 of the 96 problems (55 %) at $T = 10^{0.2}$, the temperature at which the soft trace still matters. At this precision even the forced greedy baseline changed in 10 of the 96, the known batch-variance effect. The excess over greedy decoding is 17.7 points ($[9.3, 25.6]$, $p = 2 \times 10^{-4}$). In float64, at that temperature, the 7B's loop amplifies an injected perturbation (median +0.034 nats per step), injected kicks change answers (2 of 18 kicks: six problems and three kick sizes), and the rounding-level difference between the layouts changed 5 % of the answers within the 32 steps. Two horizons thus matter. Within a short budget, continuous feedback in low precision makes answers depend on the batch, and where the loop expands, even float64 rounding does. Over longer runs, a chaotic regime amplifies float64 rounding into different readout tokens, on the schedule its Lyapunov exponent sets.

Where a steering vector enters decides how much chaos it causes. Steering vectors are a common tool for controlling and probing models [53, 72], usually added at one layer chosen by hand. In Qwen2.5-0.5B the philosophy direction, built with the same recipe at blocks 4, 8, 12, 16 and 20 and scaled to each block's residual norm, makes 5.9, 8.1, 6.8, 5.0 and 2.3 % of the same 128×128 plane chaotic (Figure 11). Early steering carries the chaos down to $\log_{10} T = -0.95$ (block 8) and -0.91 (block 4), while at block 20 the chaotic cells lie between $\log_{10} T = -0.09$ and 0.22,

comparison	cells	reach 10^{-2}	median step	end with different tokens
batch layout, float64	chaotic	190 / 192	102	90.1 %
	periodic	10 / 192		3.6 %
batch layout, float32	chaotic	192 / 192	20.5	92.2 %
	periodic	72 / 192		22.9 %
float32 against float64	chaotic	192 / 192	20	92.7 %
	periodic	75 / 192		19.3 %
batch layout, bfloat16	chaotic	192 / 192	2	99.5 %
	periodic	192 / 192	2	95.3 %
7B unsteered, float64	chaotic	24 / 24	56	95.8 %
	periodic	0 / 24		0 %
7B unsteered, float32	chaotic	24 / 24	16	95.8 %
	periodic	4 / 24		8.3 %
7B unsteered, bfloat16	chaotic	24 / 24	2	75.0 %
	periodic	24 / 24	2	50.0 %
Llama 8B unsteered, float64	chaotic	20 / 20	68	100 %
	periodic	1 / 20		5 %
Llama 8B unsteered, float32	chaotic	20 / 20	10.5	100 %
	periodic	1 / 20		0 %
Llama 8B unsteered, bfloat16	chaotic	20 / 20	1	100 %
	periodic	18 / 20	2	45 %
Llama 70B steered, float64	chaotic	24 / 24	114.5	87.5 %
	periodic	0 / 24		0 %

Table 2: Divergence of two otherwise identical runs, by the cells' float64 labels (1500 steps; "end with different tokens": the most probable tokens differ at some step of the last 500). Top: Qwen2.5-0.5B, philosophy direction at layer 12. Middle: Qwen2.5-7B unsteered, 24 chaotic and 24 periodic temperatures of its long sweep. Bottom: Llama-3.1-8B-Instruct unsteered, 20 and 20. Last block: Llama-3.3-70B-Instruct, hostility-vs-kindness at block 20, 24 and 24 (1000 steps; tokens compared over the last 300). In float32 the periodic cells that diverge may change phase on the same cycle or switch between coexisting attractors; the run does not distinguish the two.

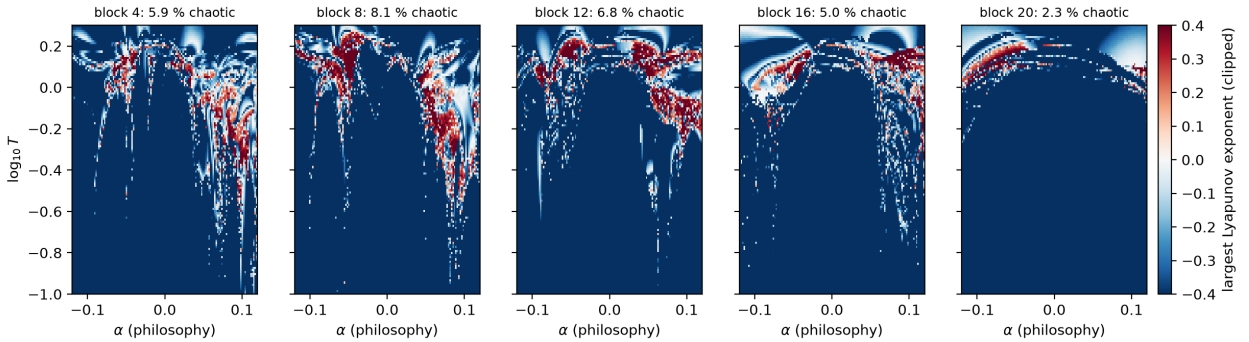


Figure 11: The philosophy direction of Qwen2.5-0.5B-Instruct, built with the same recipe at each of five blocks of 24: largest Lyapunov exponent on 128×128 float64 cells over the steering strength and $\log_{10} T \in [-1, 0.3]$. The share of chaotic cells is given above each panel.

three fifths of them above 0.1. In the strip $|\alpha| < 0.02$ every block gives 0.8 to 2.1 %. Llama-3.3-70B behaves the same way. Steered along its hostility direction, it is chaotic in 8.0 % of a 24×24 map at block 20 of 80, down to $\log_{10} T = -0.14$, against 4.3 % at block 35, where the chaos starts at $\log_{10} T = 0.01$, 4.3 % at block 50 (32 by 32 cells), and 0.7 % at block 65, all above $\log_{10} T = 0.19$. These 70B labels are those of the map protocol, one twin offset, and so are candidates. A logit bias, the latest possible point of entry, gives chaos mostly above $\log_{10} T = 0.1$ (Section 3). The gain split of Section 3, repeated along steering lines at all five blocks of the 0.5B model, shows where the depth enters. At low temperature ($\log_{10} T \leq 0$) the chaotic cells of blocks 4, 8 and 12 have expanding readout factors (medians +0.74, +0.46 and +0.46 nats per step over 54, 56 and 21 cells) and contracting network factors (−0.47 to −0.21). At block 16 the readout factor of such cells is only +0.15, and block 20 gives just 6 of them. Early injection creates token competition at low temperature more readily; late injection mostly does not. Its chaos lives at higher temperature, where the network factor carries it (block 20: network +1.09, readout −0.78; Figure 4e).

Two steering vectors interact. Steering at two blocks at once, gratitude at block 4 and hostility at block 16 of Qwen2.5-0.5B, makes 21.7 % of a 96×96 plane chaotic at $\log_{10} T = -0.1$. The grid has no row or column at exactly zero, so we computed the single pushes separately, each with the other exactly absent and on the same grid: they make 12.5 % (gratitude at block 4 alone) and 5.2 % (hostility at block 16 alone) of their lines chaotic. Large chaotic regions lie where neither push alone is chaotic (Figure 12): 1561 cells, 16.9 % of the plane and 78 % of its chaotic cells, are chaotic although each push alone, at the same strength, is not, and in 1525 of them each push alone is periodic. Over the four planes (two block orders, $\log_{10} T = \pm 0.1$) this share is 8.3 to 16.9 % of the plane, and 7.8 to 16.5 % with both single pushes periodic. With the blocks exchanged (hostility at 4, gratitude at 16) the plane is 23.4 % chaotic, and at $\log_{10} T = 0.1$ both pairs give 33 and 42 %. Each plane keeps a periodic hole around zero steering. Combining steering vectors is therefore not additive in its effect on the loop's stability: a composition of two pushes can be chaotic where each push alone is periodic.

Does a chaotic loop predict worse answers in ordinary decoding? Not beyond the model's own uncertainty. We looked for a behavioural signature of the regimes outside the loop. Two exploratory designs found none that survives controls. First, ordinary sampled answers of Qwen2.5-0.5B under steering at the strengths of chaotic and periodic map cells (30 matched pairs; arithmetic, open questions, short programs) showed no measurable accuracy contrast: at those

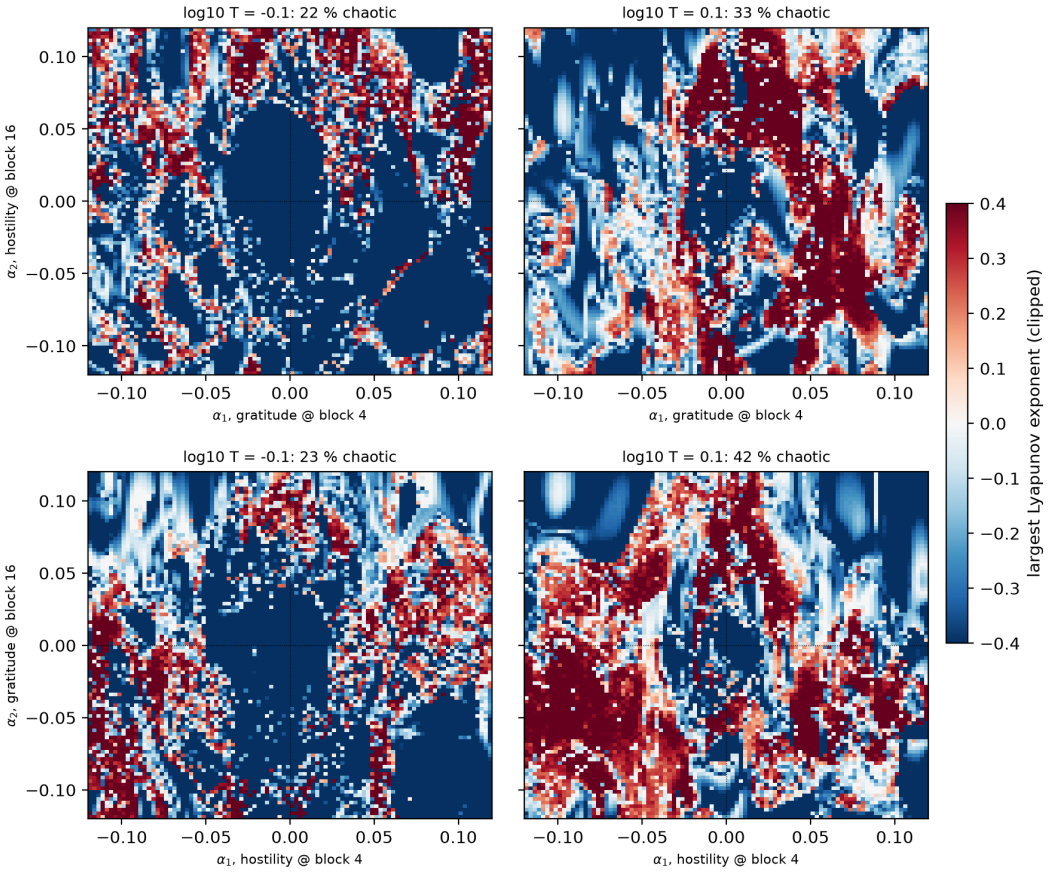


Figure 12: Two steering directions at two blocks of Qwen2.5-0.5B-Instruct: largest Lyapunov exponent on 96×96 float64 cells over the two strengths, at $\log_{10} T = -0.1$ (left) and 0.1 (right). Top: gratitude at block 4, hostility at block 16; bottom: the blocks exchanged. Dotted lines mark the zero axes; the grid does not sample them, and the single-push controls quoted in the text were computed separately on exact zero lines.

strengths, steering every position turns every reply into philosophy-flavoured text, so accuracy sits at the floor in both zones. Second, labelling each of 200 prompts by the regime of its own unsteered loop in Qwen2.5-7B, prompts with a chaotic loop were answered far less accurately, but mostly because chaotic labels fell on the harder task and on flatter answer distributions.

A frozen test addresses the second design. Its design, covariates and analysis were pushed before any data. It uses 4000 fresh products of two two-digit numbers and the unsteered Qwen2.5-7B-Instruct at $\log_{10} T = 0.40$: 1484 prompts have a chaotic loop and 2509 a periodic one (seven with other labels are excluded). Chaotic-loop prompts are again answered less accurately by ordinary sampling, by 7.1 points (95 % interval $[4.8, 9.3]$; 10 samples per prompt). The gap vanishes once seven difficulty covariates, all fixed before sampling, enter the fit. Beyond them the label's coefficient is $+0.15$ points ($[-1.4, 1.7]$, permutation $p = 0.84$), and the label adds no held-out predictive value. The first-token entropy alone absorbs nearly all of the gap (post hoc: -0.8 points remain). This is what the mechanism of Section 3 predicts. The loop turns chaotic where next tokens compete, and a model unsure of its first token also answers worse. Beyond that uncertainty, no association of the loop's regime with accuracy is detected at this precision; this does not show that the regime has no effect.

Within a prompt no association is detected either. We steered the first 1000 of these prompts slightly, at block 14 along philosophy, hostility-vs-kindness and a random direction. The strength, 1 % of the residual norm, is the largest at which greedy accuracy stayed within 5 points of the unsteered model on the first 200 of these prompts (so the calibration overlaps the test prompts); on the 1000 prompts it is 0.87 to 0.89, against 0.89 unsteered. Under these pushes the loop's label differs between conditions for 508 of the 1000 prompts. With prompt and condition fixed effects (3992 prompt-condition rows), in an analysis fixed before the data, a chaotic label is associated with a change of $+0.36$ points in the accuracy of the prompt's ordinary sampled answers (95 % interval $[-0.36, 1.08]$; permutation within prompts, $p = 0.34$). This does not identify a causal effect of entering chaos: the pushes can act on the answers through other paths as well.

Nor does accuracy peak at the edge of chaos. Adjusted for the covariates, three groups of prompts (3603 of the 4000; the rest have intermediate exponents) are answered alike:

- deep periodic ($\lambda \le -0.3$, 1961 prompts): $+0.3$ points;
- edge ($|\lambda| \le 0.1$, 242 prompts): -0.1 points;
- deep chaotic ($\lambda \ge 0.3$, 1400 prompts): $+0.3$ points.

All three intervals lie within $[-3.1, 3.0]$. The edge contrast is -0.4 points ($[-3.5, 2.6]$), and the curvature in λ is positive, the opposite of an edge optimum. Post hoc, the curvature comes from the most strongly contracting tenth of prompts ($\lambda \le -3.42$), answered 5.3 points ($[3.0, 7.6]$) better than the covariates predict. That is a lead for a future test, not a finding.

7 Period adding written in text

Between a period-2 region and the chaos lies a fan of stripes of periods $6, 8, 10, \ldots$ Their orbits, numerically validated in float64, can be read as text (Table 3). Each is ". This" followed by k pairs "is X", with period $2k + 2$, so each stripe adds one pair. The period-2 region itself is the 2-cycle "is it". At the stripe centres the orbits are effectively one-dimensional: the leading multiplier is 0.0025 to 0.086, and the next falls from 3×10^{-6} (period 6) to 2×10^{-14} (period 20). This compression belongs to these orbits, not to the map. The 2-cycle's multipliers are -0.271 and -0.058 (the second

modulus is about 0.037 near its fold), the next modulus at the period-3 flip of Section 10 is 0.088, and one period-6 orbit in Section 9 has a complex leading pair.

period	orbit (leading tokens)	μ_1	$\lvert\mu_2\rvert$
6	. This is a is a	$+0.0052$	3×10^{-6}
8	. This is there is a is a	$+0.0110$	3×10^{-7}
10	. This is there is it is a is a	$+0.0025$	2×10^{-7}
12	. This is there (is it)^2 is a is a	$+0.0163$	4×10^{-8}
14	. This (is it)^4 is a is a	$+0.0369$	2×10^{-9}
16	. This (is it)^5 is a is a	$+0.0716$	8×10^{-11}
18	. This (is it)^6 is a is a	$+0.0792$	1×10^{-12}
20	. This (is it)^7 is a is a	$+0.0857$	2×10^{-14}

Table 3: Periodic orbits of the stripe family at the stripe centres along $\log_{10} T = -0.115$ ($\alpha = 0.077872$, 0.071674, 0.067766, 0.065550, 0.064443, 0.063857, 0.063534, 0.063335), computed with the current all-float64 map (norms and rotary tables patched): Newton residuals 6×10^{-15} to 8×10^{-13}, leading multiplier μ_1 and next modulus $\lvert\mu_2\rvert$ from Rayleigh–Ritz values. Each step in period adds one "is X" pair to the text. The 2-cycle "is it" itself has multipliers -0.2706 and -0.0585 at $(\log_{10} T, \alpha) = (-0.17263, 0.061579)$. The period-2 and period-8 orbits were also reproduced by an independent implementation, and the period-8 orbit is certified by a computer-assisted proof (Section 18).

Table 4 shows what the map writes: two of these loops, cycle by cycle, next to chaotic cells between the stripes. The chaotic cells reuse the phrases of the neighbouring loops but do not settle into a fixed order.

α	regime	H	leading token at 40 consecutive steps
0.071674	period 8	1.9	` is a is a. This is there| is a is a. This is there| is a is a. This is there| is a is a. This is there| is a is a. This is there`
0.072940	chaotic ($\lambda = 0.13$)	2.4	` a is a. This is there is a is`↩`The is there is a is`↩↩`The is`↩↩`The is`↩↩`The is a is a. This is there is a is`↩`The is there`
0.067766	period 10	1.8	` is a. This is there is it is a| is a. This is there is it is a| is a. This is there is it is a| is a. This is there is it is a`
0.069060	chaotic ($\lambda = 0.20$)	2.4	`. - certainly`↩↩`The is there is it is a.`↩↩`The is a is a. This is there is it is a is`↩↩`The existence isThe is it is it is a. This`

Table 4: What the attractors write. Qwen2.5-0.5B-Instruct soft-token map steered along philosophy at layer 12, at $\log_{10} T = -0.115$: the leading token at 40 consecutive steps after 600 transient steps, for the period-8 and period-10 loops of Table 3 (bars mark the start of each cycle) and a chaotic cell beside each. λ: largest exponent over 2000 steps after 1000 transient steps, the same to two decimals at twin offsets 10^{-6} and 10^{-9}. H: mean entropy of the next-token distribution in nats. ↩: line break.

8 Mechanism: a saddle-node bottleneck

The fold. At $\log_{10} T = -0.115$ the 2-cycle "is it" dies in a fold at $\alpha_c = 0.06279827$ (float64 multiplier 1 to 3×10^{-13}, next modulus 0.037). The normal form of G has $a = -4.834$ per unit α and $b = -25.4285$, so $ab = 122.93$. The stable branch gives an independent estimate: extrapolating

its multipliers through $(1-\mu)^2/(4(\alpha_c-\alpha)) \to ab$ yields 123.1 with a quadratic fit (120.6 with a linear one). An independent implementation of the forward map and solver recovers α_c to 6×10^{-15} and a to 3×10^{-12}. Its b, from nested automatic differentiation, is -25.42846; our finite-difference value differs from it by 2.2×10^{-6} relatively, which is the finite-step bias of the difference table. The resulting $ab = 122.927$ leaves every passage prediction of Table 5 unchanged.

Controlled passages. In the oriented coordinate $v = \mathrm{sign}(b)\,u$ of Section 4, we start trajectories at $v = -U$ or 0 ($U = 0.1/|b|$) and count G-iterates until $v > U$. At four distances from the fold, the measured counts match the iterated normal form within one iterate (Table 5, Figure 13), and a transverse displacement of the same norm changes the count by at most one.

$\varepsilon = \alpha - \alpha_c$	full: measured / predicted	midpoint	full + transverse kick
3×10^{-6}	144 / 144	73 / 74	143 / 144
1×10^{-5}	71 / 71	36 / 37	70 / 71
3×10^{-5}	34 / 34	17 / 18	33 / 34
1×10^{-4}	14 / 14	7 / 7	13 / 14

Table 5: Bottleneck passages (G-iterates). Predictions iterate $u' = u + a\varepsilon + bu^2$ with a, b computed from the network's derivatives; no parameter is fitted.

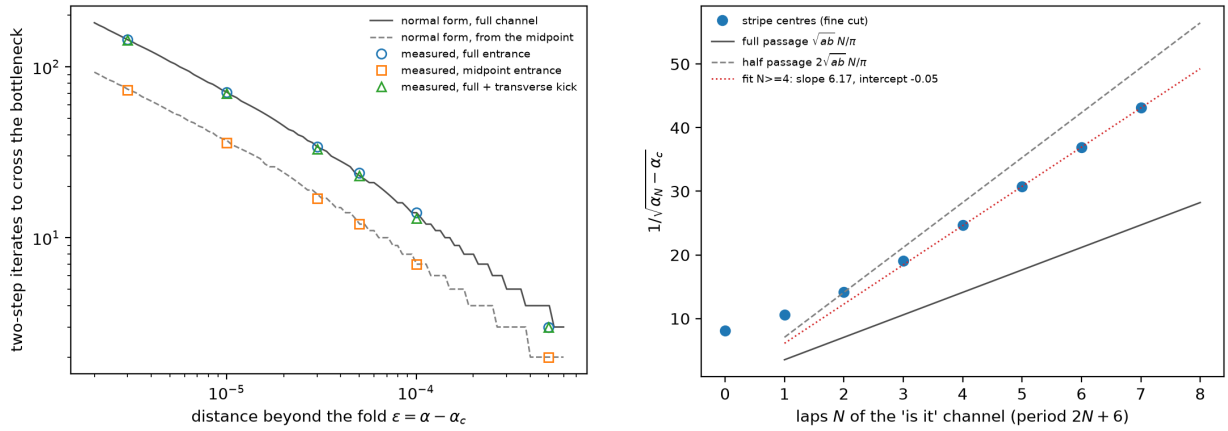


Figure 13: Bottleneck passages and the window law at the fold of the 2-cycle "is it" (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12, $\log_{10} T = -0.115$). Left: measured passage times against the normal-form predictions. Right: the window law, with stripe centres plotted as $1/\sqrt{\alpha_N - \alpha_c}$ against the lap count N, together with the full- and half-passage predictions.

Natural reinjection and the window law. Free trajectories re-enter the channel slightly upstream of the bottleneck's middle ($v \approx -3\times10^{-4}$ to -7×10^{-4}, in the same oriented coordinate). From each measured entry point, the normal form predicts the natural laminar durations within 4–8% for $\varepsilon \le 10^{-4}$, and a natural passage lasts about 0.6 of a full one. The stripes therefore follow the N^{-2} law of a saddle-node ghost, with about a third of the full-passage prefactor. Stripe centres located on a float32 cut along $\log_{10} T = -0.115$ give $d_N N^2 = 0.0263$–0.0265 for $N = 4$–7 (log-log slope -2.00), compared with $\pi^2/(4ab) = 0.0201$ for a half passage. Two-step return maps of p("it") (Figure 14) show the near-tangency.

Numerically estimated stripe edges along $\log_{10} T = -0.115$ (Table 6) sharpen this picture. Each window is bounded by a fold, where its orbit is born, and by a flip. For periods 6–12 the fold faces away from the accumulation point. For periods 16–20 the sides are exchanged, and the period-14 window is bounded by flips on both sides. The fold edges of periods 16–20 follow $\alpha_N - \alpha_c = C/(N+\delta)^2$ with $C = 0.0266$ and $\delta = 0.15$. Their slope, a rise of 6.13 per lap in $1/\sqrt{\alpha_N - \alpha_c}$, is 1.74 times the full-passage slope $\sqrt{ab}/\pi = 3.53$ of the normal form. That corresponds to an effective passage of 0.58 of a full one, which the measured natural reinjection gives independently.

period	edge toward larger α	edge toward smaller α
6	fold 0.0790337	flip 0.0766947
8	fold 0.0726585	flip 0.0706525
10	fold 0.0684572	flip 0.0670594
12	fold 0.0658982	flip 0.0651944
14	flip 0.0645850	flip 0.0642958
16	flip 0.0639284	fold 0.0638001
18	flip 0.0635683	fold 0.0635034
20	flip 0.0633535	fold 0.0633183

Table 6: Edges of the stripe windows along $\log_{10} T = -0.115$ ($\alpha_c = 0.0627983$), from the window centres of Table 3. Flips: critical multiplier a real -1 to 3×10^{-10}. Folds: the multiplier nearest $+1$ extrapolated to 1 through $(1-\mu)^2$, which is linear near a fold; last converged multiplier 0.991–0.998.

We tested whether the measured reinjection predicts windows beyond the fitted range. A model with one fixed entry point (the median measured one) and one lap offset fitted to the fold edges of periods 16–20 places the fold edges of periods 22–30 3 to 9 % closer to the fold than the constant-prefactor law. Both predictions were frozen before a float64 line of 3200 cells (spacing 10^{-7}) located the windows. The measured edges follow the law within 3 % through period 30 (within 1 % at periods 22, 24 and 28) and fall outside the model's range except at period 30. So the law extrapolates, but one fixed reinjection point does not predict it. The measured entry points move with ε (-3×10^{-4} to -7×10^{-4}).

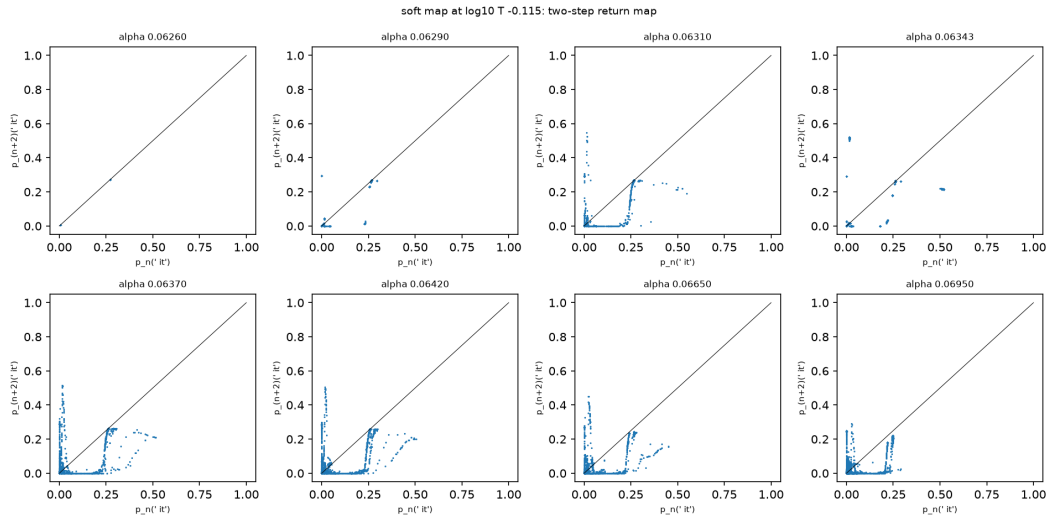


Figure 14: Two-step return maps of the probability of "it" near the fold of the 2-cycle "is it" (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12), at increasing distance from the fold.

A prospective test. To test the normal form at folds that played no part in shaping the mechanism, an independent implementation located two new folds and computed their normal-form coefficients by nested automatic differentiation. All 24 passage predictions were pushed in a frozen design before any test trajectory was run. Target A is the 2-cycle's fold at a new temperature: $\log_{10} T = -0.140$, $\alpha_c = 0.0670549$, $a = -7.055$, $b = -24.927$. Target B is the fold of the period-8 orbit of Table 3, continued in temperature at $\alpha = 0.0717$: $\log_{10} T_c = -0.1033462$, $G = F^8$, $a = -11.900$, $b = -152.05$ per unit $\log_{10} T$. Of the 24 forecasts (four distances and three entrances per target), 23 fell within one iterate of the measured passages (Table 7). The pre-specified criterion, that every row pass, failed by one row: for target A, midpoint entrance, $\varepsilon = 3 \times 10^{-6}$, the forecast was 61 and the measurement 59. All full-channel passages and all passages after a transverse kick were within one iterate.

ε	target A ($G = F^2$): measured / predicted	target B ($G = F^8$): measured / predicted
3×10^{-6}	118/118, **59/61**, 117/118	26/26, 13/14, 25/26
1×10^{-5}	56/56, 28/29, 56/56	10/10, 5/5, 10/10
3×10^{-5}	26/26, 13/14, 26/26	4/4, 2/2, 4/4
1×10^{-4}	10/10, 5/6, 10/10	2/2, 1/1, 2/2

Table 7: Prospective passage test at two folds that did not shape the mechanism. Each cell lists the full-channel, midpoint and full-plus-transverse entrances as measured / predicted G-iterates. Predictions come from the iterated quadratic normal form, with coefficients from an independent implementation, frozen before the test runs. Bold: the one row outside the one-iterate criterion. The shortest period-8 rows give only weak relative accuracy.

9 Bifurcation curves and cusps

Period-8 stripe. Continued for 80 points with a float64 test function, the fold curve of the period-8 orbit keeps its critical multiplier within $1 \pm 6 \times 10^{-6}$, follows the stripe's upper edge and caps its top near $\log_{10} T = -0.065$ (Figure 7). At $\alpha = 0.0717$ the stripe lies between a flip at

$\log_{10} T = -0.128955$ and the fold at -0.1033462. Continued downward, the flip curve keeps its multiplier at -1 for 115 points to $(\alpha, \log_{10} T) = (0.0776, -0.402)$, with two reversals of its parameter tangent near $(0.0750, -0.219)$ and $(0.0746, -0.206)$, and the fold curve keeps its multiplier at $+1$ for 120 points to $(0.0758, -0.200)$. Located by bisection along the curve (Section 4), the two reversals of the flip curve, at $(0.074955, -0.219336)$ and $(0.074608, -0.206007)$, carry no second multiplier near the unit circle (the next ones are 3×10^{-7} and 1×10^{-5}). On a flip curve the parameter tangent can vanish only where a second multiplier reaches $+1$, so both are tight but smooth turns of the curve in the parameter plane, not codimension-two points.

The accumulation boundary. The fold curve of the 2-cycle "is it", continued from $(\alpha, \log_{10} T) = (0.0628, -0.115)$ in both directions (38 and 126 points, critical multiplier 1 to 10^{-7}, no cusp), traces the boundary of the period-2 region in Figures 7 and 8. The stripe fan accumulates on this boundary, as the mechanism of Section 8 requires. Further along one direction, outside the plotted window, another multiplier leaves the unit circle (moduli up to 11). Located along the curve, this second real multiplier reaches $+1$ at $(\alpha, \log_{10} T) = (0.035628, -0.341724)$, where the eigenvectors of the two multipliers at $+1$ coincide to 3×10^{-4} degrees: a double multiplier $+1$ with a Jordan block, the 1:1 resonance of maps (the analogue of a Bogdanov–Takens point).

A period-6 window with shrimp-like anatomy. The hooks of Figure 9 (bottom) belong to the period-6 loop ". -\n\nThe is a". Float64 orbits (Newton residuals 10^{-15}–10^{-13}) show that its curved body and its thin leg are one window: the multipliers are -0.061 and $+0.048$ at the body, -0.564 lower down and -0.645 on the leg. The foot is its period-doubled loop (period 12, leading multiplier -0.156). Inside the body lies a different 6-cycle, ".The\n\nThe is a", whose leading multipliers are a complex pair, $-0.031 \pm 0.152i$. One-parameter branches from the body end in folds in every direction: at $\log_{10} T = -0.0558$ and -0.0721 in temperature, at $\alpha = 0.08074$ in steering, and, toward smaller α, in two folds only 10^{-5} apart. Such a close pair is the signature of a nearby cusp. A flip at $\alpha \approx 0.0797$ then opens the period-12 foot. The body is not effectively one-dimensional either. At two body cells the leading multiplier is a complex pair, $-0.020 \pm 0.264i$, or has a second multiplier of comparable size (-0.261 against 0.099). Below the body, where the window's strip appears to cross a period-5 strip near $(\alpha, \log_{10} T) = (0.0773, -0.0774)$, both orbits continued from their own strips are unstable at the crossing (leading multipliers -1.88 and -6.59). The crossing is therefore neither a centre nor a point where two attractors coexist.

Fold curves that bound a window and meet in cusps, with flip curves inside, form the codimension-two organization of a shrimp [59] (complex dynamics of small recurrent networks has been studied in the same spirit [73]). In the continued curves (Figure 15), a single fold arc bounds the body on its outer side and a flip arc on its inner side, and the leg lies between them where they converge. The outer fold arc extends far to the left as a thin arm and ends in a cusp at $(\alpha, \log_{10} T) = (0.069106, -0.039750)$. The fold curve through the close fold pair has a second cusp inside the body, at $(0.080097, -0.058344)$, from which two fold branches descend.

At both reversals the quadratic coefficient b changes sign between consecutive curve points while a stays away from zero (Table 8). Because the signs of a and b both flip with the orientation of the critical eigenvector, the test uses $b \operatorname{sign} a$. A cusp is nondegenerate when the cubic coefficient c of the normal form $u' = u + a\varepsilon + bu^2 + cu^3$ is nonzero where b vanishes. We compute c with the second-order centre-manifold term, $c = \ell^\top C(r, r, r)/6 + \ell^\top B(r, h_2)/2$ with $(I - M)h_2 = B(r, r) - 2br$ and $\ell^\top h_2 = 0$, where M is the monodromy matrix and B and C are the second and third derivatives of its map, obtained by central differences of float64 Jacobian-vector products. (The code reproduces c to 10^{-6} on a map with a known reduction.) At the two curve points that bracket each sign

change, c is nonzero far beyond its step-size uncertainty and has the same sign, and b changes sign with nonzero slope between them. This supports a nondegenerate cusp inside each bracket. Located along each fold curve (Section 4), $b \operatorname{sign} a$ falls to below 5×10^{-6} of its bracket values at $(\alpha, \log_{10} T) = (0.069106, -0.039750)$ on the outer arc and at $(0.080097, -0.058344)$ inside the body. At these same points $c = -1.51 \times 10^4$ and -2.38×10^5, and the unfolding matrix J defined below is regular ($\det J = 2416$ and 14.0): two numerically located cusps. (Location uses the small-step oriented coefficient $b \operatorname{sign} a$; the unfolding coefficients use a separate finite-difference ladder, and the selected plateau of c spreads by at most 4.8 % over adjacent steps.) The negative sign of c means that near each cusp, two stable period-6 orbits coexist with a saddle between the fold branches.

A generic codimension-two cusp also needs the two parameters to unfold it independently. Near the cusp the reduced map is $u' = u + \beta_1 + \beta_2 u + bu^2 + cu^3$, where β_1 and β_2 vanish at the cusp and depend on the parameters. The unfolding is regular when $J = \partial(\beta_1, \beta_2)/\partial(\log_{10} T, 10\alpha)$ has rank two ($\det J \neq 0$). We first estimated J at a fold point near the outer-arc cusp (where $b \operatorname{sign} a = 14.7$): $\det J = 2737$ (singular values 1630 and 1.7), so the estimated unfolding is nonsingular there, as it is at the located cusp ($\det J = 2416$). With these derivatives, the local cubic model predicted the number of period-6 orbits and their stability at six parameter offsets, written to disk before any orbit was solved. Newton solves started from the predicted roots and from the base state recovered the predicted number of distinct orbits, with the predicted stability, at all six. The search was seeded, not exhaustive: orbits that these starts do not reach would be missed.

	cusp on the outer arc	cusp inside the body
$b \operatorname{sign} a$ at consecutive curve points	+92.9, +42.9, +14.7, **−16.9**, −52.8	−24.7, −14.3, −2.0, **+12.8**, +28.0, +45.9, +58.9
$\lvert a \rvert$	1.2–6.9	3.13–3.35
c at the two points around $b = 0$	-1.37×10^4, -1.65×10^4	-2.38×10^5, -2.34×10^5

Table 8: Normal-form test at the two parameter-tangent reversals of the period-6 window's fold curves (bold: the flagged points; critical multiplier 1 to 10^{-7}). c: cubic coefficient at the two curve points that bracket the sign change (finite-difference steps agree to 0.1 and 0.7 %). Bisection along the curves then located both cusps (text); they are numerical, not certified.

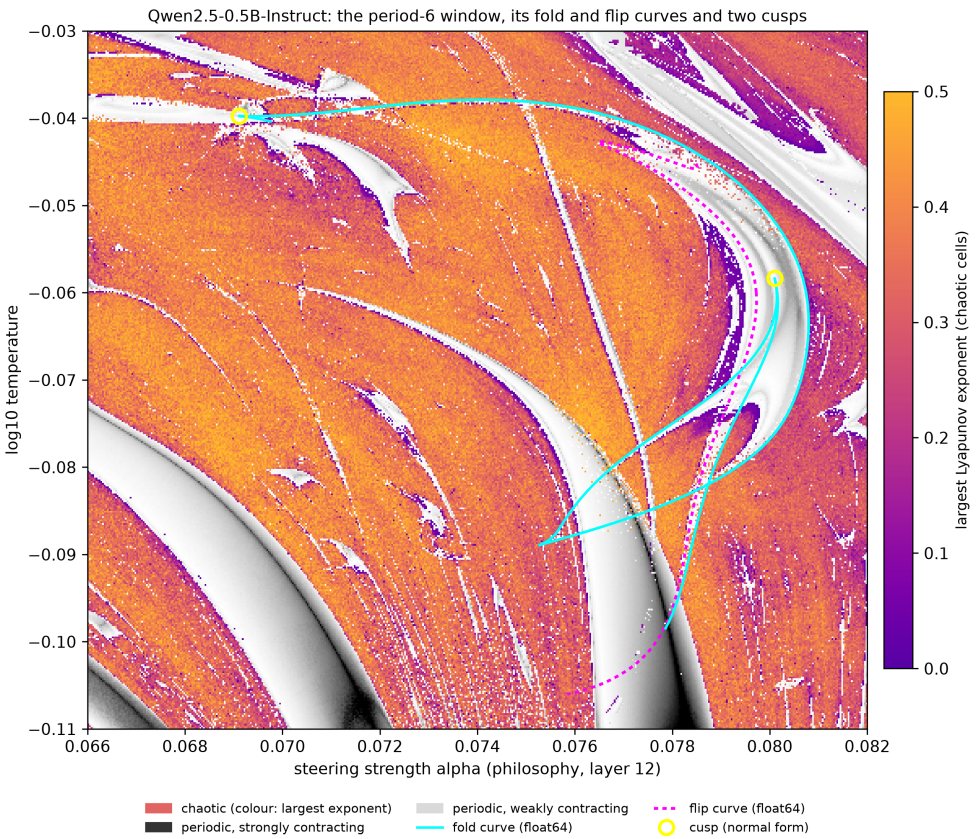


Figure 15: The period-6 window of the Qwen2.5-0.5B-Instruct map (philosophy direction, layer 12) with its float64 fold curves (cyan), flip curve (magenta, dotted) and the two located cusps (yellow circles). The background is a float32 phase map of 480×400 cells (largest exponent from 300 steps after 300 transient steps; chaotic cells coloured by the exponent, periodic cells in grey by how strongly they contract). Where it overlaps the float64 map of Figure 7, 95 % of cells carry the same chaotic or periodic label at the nearest grid point. Some fold branches were continued from two starting points and overlap. Continuation of the lower branches stopped after 60 points near $(0.0753, -0.089)$, where they approach each other; how they join there is not resolved.

A period-4 window with five cusps on one fold curve. At higher temperature, $\log_{10} T \approx 0$–0.1, periodic orbits are no longer effectively one-dimensional (second multipliers 0.06–0.09). At $\alpha = 0.06484$ the temperature branch of the period-4 orbit "\n\n - \n\n The" is S-shaped, with folds at $\log_{10} T = 0.0640$ and 0.0664. When that fold is continued in the plane, its parameter tangent reverses at three points; at the first two, the quadratic coefficient b changes sign while a stays away from zero, and the cubic coefficient c (defined above) is nonzero (Table 9). Located along the curve, these two cusps lie at $(\alpha, \log_{10} T) = (0.062959, 0.059958)$ and $(0.065574, 0.074447)$. There $b \operatorname{sign} a$ is below 1.1×10^{-5} of its bracket values, $c = -1.02 \times 10^{4}$ and $+1.37 \times 10^{5}$, and $\det J = -42.0$ and 35.5. Further traces of the same fold curve, started from other points of the period-4 branches, retrace it to within 3×10^{-7}, and one of them finds two more reversals near $\log_{10} T \approx 0.012$. All five reversals are located cusps. The other three lie at $(\alpha, \log_{10} T) = (0.083082, 0.108633)$, $(0.074072, 0.011831)$ and $(0.073220, 0.012759)$. There c is -9.7×10^{4}, -5.7×10^{3} and $+1.7 \times 10^{3}$, and $\det J$ is 25.3, -188.5 and 68.5. The last two cusps, close together, have opposite signs of c. Three period-4 orbits coexist between the two fold branches, and near the first cusp ($c < 0$) two of them are stable (Figure 16).

	cusp 1	cusp 2
b at five curve points	$+28.1,\ +18.7,\ +8.4,\ \mathbf{-1.1},\ -12.8$	$+51.8,\ +29.7,\ +9.3,\ \mathbf{-9.0},\ -23.4$
a	3.02–3.08	-2.58 to -2.57
critical multiplier	1 ± 10^{-9}	1 ± 10^{-8}
c around $b = 0$	$-1.06,\ -1.01$ ($\times 10^4$)	$+1.39,\ +1.36$ ($\times 10^5$)

Table 9: Normal-form test of the two parameter-tangent reversals (bold: the flagged points). c: cubic coefficient at the points around the sign change, nonzero at both reversals (steps agree to 10^{-4}), which supports a nondegenerate cusp inside each bracket; bisection then located both (text); at the second, $c > 0$: the orbit between the two others is the stable one.

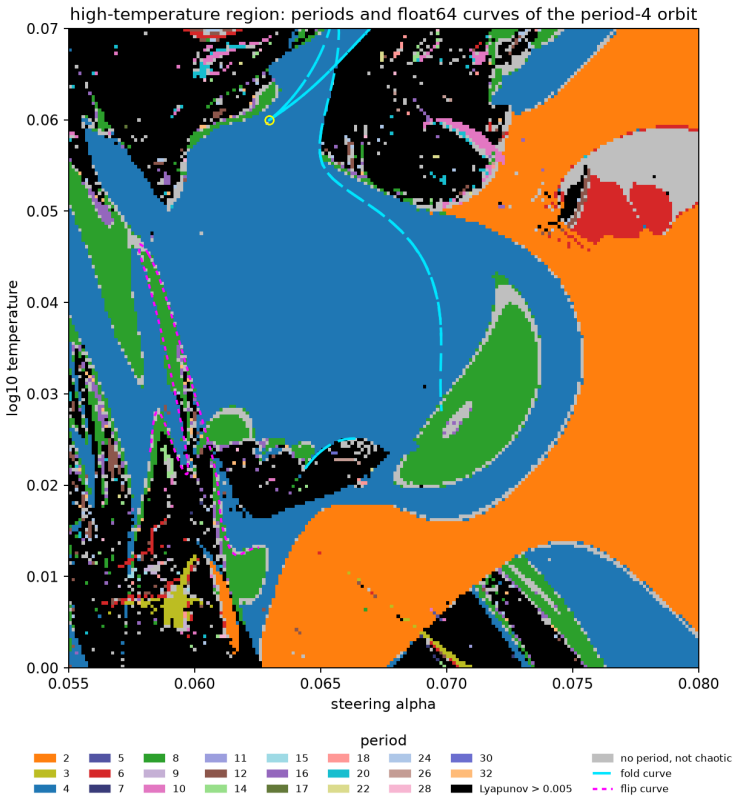


Figure 16: Period map of the high-temperature region of the Qwen2.5-0.5B-Instruct map (philosophy direction, layer 12; 192^2 float64 cells) with float64 fold curves (cyan, long dashes) and flip curves (magenta, short dashes) of the period-4 orbit. The yellow circle is the first cusp; the second lies just above the window. The flip curves run along the boundary between period 4 and period 8. The fold curves run inside the period-4 region, where three period-4 orbits coexist (near the first cusp, two of them stable). In this map a C-shaped band of the doubled loop "\n\nThe\n\nThe" (period 4) inside the period-2 region "\n\nThe" contains period 8 and an irregular core. Along the cut of Section 10 the period-8 orbit merges back into period 4 (a doubling bubble), so this band is not shown to be a second cascade. Continued from all its fold points, the region's boundary consists of an upper fold curve, with the five cusps of the text, and a separate lower fold curve, nowhere within 2×10^{-3} of it in the traced ranges; the lower curve carries the codimension-two points described below. Within the traced ranges the curves do not join, so the region is not shown to be one shrimp-organized component.

Along the lower fold curve a second real multiplier reaches $+1$ twice, at $\alpha = 0.062483$, $\log_{10} T = 0.004955$ and at $\alpha = 0.062499$, $\log_{10} T = -0.001780$. Both points are 1:1 resonances, with the

eigenvectors of the double multiplier parallel to 3×10^{-4} degrees. Close by, at $\alpha = 0.062522$, $\log_{10} T = -0.002667$, a multiplier reaches -1: a fold-flip point. We did not continue the curves that emanate from these points.

10 A period-doubling cascade to chaos

The crescent of Figure 8 is a closed region in which the period-3 loop "\n\nThe existence" doubles inward. Along the cut $\log_{10} T = -0.14798$, the map's cells show period 3 over 25 columns, period 6 over about 14, period 12 over 4 and period 24 in one column, followed by a chaotic core. The doubled loops repeat the same leading tokens twice and four times, with alternating probabilities. Locating the flips along this cut in float64 (Section 4), in the steering coordinate α, gives Table 11. The first two ratios, $\delta_2 = 2.844$ and $\delta_3 = 2.859$ (indexed as in Table 11), are far from Feigenbaum's 4.6692, even though from the second flip on the orbits are effectively one-dimensional (next modulus 0.0023, 2×10^{-6}, 2×10^{-12}). The next ratio, $\delta_4 = 3.781$, is markedly larger. The first four flip parameters agree between completed float64 computations on a CPU and an H100 to 4×10^{-15}.

Continuing a just-born doubled orbit past period 48 failed: Newton's method slides onto the half-period orbit, which is also periodic with the doubled period. We therefore also located the superstable parameters s_p, where the critical multiplier vanishes in the middle of each band (Table 10). There the orbit attracts, so it can be reached by forward iteration and cannot slide onto its half-period parent. Through period 768 the superstable ratios are 3.19, 4.02, 4.478, 4.625 and 4.660. The last three come from parameters resolved to 10^{-14} and differ from Feigenbaum's value by 0.19, 0.044 and 0.0097; from 4.02 on, the distance shrinks by factors of 3.4, 4.3 and 4.6. The flip ratios 2.84, 2.86, 3.78, 4.36, 4.60 and 4.65 rise in step (Figure 17); the last three come from runs started inside the period-96 band, which located the flips of periods 96, 192 and 384 directly. The accumulation point lies near $\alpha_\infty \approx 0.0661250$. The early superstable values are less precise. The other computed Ritz values of the period-12 and period-24 orbits blur the root, so s_{12} and s_{24} are known only within brackets of 1.24×10^{-5} and 6.9×10^{-7}, and the ratios 3.19 and 4.019 have worst-case ranges $[2.895, 3.338]$ and $[4.019, 4.101]$. s_6 is less precise still, and s_3 was not bracketed.

period p	superstable parameter s_p	bracket width	ratio $\frac{s_p - s_{p/2}}{s_{2p} - s_p}$
12	0.0659717	1.24×10^{-5}	
24	0.06608014	6.9×10^{-7}	3.19
48	0.066114132784466	3.3×10^{-15}	4.019
96	0.066122589892689	7×10^{-15}	4.4779
192	0.066124478510077	6.4×10^{-15}	4.6251
384	0.066124886853811	5×10^{-15}	4.6595
768	0.066124974490236	4.1×10^{-15}	

Table 10: Superstable parameters of the crescent's doubling sequence along $\log_{10} T = -0.14798$: the critical multiplier of the period-p orbit changes sign across the bracket, with Newton residuals below 3×10^{-13} and the other computed Ritz moduli below 10^{-20} from period 96 to 384; at period 768 the residual is 1.2×10^{-12} and the other moduli are below 3×10^{-20}. Brackets for periods 12 and 24 are the closest points where the critical multiplier is cleanly identified.

Why are the early ratios so small? We first considered a crossover. Softmax at temperature T is a smoothed argmax, so the reduced one-dimensional map might have a nearly corner-like extremum, rounded only on a scale set by T. Early doublings would then see the corner, where ratios tend toward 2, and later ones the smooth quadratic core. This predicts smaller early ratios at lower

temperature. We fixed that prediction before locating the flips along two lower cuts, and it fails. At $\log_{10} T = -0.156$ the first ratio is 14.93, five times its value at -0.148 (Table 12). At -0.152 the continued period-12 solution collapsed onto its period-6 parent near $\alpha \approx 0.0657$, and the forward search found no next flip. This is compatible with a doubling bubble, but we did not locate its reverse boundary or exclude a narrower cascade on that cut. The early ratios thus depend strongly on where a cut crosses the crescent, and we withdraw the crossover explanation. The later ratios at -0.156 behave as universality requires. The next ratios are 3.124, 4.4988, 4.6208, 4.6618, 4.6673 and 4.6688. Their deviations from $\delta = 4.6692$ are -1.55, -0.17, -0.048, -0.0074, -0.0019 and -0.00036, and from the third on each step closes the gap by a factor of 3.5 to 6.5. The last ratio uses the flip of the period-768 orbit (the $768 \to 1536$ flip), located with $|\mu + 1| = 4.1 \times 10^{-7}$. The penultimate flip, of the period-384 orbit, approaches the precision floor of its multiplier, a product of 384 step Jacobians. On an H100 the computed $\mu + 1$ scatters by 1.8×10^{-5} about a straight line near the root; under that local linear fit, the largest residual divided by the slope gives a location uncertainty of about $\pm 3.3 \times 10^{-13}$, an estimate rather than a certified bound. The workstation GPU located it at $\alpha = 0.06510347354866666$ with $|\mu + 1| = 4 \times 10^{-9}$, 9×10^{-14} from the H100 estimate. Over that range the ratio lies between 4.6672 and 4.6675. The superstable points along the same cut, through period 384, give the ratios 10.21, 3.713, 4.572, 4.643 and 4.6646. Before computing them we fixed the prediction that the last two superstable ratios at a new temperature would lie within 5 % of δ, in $[4.436, 4.902]$. At the original readout, through period 192, the point estimates 4.572 and 4.643 met it, but the bracket envelope of 4.572 did not; extending the computation to period 384 later gave the pair 4.643 and 4.6646, both inside the interval. The last, 4.6646, rests on parameters resolved to 5×10^{-15} and differs from δ by 0.1 %. Over the corners of the brackets the one before lies in $[4.639, 4.647]$, limited by s_{48}, whose bracket is 2.4×10^{-9} wide. The earlier ratio 4.572 depends on s_{24}, known only within a bracket of 5.9×10^{-6} (worst-case range $[1.01, 4.65]$).

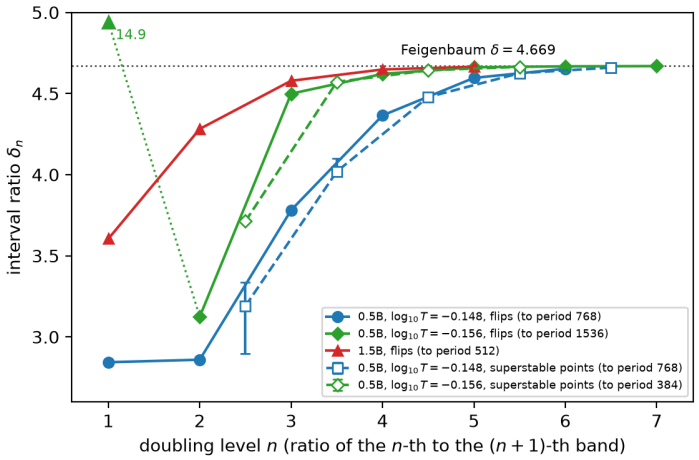


Figure 17: Finite-order interval ratios of the doubling sequences: the crescent of Qwen2.5-0.5B-Instruct (philosophy direction, layer 12), from flips and from superstable points (which reach period 768), and the route of Qwen2.5-1.5B (philosophy direction, layer 14; Figure 19), from flips. A superstable ratio spans half-bands, so it is plotted half a level after the flip ratio of its first band. Bars: worst-case ranges from the brackets of s_{12} and s_{24}. At $\log_{10} T = -0.148$, and in the 1.5B model, the ratios rise toward Feigenbaum's δ from below; the last superstable ratio at -0.148 is within 0.21 % of it, and the last 1.5B ratio (4.665) within 0.1 %. At $\log_{10} T = -0.156$ the first ratio, 14.9, lies off the scale (arrow), and the later ratios approach δ from below, the last (4.6688) within 0.01 % of it. Its first two superstable ratios carry no bars: the brackets of s_{12} and s_{24} allow $[3.40, 29.3]$ and $[1.01, 4.65]$. Each sequence ends at the deepest level computed: the flips at -0.148 at period 384, the superstable points at period 768, the 1.5B route at period 256, and the cut at -0.156, which we followed furthest, at the $768 \to 1536$ flip.

Finite-order ratios also depend on the parameter coordinate, so only the trend of later ratios can test universality [22, 45, 46].

n	orbit	flip parameter α_n	next modulus	ratio $\delta_n = \frac{\alpha_n - \alpha_{n-1}}{\alpha_{n+1} - \alpha_n}$
1	$3 \to 6$	0.0654807769499	0.088	
2	$6 \to 12$	0.0659055406325	0.0023	2.844
3	$12 \to 24$	0.0660549041785	2.4×10^{-6}	2.859
4	$24 \to 48$	0.0661071412523	2.3×10^{-12}	3.781
5	$48 \to 96$	0.0661209563174	$< 10^{-13}$	4.365
6	$96 \to 192$	0.0661241213732	$< 10^{-13}$	4.596
7	$192 \to 384$	0.0661248100141	$< 10^{-13}$	4.653
8	$384 \to 768$	0.0661249580173	$< 10^{-13}$	

Table 11: Flips of the crescent's period-doubling sequence along $\log_{10} T = -0.14798$. For the first five flips (H100) the critical Ritz value is real and differs from -1 by at most 4×10^{-10}, the Newton residuals are below 4×10^{-13}, and the other computed Ritz values lie inside the unit circle. The sixth flip was located at the precision floor of the period-96 multiplier: the critical value differs from -1 by 5.7×10^{-7}, its scatter between bracket points 10^{-13} apart is about 10^{-6}, and the Newton residual is 5.5×10^{-12}; the location is uncertain by about 2×10^{-12}, which changes δ_5 by less than 10^{-5}. The seventh flip is clean (critical value -1.000000016, Newton residual 2.8×10^{-12}). The eighth was accepted at the precision floor of the period-384 multiplier: there the critical value differs from -1 by 5.3×10^{-5} (acceptance threshold 10^{-4}), bracket evaluations within 4×10^{-12} of it scatter by about 10^{-4} in that difference, and the Newton residuals are below 4×10^{-11}. Its location is uncertain by about 4×10^{-12}, which changes δ_7 by about 10^{-4}. Feigenbaum's constant is $\delta = 4.6692\ldots$ [22]; for the logistic map the first ratios are 4.751, 4.656, 4.668. The first four parameters agree between the completed CPU and H100 computations to 4×10^{-15}. Superstable parameters (Table 10) extend the ratio sequence through period 768.

n	orbit	flip parameter α_n	next modulus	ratio δ_n
1	$3 \to 6$	0.0644766567385	0.12	
2	$6 \to 12$	0.0650493555308	0.045	14.93
3	$12 \to 24$	0.0650877194158	7.0×10^{-5}	3.124
4	$24 \to 48$	0.0650999993318	4.3×10^{-7}	4.4988
5	$48 \to 96$	0.0651027289524	1.8×10^{-13}	4.6208
6	$96 \to 192$	0.0651033196816	2.9×10^{-14}	4.6618
7	$192 \to 384$	0.0651034463988	1.5×10^{-14}	4.6673
8	$384 \to 768$	0.0651034735487	3.2×10^{-13}	4.6688
9	$768 \to 1536$	0.0651034793638	2.9×10^{-14}	

Table 12: Flips of the crescent's doubling sequence along $\log_{10} T = -0.156$, in float64. Flips 1–3 were computed on an H100 and flips 3–8 on a workstation GPU (NVIDIA RTX 2000 Ada); the two computations of flip 3 agree to 4×10^{-16}, and an H100 computation of flip 8 from its evaluations at the multiplier's precision floor lies 9×10^{-14} away. Flip 9 was computed on an H100. Every critical multiplier is real; for flips 1–8 it differs from -1 by at most 5×10^{-8}, and for flip 9 by 4.1×10^{-7}.

Not every doubling leads to chaos. In the high-temperature region (Figure 16), along $\log_{10} T = 0.0319$, the period-2 loop "\n\nThe" flips at $\alpha = 0.0753534$ and its doubled loop at 0.0736921. These orbits are far from one-dimensional (next moduli 0.38 and 0.28). Toward smaller α the period-8 orbit does not double again: near $\alpha \approx 0.0705$ it merges back into the period-4 orbit. Along this cut it is therefore a period-doubling bubble, bounded by two flips of the period-4 orbit, not the start of a cascade to chaos.

	offset	β	amplitude ratio	doubled multiplier	$1-4\beta$
period $2\to4$	2.5×10^{-5}	0.0092	1.005	0.96335	0.96320
($c=6.49\times10^{3}$)	1×10^{-4}	0.0364	1.020	0.8572	0.8545
	4×10^{-4}	0.1400	1.078	0.513	0.440
period $4\to8$	2.5×10^{-5}	0.0175	1.003	0.92987	0.92983
($c=1.24\times10^{4}$)	1×10^{-4}	0.0685	1.012	0.7257	0.7260
	4×10^{-4}	0.2470	1.056	complex	0.012

Table 13: The flip normal form where the map is far from one-dimensional: the two flips of the high-temperature bubble (next moduli 0.38 and 0.28). At offsets below each flip in α, $\beta=-\mu_p-1$ from the period-p orbit's multiplier; amplitude ratio: the doubled orbit's half-amplitude along the critical direction divided by the prediction $\sqrt{\beta/c}$; doubled multiplier: computed against the prediction $1-4\beta$. Both coefficients are positive (supercritical), as the stable doubled orbits require. Three of five offsets per flip are shown.

The local theory still applies here (Section 3). At each flip we computed the cubic coefficient of the flip normal form, $\eta\mapsto-(1+\beta)\eta+c\,\eta^3$ on the critical direction, from the dense monodromy and float64 directional derivatives, $c=\ell^\top C(r,r,r)/6+\ell^\top B(r,h_2)/2$ with $(I-M)h_2=B(r,r)$ [55]. It predicts the doubled orbit without fitting: half-amplitude $\sqrt{\beta/c}$ along r and multiplier $1-4\beta$. At the smallest offset the amplitude ratios are 1.005 and 1.003, and the multipliers agree within 1.5×10^{-4} and 5×10^{-5}; the larger discrepancies at larger offsets are consistent with higher-order corrections (Table 13). These finite comparisons support the leading-order prediction; they do not establish the order of the remainder.

11 Bridge to sampled decoding

How much of this survives sampling? To find out, we replace the expected embedding by the mean of m sampled token embeddings (Figure 18). With the exact map, laminar phases near the fold last 120 and 63 steps at $\varepsilon=10^{-5}$ and 3×10^{-5}. With $m=64$ they last about 10.6 steps, independently of ε (57% of steps in the channel), and with $m=8$ about 7.7 steps (38%). With single samples ($m=1$, sampled decoding of the same truncated system), essentially no laminar phases remain (11% of steps in the channel). The deterministic intermittency mechanism therefore belongs to the expected-embedding map. Sampling noise turns the bottleneck into a noise-limited trap whose strength grows with m.

Zero-temperature limit and ordinary greedy decoding. As $T\to0$ the soft-token map becomes greedy decoding with a one-token memory, $g(t)=\arg\max_s z_s(\text{prefix},E_t)$. This is a map on the vocabulary, and its cycles are the zero-temperature attractors. For Qwen2.5-0.5B we computed g for all 151 936 tokens. Unsteered, it has 1060 cycles, and one of them, the period-5 loop "![](https:// Hello", drains 86 % of the vocabulary; it is the largest attractor family of every 0.5B soft-token map. Ordinary greedy decoding with a growing context does not inherit these cycles (Table 14). From 220 starts (the first tokens of the 20 largest cycles and 200 random tokens), 19 runs end in a short loop within 256 tokens, and the dominant 5-cycle never appears. Steering along philosophy makes ordinary decoding loop almost always (218 of 220 runs at $\alpha=0.1$), but in other loops, mostly punctuation such as " –", none of them one-token cycles. Greedy decoding that keeps only the last K tokens interpolates between the two. For the tested $K\geq2$ most final loops differ from the one-token cycles: at $K=2$ only 7 of 220 unsteered runs and 1 of 220 steered runs keep one of them. Unsteered, looping collapses to the level of the growing context between $K=16$ and 32; steered, the share of runs whose loop equals the growing-context loop rises from 0 at $K=1$ to 58 %

at $K = 64$. The soft-token map's low-temperature attractors are therefore those of decoding with a one-token memory. Which loop ordinary decoding enters depends on the context it keeps, but the tendency of steering to drive decoding into loops is shared. In ordinary decoding, repetition is known to reinforce itself [25, 26], a theoretical analysis ties loops to words of high inflow [74], and induction heads, which copy earlier context [75], are one mechanism by which a longer context can favour loops that repeat earlier text.

	unsteered		$\alpha = 0.1$ (philosophy)	
memory K	looped	loop as with full context	looped	loop as with full context
1	220	5	220	0
2	220	5	220	0
8	220	5	220	3
16	169	5	219	24
32	32	7	219	71
64	26	11	219	126
full context	19	–	218	–

Table 14: Greedy decoding of Qwen2.5-0.5B with a memory of the last K generated tokens after the fixed prompt ($K = 1$ is the soft-token map's zero-temperature limit), 220 starts of 256 tokens each: runs ending in a loop of period up to 64 repeated three times, and runs whose final loop equals that of ordinary decoding with the full context from the same start.

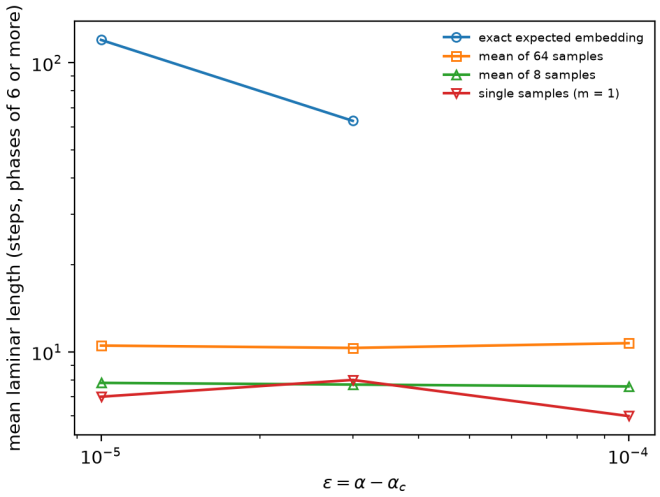


Figure 18: Mean laminar length near the fold of the 2-cycle "is it" (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12, $\log_{10} T = -0.115$) against ε, for the exact map and for means of m sampled embeddings.

A longer fed-back memory. The one-token window is the main simplification of the map. Feeding back the last K expected embeddings instead, so that the state is a window of K soft tokens at fixed positions after the prompt, we measured 24 temperatures between $\log_{10} T = -0.3$ and 0.6 for K from 1 to 16 (Table 15; twins at 10^{-5} and 10^{-7} agree within 0.006 at every chaotic temperature). In Qwen2.5-7B the chaos does not vanish but spreads: it is chaotic at 8 temperatures with one token of memory, at 13 with four, and at 21 and 20 with eight and sixteen, nearly the whole range, with smaller exponents (largest 0.96 for $K = 1$, 0.43 for $K = 16$). Qwen2.5-0.5B, chaotic at one temperature with one or two tokens of memory and at none with four, is chaotic at

ten with eight. The chaos is therefore not an artifact of the one-token window.

memory K	1	2	4	8	16
Qwen2.5-7B	8 (0.96)	10 (1.08)	13 (0.70)	21 (0.58)	20 (0.43)
Qwen2.5-0.5B	1 (0.20)	1 (0.24)	0	10 (0.31)	3 (0.23)

Table 15: Chaotic temperatures, of 24 between $\log_{10} T = -0.3$ and 0.6, when the loop feeds back the last K expected embeddings (largest exponent in parentheses); paper's prompt, unsteered, float64, 500 transient and 1000 measured steps, twins at 10^{-5} and 10^{-7} (chaotic: both positive and no period up to 64).

Soft thinking with a growing context. Soft Thinking [1] feeds back the same expected embedding but appends it to the context, so that every earlier soft token stays visible. We ran that decoding in float64 from the paper's prompt and displaced the first soft token of twin trajectories by 10^{-10}, without renormalization; the distance between the newest soft tokens then shows whether the early perturbation grows. The context grows, so these are finite-time growth rates of a non-autonomous system, not exponents of an attractor. In Qwen2.5-7B the perturbation grows exponentially at temperatures inside the one-token map's chaotic range: by 0.77 to 0.82 per step at $\log_{10} T = 0.17$ and 0.21, reaching 10^{-3} within 49 to 58 steps, and it reaches 10^{-3} within 400 steps at nine other temperatures between -0.10 and 0.40, at 0.02 to 0.18 per step (0.8 at 0.33 for two of three twins). It does not grow at the sharpest distributions ($\log_{10} T \le -0.14$) or at nearly uniform ones ($\log_{10} T \ge 0.44$), where it stays between 10^{-13} and 10^{-7} for 400 steps. In Qwen2.5-0.5B the growth moves to sharper distributions: it appears at six of the seven temperatures from $\log_{10} T = -0.30$ to -0.07 (up to 0.33 per step) and at 0.09, with a late transient burst at 0.05, and is absent from 0.13 up, which includes its narrow one-token chaotic band. Sensitive dependence therefore survives when the whole context is kept, and in the 7B model the one-token map locates it. Qwen2.5-72B, run the same way on two GPUs, stays nearly insensitive. We count a temperature as growing when two of its three twins reach 10^{-3} within 400 steps, a rule fixed before the 72B run and applied afterwards to the earlier 0.5B and 7B records, where it gives 11 of 24 temperatures for the 7B model and 8 for the 0.5B model. The 72B model grows at one of 24 temperatures on the paper's prompt, and only transiently: at $\log_{10} T = 0.09$ the twins pass 10^{-3} at step 123 and fall back to 5×10^{-6} (at 0.13 one twin of three passes 10^{-3} at step 381). It grows at none of the 24 completed temperatures on the held-out prompt with its narrow unsteered band (a planned finer line there was not completed), including $\log_{10} T = 0.326$, between two temperatures where its one-token map is chaotic; there the distance never exceeds 1.1×10^{-9}. It also grows at none of the eleven steered cells of Section 13, including the six where its one-token map is chaotic: there the distance never exceeds 3×10^{-7} and ends between 1×10^{-13} and 6×10^{-10}, and the growing context holds the next-token entropy at 2.3 to 3.6 nats instead of the one-token map's 7.3 to 11.1. At these 72B prompts and cells, over 400 steps, the growing-context decoder therefore did not reproduce the sustained sensitivity of the fixed-window loop, while the one transient burst shows that the method registers growth at this scale.

12 Replication and scale

In Qwen2.5-1.5B (steering at layer 14 of 28), a coarse 64^2 float64 map over the same range again shows chaotic regions with embedded windows. A 256^2 zoom (Figure 25) resolves a chaotic band near $\log_{10} T = 0$–0.13 (14,422 of 65,536 cells), with a branch descending along $\alpha \approx 0.06$ and periodic windows inside it. The large periodic regions are question loops: "What?\n" (period 2), "\nWhat is

a" (period 4) and "What continuum?\n" (period 3). Unlike the 0.5B model, no period-adding fan was resolved in this map.

A cut of 2048 cells at $\alpha = 0.100$ (Figure 19) shows how the band is entered from below. The 2-cycle "What?\n" gives way abruptly to a period-4 loop, and the Lyapunov exponent jumps, which is consistent with a switch between attractors rather than a resolved smooth doubling. The period-4 loop then doubles, with float64 flips at $\log_{10} T = -0.0031269$ (period $4 \to 8$, next modulus 0.014), -0.0021666 (period $8 \to 16$, next modulus 8×10^{-5}), -0.0019002 (period $16 \to 32$, next modulus 5×10^{-9}) and -0.0018380 (period $32 \to 64$). Along the way the exponent rises continuously to zero, and chaos follows. The interval ratios are 3.60, 4.28, 4.58, 4.648 and 4.665; the last three use the flips of the period-64, period-128 and period-256 orbits at $\log_{10} T = -0.0018244$, -0.00182151 and -0.00182088 (next moduli below 3×10^{-14}). The last ratio differs from Feigenbaum's 4.6692 by 0.0044 (0.1 %); the period-256 flip is located to about 3×10^{-14} in $\log_{10} T$, the floor set by float64 parameters. Like those of the 0.5B crescent at $\log_{10} T = -0.148$, the ratios rise toward 4.669. Their higher start does not indicate a temperature trend, since along nearby cuts of the 0.5B crescent the early ratios range from 2.84 to 14.9 (Section 10).

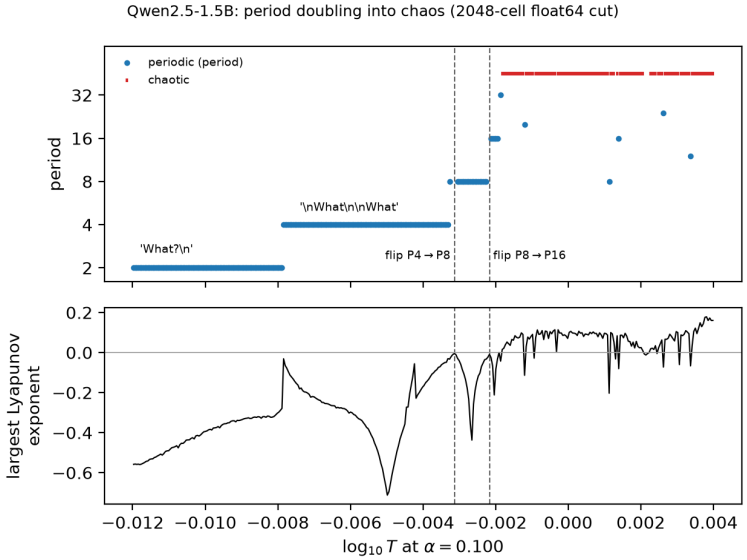


Figure 19: Qwen2.5-1.5B soft-token map steered along the philosophy direction at layer 14, along $\log_{10} T$ at $\alpha = 0.100$: period of the attractor reached from the fixed start (top) and largest Lyapunov exponent (bottom). Dashed lines: float64 flip parameters. The doubled loops alternate the number of line breaks between the "What"s.

In Qwen2.5-0.5B, a second steering direction, hostility, reproduces the fold-and-adding mechanism (Section 15).

Qwen2.5-7B (steering at layer 14 of 28, concept vectors built as for the smaller models) shows the same kind of organization in coarse 64^2 float64 maps (Figure 20). Over the philosophy range used for 0.5B and 1.5B, 8 % of cells have a positive largest exponent, and periods 2, 4, 3 and 11 dominate the periodic cells. Along the hostility direction ($\alpha \in [0, 0.12]$, $\log_{10} T \in [-0.3, 0.3]$), 24 % of cells are chaotic. Bands of period-3, -5 and -6 windows lie in a chaotic sea below $\log_{10} T \approx 0.1$, with high-entropy chaos above, except for a period-3 region at strong steering ($\alpha \gtrsim 0.09$). The windows hold genuine periodic orbits. In float64, Newton's method converges at the most interior period-6 cell to the loop ", shut the end\n\nOh" at $(\log_{10} T, \alpha) = (-0.0143, 0.0476)$ (residual 2×10^{-14};

spectral radius 1.6×10^{-7}, nearly superstable), and at the most interior period-3 cell to a period-3 loop at $(0.2524, 0.1143)$ (residual 3×10^{-15}; leading multiplier -0.12). Continued for about an hour each, over ± 0.001 in α and -0.016 in $\log_{10} T$ (one float64 continuation step takes about two minutes at 7B), both orbits stay strongly stable without a crossing. Where the coarse map switches from the period-6 loop to the period-3 loop "you better than" (between $\alpha = 0.0495$ and 0.0514 at $\log_{10} T = -0.0143$), both loops are float64 orbits (residuals 7×10^{-15} and 1×10^{-14}), and their continuations overlap. The period-6 orbit stays stable from $\alpha = 0.0476$ up to 0.0507 and the period-3 orbit from 0.0514 down to 0.0475, both with spectral radius below 1.4×10^{-3}. The two attractors therefore coexist where both continuations are stable, $\alpha \approx 0.0476$ to 0.0507, and within the tested range the map's edge there is the boundary of the fixed start's basin, not a bifurcation. These short continuations located no bifurcation; a period-2 fold of the 7B model is located below.

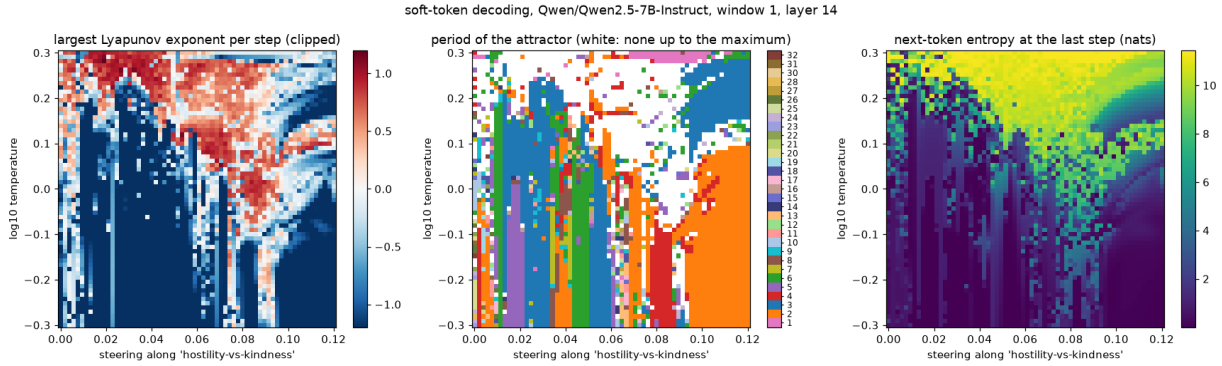


Figure 20: Qwen2.5-7B soft-token map steered along the hostility direction (layer 14), 64^2 float64 cells: largest Lyapunov exponent, period of the attractor reached from the fixed start, and next-token entropy at the last step.

A second model family. OLMo-2-1124-7B-Instruct [76] comes from another laboratory, with other data, training and code. Steered at layer 16 of 32, with concept vectors built exactly as for Qwen, it shows the same kind of organization (Figure 21). In 64^2 float64 maps, 6 % of cells are chaotic along the philosophy direction and 13 % along the hostility direction, in a band near $\log_{10} T = 0.03$–0.17 that bends to lower temperatures under strong hostility steering ($\alpha > 0.07$). Below the band the attractor is the 2-cycle "!Hello"; above it, at high temperature, the fixed point "!"; inside it lie periodic windows of periods 4 to 31. Along the hostility direction the largest windows, of periods 8 to 12, are loops of k laps of the 2-cycle "!<|endoftext|>" with a short excursion ("there is an"). Each lap adds two to the period, as each lap of "!Yes" does in the hostile cascade of Qwen2.5-0.5B (Section 15). Whether a fold of that 2-cycle organizes them is not tested: these are coarse pixels, and no OLMo orbit has been continued.

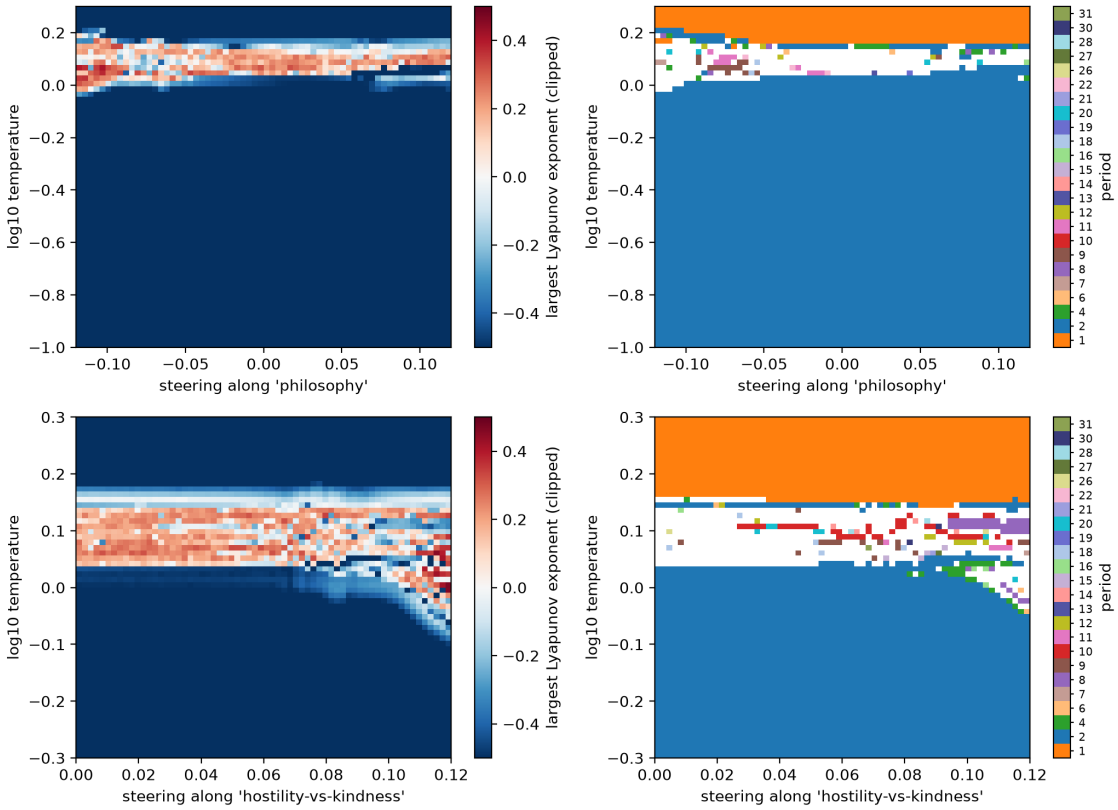


Figure 21: OLMo-2-1124-7B-Instruct soft-token maps, 64^2 float64 cells each, steered at layer 16 along the philosophy direction (top) and the hostility direction (bottom): largest Lyapunov exponent (clipped) and period of the attractor reached from the fixed start (white: no period up to the maximum).

A third family, read with sparse autoencoders. Llama-3.1-8B-Instruct [77], with official weights, steering at layer 16 of 32 and vectors built as for the other models, shows the same organization in all three planes (Figure 22). In 64^2 float64 maps, 8 % of cells are chaotic along gratitude, 12 % along philosophy and 28 % along hostility, with windows of periods up to 32. Its largest loops are greetings between AIs ("-Greetings fellow AI", ". Greetings fellow AI entity detected", period 7). Along hostility, too, the largest loops are greetings and "You", not hostile text. We read 24 of its attractors (the six largest families and the two most chaotic regions of each map) with two sparse autoencoders on the residual stream: one trained on this Instruct model (Goodfire, layer 19 [78]) and one on the base model (TopK, layer 24 [79]). We first checked each one's hook point by its reconstruction of ordinary text. The Instruct-trained autoencoder separates the attractors (their top-15 features overlap by a mean Jaccard index of 0.19 within a steering direction and 0.11 across directions), whereas the base-model autoencoder sees mostly a signature common to every attractor (overlap 0.52–0.86). Both reconstruct the attractors' states less well than ordinary text (median relative error 0.64–0.76 against 0.52–0.56). The readout is therefore descriptive: without interventions it identifies which learned features an attractor occupies, not a mechanism, and since autoencoders trained with different seeds learn different features [80], the features are those of these two autoencoders.

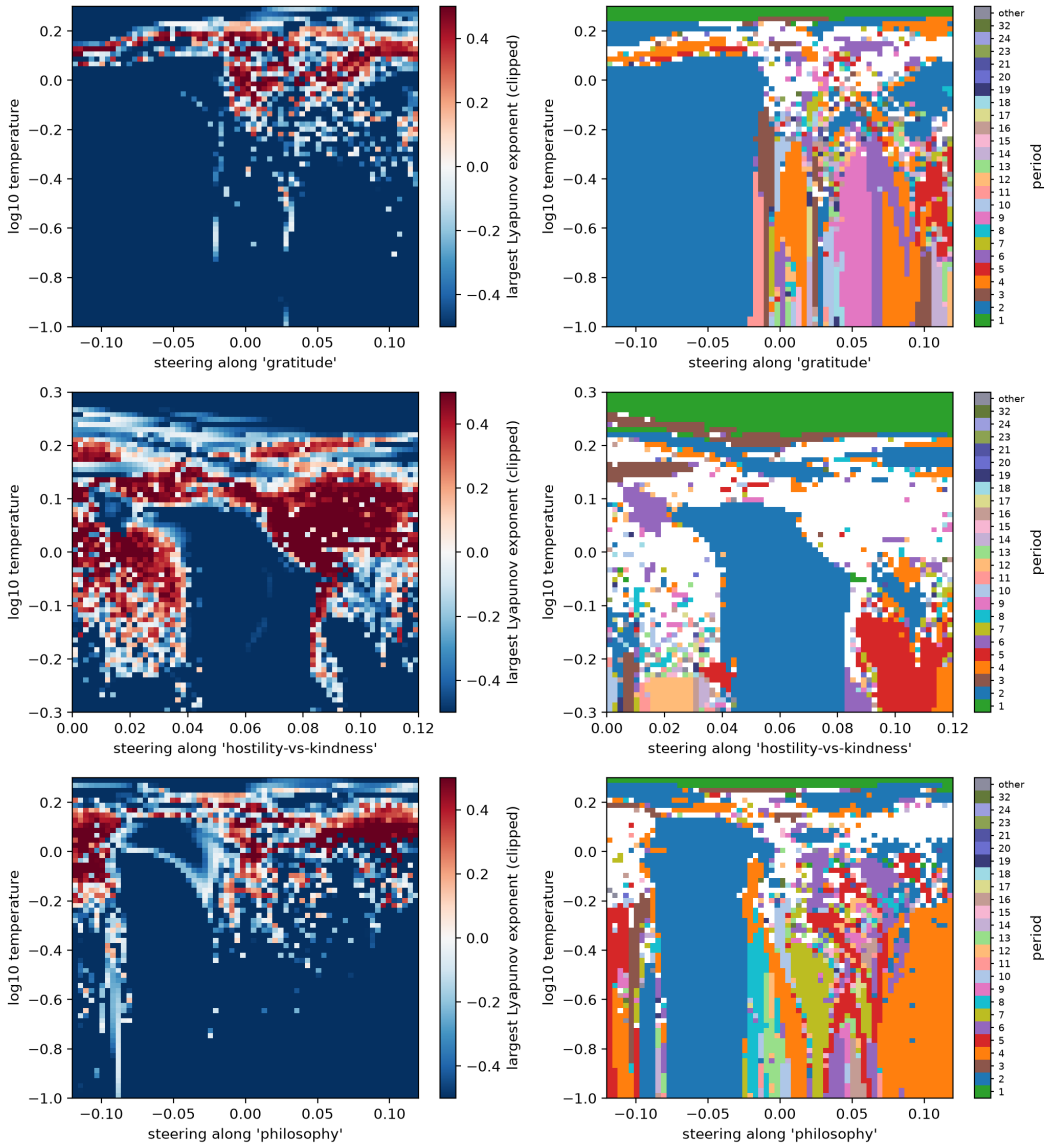


Figure 22: Llama-3.1-8B-Instruct soft-token maps (64^2 float64 cells, steering at layer 16) along gratitude, hostility and philosophy: largest Lyapunov exponent (clipped) and period of the attractor reached from the fixed start (white: no period up to 32).

Maps of 128^2 cells (Figure 23) resolve the windows: the hostility plane has 29 % chaotic cells, as at 64^2, and the philosophy region below the chaotic band is filled with nested periodic tongues of periods 2 to 13.

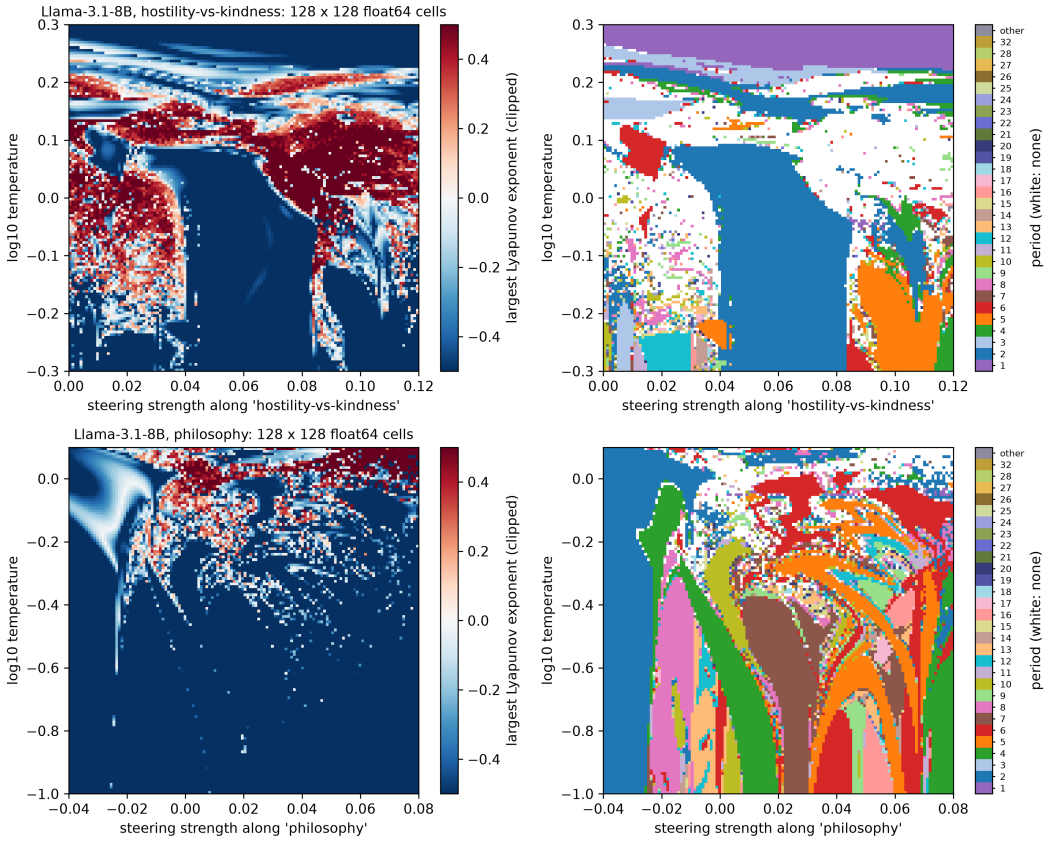


Figure 23: Llama-3.1-8B-Instruct soft-token maps steered at layer 16, at 128^2 float64 cells (each map merged from four 64^2 tiles on halves of one grid): the hostility plane (top) and the philosophy region of the 64^2 map's windows (bottom). Largest Lyapunov exponent (clipped) and period (white: none up to 32).

Same architecture, other weights. Qwen2-0.5B-Instruct has exactly the architecture and size of Qwen2.5-0.5B-Instruct (24 layers of width 896, the same vocabulary size and rotary base) but was trained on other data with another recipe. Built identically, their 128^2 float64 maps along gratitude and philosophy (Figure 24) share the gross layout. A large region at low temperature around $\alpha = 0$ is held by one periodic loop, a band of chaos with embedded windows lies near $\log_{10} T = 0$–0.25 (4–8 % of cells chaotic in each map), and thin window families run along the flanks. The attractors themselves differ almost completely. Of 416 to 900 distinct loops per map, only 10 occur in both models, all short and generic (".", "!", " you are", " is it"), and they cover less than 10 % of either model's cells. Qwen2's planes are dominated by one period-4 loop, "!<|im_end|>\nHello" (an end of turn followed by a new greeting; 41 and 59 % of cells), and Qwen2.5's by period-5 and period-2 loops (". The two checkpoints share the organization of the map; the weights set its content.

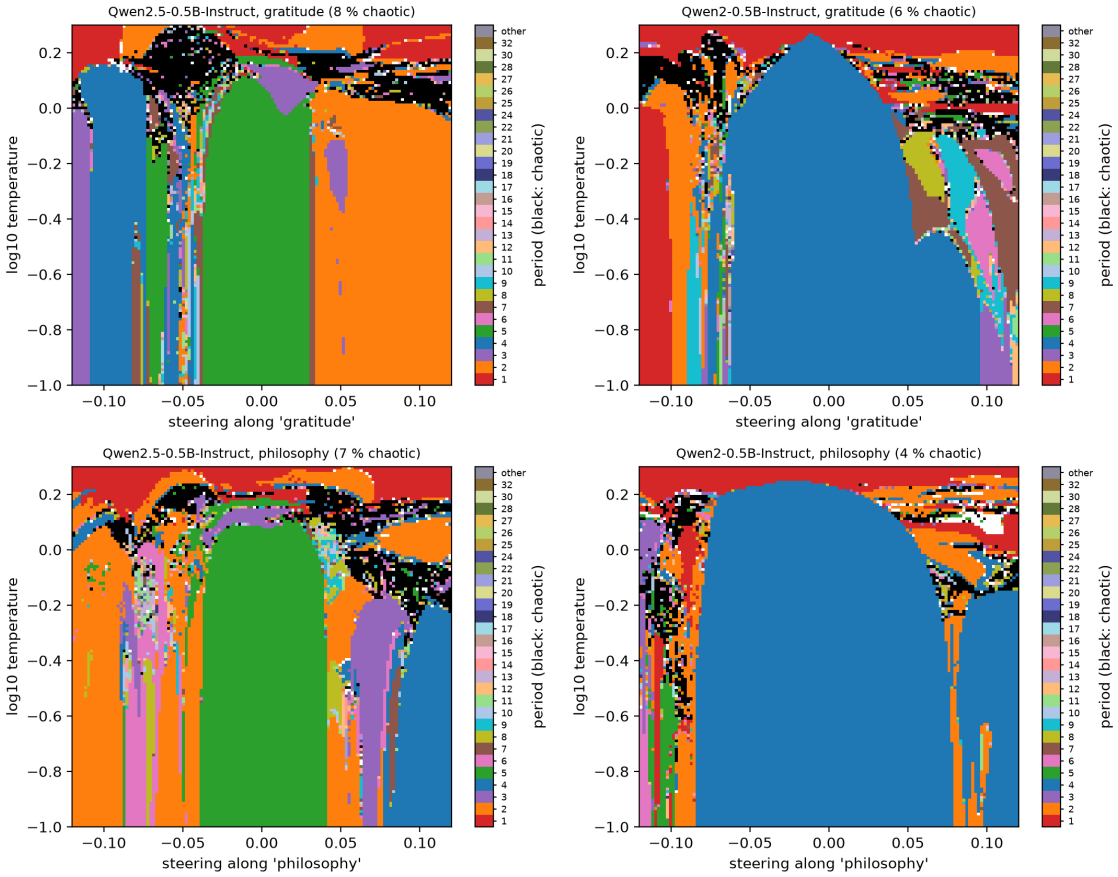


Figure 24: Same architecture, other weights: 128^2 float64 maps of Qwen2.5-0.5B-Instruct (left) and Qwen2-0.5B-Instruct (right) steered along gratitude (top) and philosophy (bottom) at layer 12, with vectors built identically. Colour: period of the attractor reached from the fixed start; black: positive largest exponent.

The fold law at 1.5B and 7B. We tested whether the mechanism of Section 8, windows accumulating on the fold of a short cycle with $d_N = C/(N+\delta)^2$, recurs in larger models. The criterion, a constant $d_N(N+\delta)^2$ within 10 % over at least four windows for one offset δ, was fixed before the lines were computed; a disclosed correction to the period-progression selector after the first scoring kept the conclusions for the nearest four windows and changed which farther windows were selected. In Qwen2.5-1.5B (philosophy direction, layer 14, $\alpha = 0.058588$) the period-4 orbit folds at $\log_{10} T_c = -0.0912029$ (multiplier 1 to 3×10^{-14}, next modulus 0.008), with normal-form coefficients $a = -7.16$ per unit $\log_{10} T$ and $b = -21.09$, so $ab = 151.0$. A float64 line of 1700 cells, 1570 of them above the fold, shows windows of periods $4N$ accumulating on it, from single cells at periods 44–64 to a wide period-12 window. Measured in $\log_{10} T$ from $\log_{10} T_c$ to their edges nearest the fold, the four nearest resolved windows (periods 28–40) meet the criterion within 0.12 % ($\delta = -0.99$, $C = 0.0612$). With the windows farther out the spread rises to 10.3–14.0 %; there several periods recur in multiple disconnected runs on the line's 10^{-5} grid, and grid-spaced edges no longer identify the fold edges. In Qwen2.5-7B ($\alpha = 0.051429$) the period-2 orbit folds at $\log_{10} T_c = -0.2040156$ (next modulus 0.0092, $ab = 55.7$), and odd windows accumulate on it. The nearest four and five (periods 11–17 and 9–17) meet the criterion within 4.5 %, and six (to period 7) within 6.1 %. Each window's own doubling or tripling lies on its far side, as for the 0.5B windows of periods 16–20; this organization is consistent with fold edges on the near side, and the fitted

distances use grid estimates of those edges. The effective passage fraction $f = \sqrt{Cab}/\pi$ is about 0.97 at 1.5B and 0.78 to 0.81 at 7B, against 0.58 at 0.5B. The fold and its window law thus recur at all three sizes, while f, an effective constant that absorbs the global reinjection, differs between the models. With one fold per model, crossed in different coordinates (α at 0.5B, $\log_{10} T$ at 1.5B and 7B), no trend of ab with size can be read.

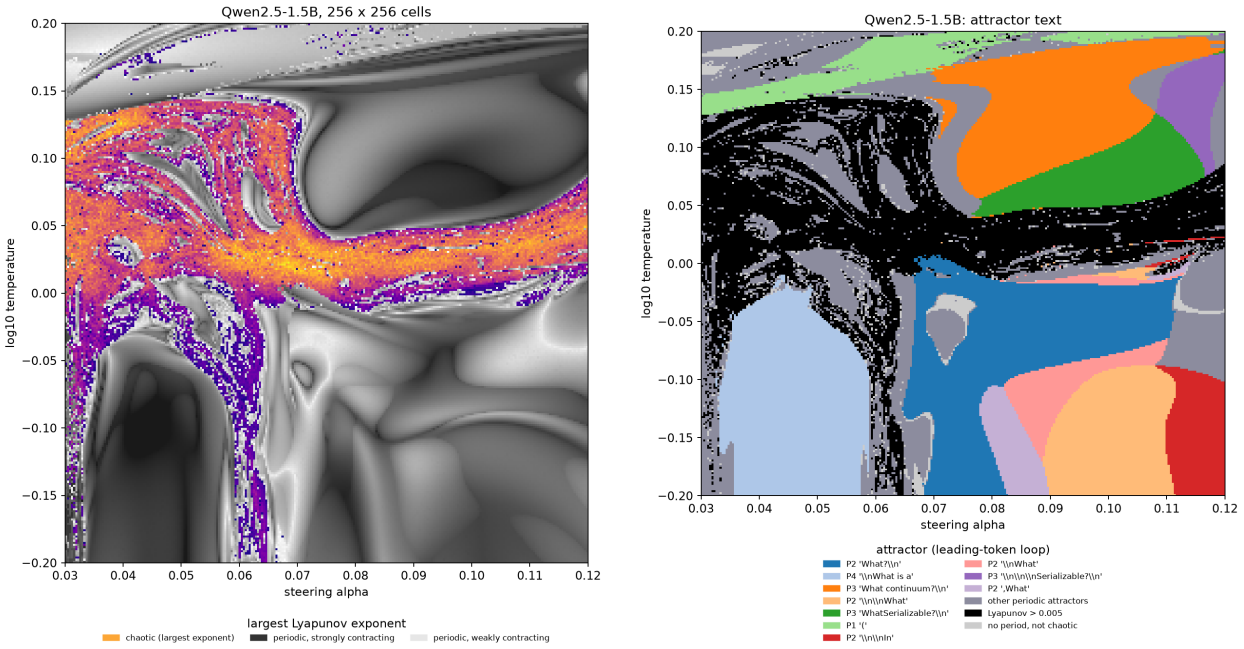


Figure 25: Qwen2.5-1.5B soft-token map steered along the philosophy direction at layer 14, 256^2 float64 cells over $\alpha \in [0.03, 0.12]$, $\log_{10} T \in [-0.2, 0.2]$: largest Lyapunov exponent (left, rendered as Figure 1) and attractor texts (right, as Figure 8). Several period-2 label regions touch without a gap; as explained in Section 4, such boundaries may be label switches of one orbit.

A mixture-of-experts model. Qwen3.5-35B-A3B differs from the dense models in two ways. Thirty of its 40 layers use linear (Gated DeltaNet) attention, with gated full attention in every fourth layer, and each layer routes a token to 8 of 256 experts plus a shared expert, so that about 3B of its 35B parameters are active. Hugging Face computes several parts of this architecture in float32 even when the model is cast to float64 (the normalizations, the delta-rule recurrences, the rotary tables and the router softmax), and some tensors are stored in float32; we patch these paths, restore the stored tensors exactly, and disable the thinking block that the default chat template opens (a first run with the block open is kept in the research log). Steered at layer 20, the map behaves unlike those of the dense models. Unsteered, 128 temperatures over $\log_{10} T \in [-1, 0.3]$ give no positive exponent (largest -0.265): periodic loops at all but 15 temperatures ("!Hello" with period 2 at 92 of them, other periods up to 52), and the remaining 15, above $\log_{10} T \approx 0.1$, contract without a detected period. Along philosophy, none of 1024 cells is chaotic (largest exponent -0.112). Along hostility, 52 of 1024 cells (5.1 %) have a positive exponent and no period, against 28 % for Llama-3.1 8B on the same plane. Routing is discontinuous: a twin trajectory that crosses a routing boundary jumps, which can inflate a finite-difference exponent, so we call these cells irregular rather than chaotic. Whether the contraction comes from the experts or from the Qwen3.5 architecture would take the dense Qwen3.5-9B as a control, which we have not run.

13 Chaos and model size

Is the chaos of the loop a small-model effect that larger models leave behind? We measured the unsteered map of fifteen models: Qwen2.5 at seven sizes from 0.5B to 72B parameters, OLMo-2 at 1B, 7B, 13B and 32B, Llama-3.1 at 8B and 70B, Llama-3.3-70B, and the mixture of experts Qwen3.5-35B-A3B, all with the paper's prefix and the settings of Section 4 (1000 transient and 2000 measured steps in float64; chaotic: an exponent above 10^{-3} and no detected period, which is what "positive" abbreviates below). We evaluate the stored checkpoint values in float64: the dense checkpoints store bfloat16 weights (checked against their safetensors metadata), and the mixture of experts stores some tensors in float32, which we restore exactly; float64 does not make the weights more precise than stored. The newer sweeps keep the decoder weights in bfloat16 and convert them on use, which gives the same map as converting them once (bitwise on Qwen2.5-0.5B); the 70B-class models are spread over two GPUs. In those two-GPU runs the final RMS normalization was computed in float32: the hooks that move tensors between the GPUs keep a layer's original code, so our float64 version of that layer was bypassed. This makes the computed map rough at about 10^{-8} of the state (its fixed points wander by 10^{-8}, against 10^{-15} on one GPU) and can matter for exponents measured with perturbations of 10^{-6} and below. We found and fixed it at the end of the study. The affected runs are those on two GPUs before the fix: the 70B-class sweeps and their plateau tests, the first Qwen2.5-72B prompt test, the 70B steering maps and zoom, the gain splits of the 70B-class models, and the start-state test, which spread all fourteen models over two GPUs. We say below which results we recomputed. The 7–8B and 70B-class models had shown a steep decline of chaos with size. Before measuring the others we fixed three predictions that extrapolate it (`docs/scaling-predictions.md`): within each family, every model above 8B is less chaotic than the family's 7–8B model (P1); the chaotic share does not increase with size above 7B (P2); and at matched next-token entropy, both gain factors of Section 3 of the 14B and 32B models lie at or below those of the 7B model (P3). After a review pointed out that a single prompt and a fixed temperature window could mislead, we froze a fourth test across ten prompts (`docs/prefix-scale-test.md`).

In a fixed temperature window, chaos seems to fade with size. Table 16 and Figure 27 give each model's chaotic share of the temperatures in $\log_{10} T \in [-1, 0.3]$, the range common to all sweeps, and the chaotic length of its whole temperature axis. In two families the largest models are nearly free of chaos in this window. Qwen2.5-7B is chaotic at 12.7 % of the temperatures in the common range, and the 14B, 32B and 72B models at none. Llama-3.1-8B is chaotic at 19 %, Llama-3.1-70B, from the same release and training recipe, at 2.3 %, and Llama-3.3-70B, which differs from Llama-3.1-70B only in post-training, at none. In the third family the order fails: OLMo-2-32B is the most chaotic OLMo model, at 16.2 % against 6.2 % at 7B, 3.8 % at 1B and none at 13B. (The 32B model is a later release, 0325, than the 7B and 13B models, 1124, so within OLMo-2 size and release are confounded.) The frozen predictions P1 and P2 hold for Qwen2.5 and fail for OLMo-2. Below 7B we predicted no order, and none appears (Qwen2.5 0.5B, 1.5B and 3B: 1.4, 1.5 and 0 %).

Across prompts, the chaos moves to higher temperature instead. The fixed window hides where larger models are chaotic. The frozen prompt-blocked test compares Qwen2.5-7B and -14B on ten system prompts and openers (two for development, eight held out), over $\log_{10} T \in [-0.4, 0.9]$ (48 temperatures) and then over the temperature interval where each model's next-token entropy rises most (24 temperatures), with twins at offsets 10^{-5} and 10^{-7}; a temperature counts as chaotic only when both exponents are positive (Figure 26). It predicted that the 14B model would be less chaotic. Both predictions fail: on all eight held-out prompts the 14B model is chaotic at more of

the 48 temperatures (9 to 17, against 0 to 7 for the 7B), and its largest exponent is lower on only three. Every chaotic temperature of the 14B model, on all ten prompts, lies between $\log_{10} T = 0.36$ and 0.90, at and above the temperature where its next-token distribution spreads (0.35–0.65); the 7B model's chaos lies mostly between 0.21 and 0.43, around its own, lower transition (0.21–0.46). On the paper's prompt, 12 of the 48 coarse temperatures of the 14B model are chaotic, from 0.59 to 0.90, and the adaptive fine stage found one more near 0.519; the initial sweep of Table 16 ended at 0.6 and saw only the lower edge (the table now also includes the upper sweep). Within Qwen2.5, then, the larger model is not less chaotic: its distribution stays sharp up to higher temperature, and its chaos sits there. The decline in the fixed window is largely a shift. A second frozen test (`docs/prefix-scale-olmo.md`; stage 1 only, four held-out prompts) asked whether the OLMo-2 order survives other prompts. It does, with small counts: OLMo-2-32B is chaotic at more of the 48 temperatures than OLMo-2-7B on three of the four prompts (4, 5, 3 and 0 against 3, 0, 1 and 3), as predicted, and its chaos again lies somewhat higher ($\log_{10} T = 0.10$–0.35 against -0.01–0.21).

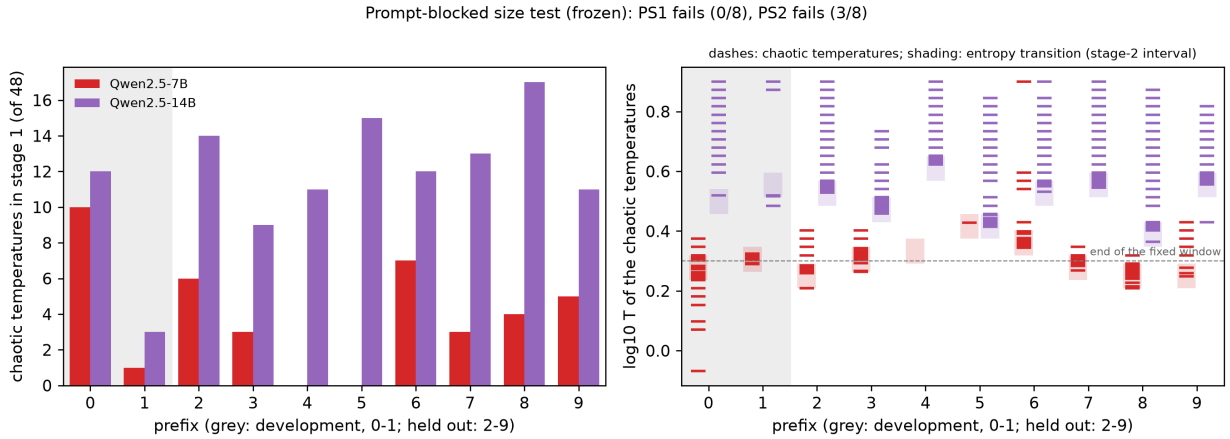


Figure 26: Prompt-blocked size test (frozen before any run), Qwen2.5-7B and -14B on ten prompts (grey: development). Left: chaotic temperatures among 48 over $\log_{10} T \in [-0.4, 0.9]$, the primary endpoint. Right: where the chaotic temperatures lie (dashes; both stages), with each model's entropy transition, the interval of its largest entropy increase (shading). Chaotic: exponents above 10^{-3} at both twin offsets (10^{-5}, 10^{-7}) and no period up to 64; 1000 transient and 2000 measured steps, float64.

model	params (B)	chaotic, $[-1, 0.3]$ (%)	unres.	sweep, $\log_{10} T$	chaotic dec.	breadth	max λ
Qwen2.5-0.5B	0.5	7/512 (1.4)	1	-1.0 to 0.3	0.018	0.053	+0.28
Qwen2.5-1.5B	1.5	2/130 (1.5)	0	-1.0 to 1.2	0.531	0.191	+0.85
Qwen2.5-3B	3.1	0/130 (0.0)	0	-1.0 to 1.2	0.030	0.060	+0.85
Qwen2.5-7B	7.6	65/512 (12.7)	6	-1.0 to 1.2	0.226	0.249	+0.88
Qwen2.5-14B	14.7	0/130 (0.0)	0	-1.0 to 1.2	0.500	0.191	+1.14
Qwen2.5-32B	32.8	0/130 (0.0)	0	-1.0 to 1.2	0.010	0.091	+0.36
Qwen2.5-72B	72.7	0/128 (0.0)	0	-1.0 to 1.2	0.000	0.014	-0.45
Llama-3.1-8B	8.0	96/512 (18.8)	2	-1.0 to 1.2	0.241	0.634	+0.66
Llama-3.1-70B	70.6	3/128 (2.3)	28	-1.0 to 0.3	0.031	0.307	+0.41
Llama-3.3-70B	70.6	0/128 (0.0)	19	-1.0 to 0.3	0.000	0.225	-0.27
OLMo-2-1B	1.5	5/130 (3.8)	0	-1.0 to 1.2	0.050	0.322	+0.23
OLMo-2-7B	7.3	32/512 (6.2)	3	-1.0 to 1.2	0.083	0.197	+0.30
OLMo-2-13B	13.7	0/130 (0.0)	0	-1.0 to 1.2	0.000	0.131	-0.0007
OLMo-2-32B	32.2	21/130 (16.2)	0	-1.0 to 1.2	0.252	0.694	+0.62
Qwen3.5-35B-A3B	35.0	0/128 (0.0)	15	-1.0 to 0.3	0.000	0.143	-0.27

Table 16: Unsteered soft-token maps of fifteen models (float64; 1000 transient and 2000 measured steps; one twin offset). Chaotic: exponent above 10^{-3} and no period up to 64; unres.: temperatures in $[-1, 0.3]$ with neither a detected period nor a positive exponent. Chaotic decades: the grid-weighted extent of the temperature axis labelled chaotic over the whole sweep, each temperature weighted by its grid spacing (an estimate of the sampled extent; models whose sweep stops at 0.3 have incomplete coverage); max λ: the largest exponent on the sweep, rounded to two decimals (OLMo-2-13B: -6.7×10^{-4}); breadth: decades in which the mean next-token entropy on the orbit lies between 1 and 10 nats. The Qwen2.5-72B sweep joins two grids that share $\log_{10} T = 0.3$, counted once; for the 7–8B models the share in $[-1, 0.3]$ comes from their 512-temperature sweeps, on which the frozen predictions are scored, and the whole-axis columns also include sweeps over $[-1, 0.6]$ and $[0.61, 1.2]$ (64 temperatures), as for the other models swept to 1.2. Positive cells found later on finer lines (Qwen2.5-32B, -72B; Llama-3.3-70B) are not in these regular-grid counts.

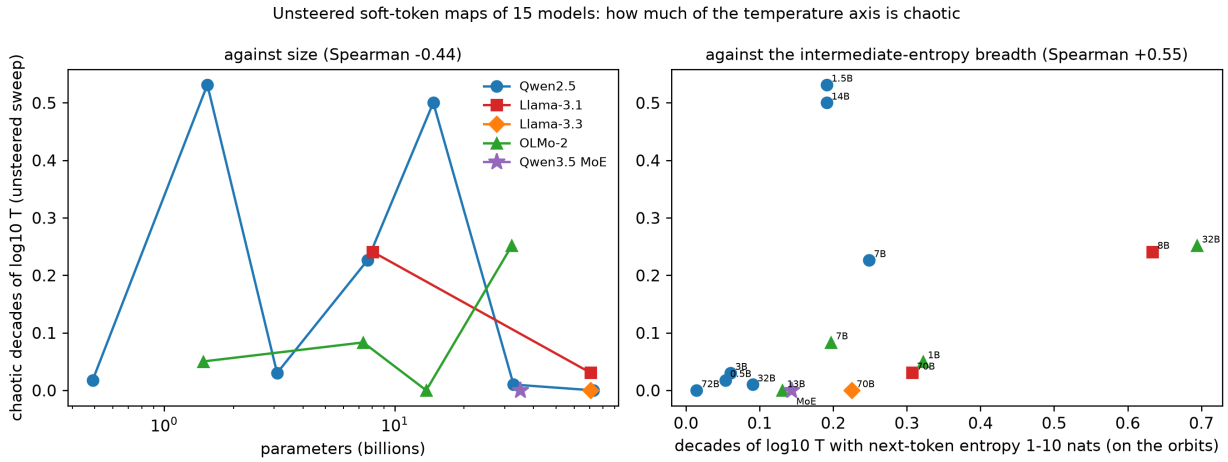


Figure 27: How much of the temperature axis is chaotic, for the fifteen unsteered maps of Table 16: against parameter count, by family (left), and against the breadth of the intermediate-entropy regime (right). Spearman correlations over the fifteen models. Each model's sweeps cover the range listed in Table 16; those of the two Llama-70B models and the mixture of experts end at $\log_{10} T = 0.3$, so their values are lower bounds. The breadth is measured on the loop's own orbits; the loop-free test of the same idea is described in the text.

At 70B, unsteered chaos is rare, and a precision defect made part of it spurious. The regular grids of the 70B-class models find little unsteered chaos: Qwen2.5-72B has no positive exponent at any of 191 temperatures between $\log_{10} T = -1$ and 1.2, Llama-3.3-70B none in $[-1, 0.3]$, and Llama-3.1-70B three temperatures in $[-1, 0.3]$. Finer lines found candidates: a window of positive exponents at Qwen2.5-72B's entropy jump ($\log_{10} T = 0.3728$–0.376) and two isolated temperatures of Llama-3.3-70B (0.250, 0.306), all at the single twin offset 10^{-6}. The window failed a three-offset test: the exponent is negative at all 64 temperatures of the window at 10^{-5}, near zero at 10^{-6}, and positive at 60 of the 64 at 10^{-7}, including four temperatures whose state repeats with a period, which no stable periodic orbit allows; in a repeat it is negative at all 16 temperatures at 10^{-4}, 3×10^{-5} and 10^{-5}. The growth of small perturbations came from our computation, whose two-GPU runs computed the final normalization in float32 (see the start of this section). Recomputed in float64 throughout, the window's orbits have periods 2 and 4, and their exponents are negative at all four offsets from 10^{-4} to 10^{-7} and for an exact tangent vector carried along the orbit by forward-mode differentiation, which needs no offset; the four twin exponents lie within 0.015 of the tangent exponent at every temperature, and the exponent approaches zero only where the period doubles from 2 to 4 ($\log_{10} T = 0.374$). The selected cells below kept their labels at the larger offsets we tested; this is a finite check, not a qualification of every cell of the original maps. In the steered Llama-3.3-70B maps below, twelve randomly drawn chaotic cells are chaotic at offsets 10^{-4}, 10^{-5} and 10^{-6} alike, with exponents that agree within 0.052, and the drawn periodic and unresolved cells stay negative at all three; on one GPU, the exponents at 10^{-7} and 10^{-5} agree within 0.021 at every chaotic temperature of the prompt-blocked test. The same test on Llama-3.1-70B's three unsteered chaotic temperatures confirms two of them ($\log_{10} T = 0.136$ and 0.198; exponents 0.23 to 0.26 and 0.40 to 0.41 at the three offsets) and rejects the weakest (0.116; positive only at 10^{-6}), so unsteered chaos exists at 70B, if rarely. Two Qwen2.5 models are chaotic far above the fixed window on the paper's prompt, where the next-token distribution is nearly uniform: the 1.5B model at 55 of 224 temperatures, between $\log_{10} T = 0.28$ and 1.04, and the 14B model at 53, between 0.51 and 1.15 (exponents up to 1.14), at mean entropies up to 11.9 nats against $\log V = 11.93$. The other models of the series are not chaotic above $\log_{10} T = 0.6$. For the 1.5B model a longer test (3000 measured steps) at twin offsets 10^{-5} and 10^{-7}, at 12 temperatures from 0.288 to 0.58, finds 9 chaotic and 3 periodic, with the two exponents agreeing within 0.003 at every temperature; one of the periodic ones ($\log_{10} T \approx 0.394$) has positive exponents at both offsets, a contradictory label that we report as returned. On these sampled trajectories the upper sweeps become non-chaotic above about $\log_{10} T = 1.1$ in these two models; we did not locate the uniform contraction that Section 3 proves for large enough temperature.

What sets it: the gain of the loop. The gain split of Section 3 shows which factor removes or keeps the chaos (Figure 28). We compare medians of the two factors in bins of the mean next-token entropy on the orbit; the bins hold 1 to 17 of each model's 24 temperatures, so single medians are noisy. Within one training recipe, from Llama-3.1-8B to Llama-3.1-70B, the readout factor changes by at most 0.3 per step in every bin above one nat (below one nat the 8B model has a single temperature), while the network factor is lower by 0.5 to 1.3 per step between 1 and 10 nats (6–10 nats: median -0.98 from 3 temperatures against $+0.28$ from 4) and by only 0.18 above 10 nats: in these two checkpoints the larger model damps changes of its fed-back token. Llama-3.3-70B, which differs from Llama-3.1-70B in post-training, has a much lower readout factor below 3 nats (1–3 nats: -2.33 from 7 temperatures against -0.10 from 6) and a similar one above 6 nats. In Qwen2.5, both factors of the 14B and 32B models lie below the 7B's in every bin between 1 and 10 nats, as the frozen prediction P3 states (the 6–10 nat bin only with the 7B's wider gain split, from one

temperature), and both factors of the 72B lie below the 7B's in every bin. Above 10 nats, where the distribution is nearly uniform, the 14B and 32B models instead have the larger network factor (+2.98 and +1.17 against +0.19), and those are the bins of their chaotic temperatures above $\log_{10} T = 0.3$. Below 3 nats the readout factor dominates at 32B and 72B, because the next-token distribution stays sharp up to high temperature (for the 72B, entropy at most 0.15 nats up to $T = 2$). In OLMo-2, P3 fails, and the readout factor tracks the chaos: OLMo-2-13B, with no chaotic temperature, has a lower readout factor than OLMo-2-7B in all three bins between 1 and 10 nats (-1.0 to -1.4 against $+0.06$ to -0.72 per step) although its network factor is higher in two of them, whereas OLMo-2-32B, the most chaotic OLMo model, has a higher readout factor than the 7B in all three (+0.10 to +0.35), positive as in the chaotic 7–8B models. These are descriptive comparisons of trained checkpoints, not interventions on size or post-training: the split is a decomposition along the evolving tangent direction, and equal entropy does not imply equal embedding covariance. Entropy clamping tests at 72B whether forcing an intermediate entropy suffices to destabilize the loop. If its loop were stable only because its distribution jumps from sharp to flat, holding the distribution in between should destabilize it. We therefore choose each step's temperature so that the next-token entropy equals a target and sweep the target instead of the temperature (entropy-clamped decoding; the map stays deterministic, and the entropy holds to 10^{-14}). The temperature then depends on the state, $T = \tau(x)$ with $H(x, \tau(x)) = H_\star$ for the entropy H, so the derivative of the clamped map $\hat{F}(x) = F(x, \tau(x))$ gains a rank-one term, $D\hat{F} = D_x F - \partial_T F \, D_x H / \partial_T H$. We therefore take its exponents from twins of the clamped map itself, temperature solve included, and use no fixed-temperature derivative or gain split for it. The safeguarded solve is only piecewise smooth, but on these orbits one-step central differences at 10^{-5} to 10^{-7} agree with those at 10^{-4} within 6×10^{-6}. Held at any of 16 targets between 1 and 11 nats, Qwen2.5-72B has only periodic orbits, the echo of period 2 on the paper's prompt and periods 1 to 19 on a held-out prompt, with twin exponents between -0.08 and -4.5 that agree within 0.004 across offsets 10^{-5} to 10^{-7}, whereas Qwen2.5-0.5B under the same clamp is chaotic at 6.1 nats. The clamped 72B loop thus stays stable even where its readout is spread. The clamp changes the map, so this is consistent with damping by the network in the original map but does not identify it. The pretrained checkpoint is also mostly non-chaotic. A test frozen before its data (`docs/prefix-scale-72b-base.md`) predicted that the pretrained base model of the same size would be chaotic at three or more temperatures more than Qwen2.5-72B-Instruct on at least three of the four held-out prompts. It fails on all four: the base model is chaotic at 1, 0, 0 and 0 of the 48 temperatures (and at 2 on the paper's prompt; computed after the precision fix, with 500 transient and 1000 measured steps, half the Instruct model's horizon), although its next-token distribution spreads over 2.5 to 5.5 times the temperature range (intermediate-entropy breadth 0.28 to 0.61 decades against 0.11 to 0.22) and from lower temperatures on. At 72B, then, the sharpness that post-training adds does not explain the stability, and post-training is not necessary for it; the failed prediction does not quantify post-training's other effects. The gain split points to the network's damping of the fed-back token (the 72B's network factor lies below the 7B's in every entropy bin), and the breadth that orders the fifteen models fails for this pair. The damping holds when the whole context is kept as well: in Soft Thinking's growing context (Section 11), an early perturbation grows at none of eleven steered cells, six of them chaotic as one-token maps, and at one of 48 unsteered temperatures on two prompts, transiently.

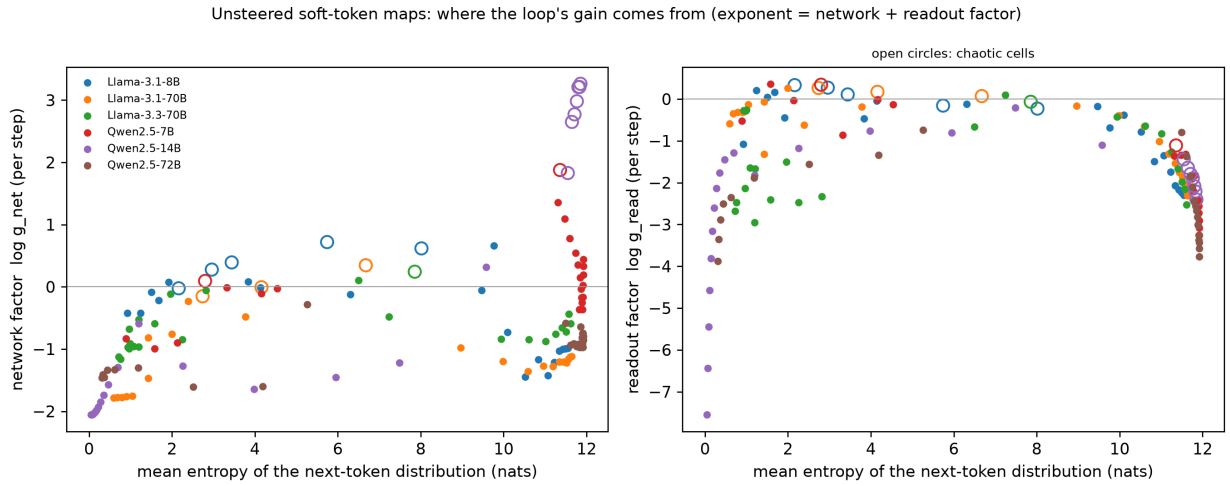


Figure 28: Gain split of the unsteered loop (Section 3) at 24 temperatures per model (500 transient and 1000 measured steps, relative twin offset 10^{-6}): network factor (left) and readout factor (right) against the mean next-token entropy on the orbit; open circles: chaotic temperatures. The two factors add up to the largest exponent up to a boundary term. Medians per entropy bin are given in the text.

Can a cheap property predict it? Motivated by the saturation and high-temperature limits of Section 3, we tested the breadth of intermediate entropy as a heuristic predictor: a model whose next-token distribution jumps from sharp to flat over a small range of temperature might leave little room for chaos. The breadth of this intermediate regime, the decades of $\log_{10} T$ in which the mean entropy on the orbit lies between 1 and 10 nats, ranges from 0.014 (Qwen2.5-72B) to 0.69 (OLMo-2-32B). It orders the fifteen models' chaotic length (Spearman +0.55) somewhat better than their parameter count does (−0.44; Figure 27, right), but it misses the chaos of Qwen2.5-1.5B and -14B at nearly uniform distributions, above its 10-nat bound. This analysis was adapted after the OLMo-2-32B result, and the breadth is measured on the loop's own orbits, which chaos itself spreads, so the test is partly circular. A loop-free version, frozen before it was run (`docs/start-state-predictor.md`), uses one forward pass per model and prompt: the entropy profile of the logits at the loop's start state, and the one-step gain there. Its primary prediction fails: the start-state breadth on the paper's prompt orders the fourteen dense models' chaotic length with Spearman +0.30, below the frozen threshold of 0.5. The secondary criterion, its median over ten prompts, narrowly passes on the frozen outcome (+0.52; $p = 0.06$) and weakens on the expanded sweeps (+0.44), and the one-step gain at the start state is negative in every model, the most chaotic ones included (−0.5 to −3.4 per step), and orders nothing (−0.32). These features ran on two GPUs before the precision fix, which changes them only at about the 10^{-7} level. We have not established a reliable predictor from a single forward pass: as for the period-8 orbit of Section 3, it has to be measured on the loop.

Steering at 70B. Under steering, the 70B maps show chaos, rarely and not independently of the direction (Figure 29; labels at the single offset 10^{-6}, which the plateau test above supports for this computation). On Llama-3.3-70B's hostility plane, at the layer of Goodfire's 70B SAE and at equal norm, 4.3 % of cells are labelled chaotic along the concept vector, 3.5 % along the frozen-rule hostility feature, 1.1 % along a Gaussian direction and 0.2 % along a random live feature, against 28, 29, 39 and 27 % for the same four directions in Llama-3.1-8B (on one GPU); Qwen2.5-72B along hostility gives 0.8 %: eight cells, all at $\log_{10} T = 0.26$–0.30. Recomputed in float64 throughout, six of the eight are chaotic, with exact tangent exponents of 0.23 to 0.36 that the twins at all four offsets

from 10^{-4} to 10^{-7} reproduce within 0.01; five have nearly uniform next-token distributions (11.0 to 11.1 nats) and one alternates between sharp and flat behind a period-2 leading token. The same temperatures without steering are period-2 orbits with exponents between -3.8 and -3.6, and the steered cells above $\log_{10} T = 0.325$ are near-uniform fixed points, so steering drives Qwen2.5-72B into chaos too, in a narrow band. The chaotic cells of the hostility directions lie on a ridge that steering moves from $\log_{10} T \approx 0.3$ to 0.1. A 64×64 zoom of the concept plane at the same offset (Figure 30; $\alpha \in [0.06, 0.12]$, $\log_{10} T \in [0.12, 0.27]$, four bands computed separately with identical vectors and joined) resolves it: a chaotic band, with periodic cells interspersed, whose centre steering moves from $\log_{10} T = 0.25$ at $\alpha = 0.06$ to 0.19 at $\alpha = 0.12$ while its extent grows from about 0.02 to 0.05 decades, and a thinner chaotic layer near $\log_{10} T = 0.15$–0.16 at strong steering; 22 % of the 4096 cells are chaotic (largest exponent 0.59). Above the band, 90 % of the cells are periodic, half of them fixed points or 2-cycles, and below it bands of period-6 windows cross the plane. A further 44 % of all cells, mostly below the band, have neither a detected period up to 32 nor a positive exponent. At 70B the state tolerance of the period test (10^{-8}) lies near the noise floor of the computation, and in a finer unsteered line such cells had repeating leading tokens, so many of them are probably periodic, but we leave them unresolved.

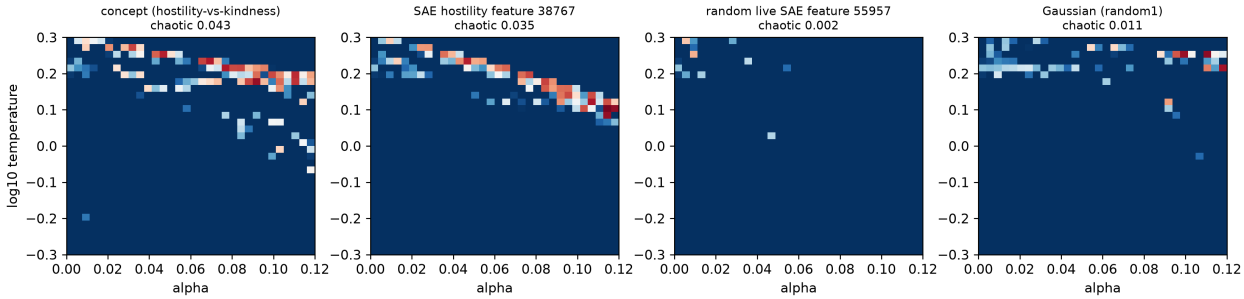


Figure 29: Steering controls at 70B: Llama-3.3-70B-Instruct soft-token maps (32^2 float64 cells, 300 transient and 300 measured steps, maximum period 32) steered after zero-based block 50, the hook point of Goodfire's 70B sparse autoencoder, along the hostility concept vector, the autoencoder's hostility feature selected by a rule frozen before any map, a random live feature and a Gaussian direction, all at the same norm. Chaotic cells: 44, 36, 2 and 11 of 1024 (concept, feature, random feature, Gaussian); cells with neither a period nor a positive exponent: 468, 358, 223 and 424.

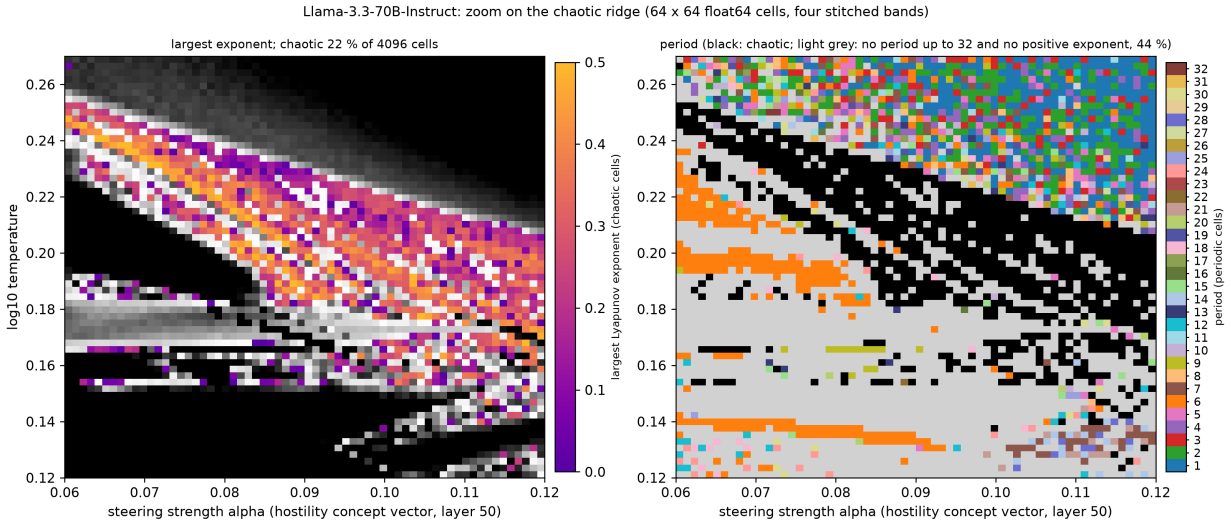


Figure 30: Zoom on the chaotic ridge of Llama-3.3-70B-Instruct steered along the hostility concept vector (after zero-based block 50): 64×64 float64 cells (300 transient and 300 measured steps, maximum period 32), joined from four bands of 16 temperatures computed on separate pods with identical vectors. Left: largest Lyapunov exponent (chaotic cells coloured by the exponent; other cells in grey by how strongly they contract). Right: period of the attractor reached from the fixed start (black: chaotic; light grey: neither a detected period nor a positive exponent).

What this implies for larger models. Size does not set this chaos in a simple way. In Qwen2.5, the 14B model's chaos did not vanish but moved to higher temperature, to where its next-token distribution spreads; in OLMo-2 the largest model was the most chaotic; two 70B Llama checkpoints with different post-training gave different results on the same grid; and at 70B, chaos that passes a three-offset test remains, rarer, with and without steering. The shift does not continue to Qwen2.5-72B. A prompt-blocked test frozen before its data (`docs/prefix-scale-72b.md`) predicted that the 72B model would be chaotic at fewer of the 48 temperatures than the 14B on at least three of four held-out prompts. It is chaotic at 0, 1, 0 and 0 of them, against 14, 9, 11 and 15, so the prediction holds on all four. These lines ran before the precision fix (1000 transient and 2000 measured steps, twins at 10^{-5} and 10^{-6}, chaotic only when both exponents are positive); no temperature on them is positive at only one offset, and the one chaotic temperature ($\log_{10} T = 0.319$, exponents 0.35 and 0.34) is chaotic again when recomputed in float64 throughout, with an exact tangent exponent of 0.34. With the paper's prompt, where it showed no robust chaos up to $\log_{10} T = 1.2$, that makes five prompts on which the 72B model is nearly free of unsteered chaos. Its next-token distribution jumps from sharp to nearly uniform within less than the grid spacing (0.03 decades), so a narrow chaotic band at the jump would be missed by the grid. Fine lines through the jump, computed in float64 throughout with exact tangent exponents, find only periodic orbits on the paper's prompt and on three of the four held-out prompts, and on the fourth a band of chaos about 0.02 decades wide just above the jump ($\log_{10} T = 0.319$–0.336; exponents up to 0.40, which twins at 10^{-5} to 10^{-7} reproduce within 0.004), at nearly uniform distributions, as the 14B model's chaos but far narrower. A fixed temperature window is therefore the wrong yardstick, and so is size alone. A model's chaos follows the gain of its loop where its own next-token distribution is spread, no single forward pass we tested predicts it, and to find it in a larger model one has to run the loop there, across prompts. For frontier models this means two things: their unsteered chaos may be rare, if, like the largest models here, their distributions spread over a narrow range of temperature; and steering still drives

both 70B-class models that we steered, Llama-3.3-70B and Qwen2.5-72B, into chaos.

14 A second system: recurrent depth

Huginn-0125 iterates a recurrent block at fixed input tokens [3]. For a GSM8K-style [81] prompt, the native loop converges to a fixed point whose leading Jacobian pair is $0.588 \pm 0.771i$ (modulus 0.970). For the relaxed loop $h \leftarrow (1-\lambda)h + \lambda F(h)$, whose multipliers are $1 - \lambda + \lambda\mu$, this predicts that the pair crosses the unit circle at $\lambda_c = 2(1 - \mathrm{Re}\,\mu)/|1-\mu|^2 = 1.0777$, transversally and away from the strong resonances (at 0.156 of a revolution per step): a Neimark–Sacker bifurcation. Below it the latent state decays; above it the state settles on a closed curve. A fit of the squared amplitude gives $\lambda_c = 1.0778$, with $A/\sqrt{\lambda - \lambda_c} = 0.52$–$0.54$. A stable circle that grows like the square root of the distance is consistent with a supercritical Neimark–Sacker bifurcation; we did not compute its first Lyapunov coefficient, so its nondegeneracy is not established. Causal attention makes the loop block lower-triangular in token positions. The first position to oscillate is the "." of the prompt, and all later positions follow. On the invariant circle the motion turns by 0.156 of a revolution per step, matching the argument of the critical pair. The rotation number changes smoothly, by about 10^{-4} over $\lambda \in [1.079, 1.092]$, and no return within 60 steps was found at 10^{-3} of the radius. This is consistent with quasi-periodic motion, although locking in narrow high-order tongues cannot be excluded. From $\lambda \approx 1.094$ the circle deforms or a second frequency appears, near where a second complex pair destabilizes the fixed point ($\lambda = 1.0814$). Fixed-point and Jacobian stability of looped models has been studied before [16]; what this adds is the predicted crossing, its amplitude law and the rotation number.

The spectrum predicts the relaxation response. With the full instruction template, the GSM8K prompts' fixed points look different: the leading multipliers surround the origin, a real 0.914, complex pairs of modulus 0.85–0.89 at many angles and negative real values near -0.8, with no near-critical pair. Because the prefill Jacobian is block lower-triangular in positions, its spectrum is the union of per-position blocks, and template positions contribute identical eigenvalues to every prompt. Such a spectrum makes three predictions before any task is scored. First, within the computed spectral estimates of these prompts, no fixed relaxation can speed convergence much. With leading multipliers spread around the origin at many angles, any λ that shrinks one group enlarges another, and the best $\lambda^* = 1.03$–1.06 lowers the rate from 0.914 to only 0.908–0.911. (A finite computed spectrum does not show that no other polynomial scheme could do better; Anderson acceleration, tried below, did not.) Second, under-relaxation slows convergence ($\lambda = 0.55$: predicted 0.953, measured 0.951 on one prompt). Third, over-relaxation destabilizes the negative modes first, at $\lambda_c = 2\,\mathrm{Re}(1-\mu)/|1-\mu|^2 = 1.074$–$1.126$ (five prompts), in a flip or a Neimark–Sacker bifurcation close to -1. The approach to it is visible: on the prompt with $\lambda_c = 1.074$, the measured rate at $\lambda = 1.05$ is 0.955 against a predicted 0.956 (native 0.864). Elsewhere the spectral radius bounds the rate measured over the accessible window from above, by up to 0.05, because the slowest modes carry little amplitude there. Independently, Wang et al. [82] report a Huginn GSM8K drop from 19.41 % to 8.34 % at a constant step of 1.2; at 0.8 GSM8K is 19.79 %, while their four-task mean falls slightly. Their trajectory-based scheduling analysis does not compute the prompt-specific spectral thresholds studied here.

Does the task notice? (development data). On ARC-Challenge [83] questions scored by log-likelihood, where one prefill per answer choice is exactly the analysed recurrence, the spectra again predict convergence. Rates agree to 0.002–0.006 where the slowest mode dominates, and

beyond each question's λ_c (1.075–1.149 for four questions) the residual stops decaying. For 6 of 30 development questions the native fixed point is itself unstable ($\rho(1) = 1.009$–1.273; the fixed point, located with a relaxed transient, has a real multiplier beyond -1 or a complex pair outside the circle). The native loop then oscillates (relative amplitude 2×10^{-2} for one of them), and relaxation to 0.55 or 0.75 restores convergence to 10^{-6}. For the other 24, no unstable mode was detected in the computed spectral estimates up to $\lambda = 1.020$–1.149 (median 1.095); partial Ritz spectra cannot exclude an undetected unstable mode. Question by question, an estimated spectral feature ranks the question's residual. This feature, $\hat{\rho}_q(\lambda)$, is the largest $|1 - \lambda + \lambda\mu|$ over the computed Ritz values of the question's gold-answer prefill, and its Spearman correlation with the residual is 0.73 at $\lambda = 1$ and 0.75 at 1.10 over the 30 questions (development data). In this historical analysis the residual was the all-choice aggregate entering the 32nd recurrence call, so this is a predictive association across related maps and precisions, not a same-state contraction identity. It is consistent with the latent orbits that Geiping et al. [3] observed on some tokens of harder questions, here traced to the fixed point's spectrum. The item-level part of the frozen test below uses matched gold-row token IDs and a per-row residual with an explicit clock.

Accuracy, however, tolerates the instability (300 training questions, 32 steps): it is 0.367 native, 0.387 at $\lambda = 1.10$ and 0.373 at 1.15, where the latent state oscillates with relative size 0.08 and 0.23. Only at 1.25 (size 0.68) does it fall, to 0.300. Yet the oscillation does reach the answers. Reading every step from 24 to 32 on 100 questions, the margin between the correct choice and its best rival moves by 0.11 nats per step natively, 0.37 at 1.10 and 0.95 at 1.15, in alternating directions, and the chosen answer changes within these nine steps for 12 %, 16 % and 33 % of the questions. The changes go both ways, so the accuracy at any one step stays the same on average. Under-relaxation instead costs 4–5 points at 8–16 steps and catches up by 32 (a slower transient), and Anderson acceleration neither speeds the transient nor gains accuracy. Testing the accuracy consequence therefore needs a task in which such step-to-step changes can compound.

GSM8K generation is such a task. With eight chain-of-thought examples in the lm-evaluation-harness format, greedy decoding and 32 recurrence steps per token (development problems 0–119, every arm on the same problems and initial states), accuracy is 0.375 natively, 0.425 at $\lambda = 1.05$ and at 1.10, 0.29 at 1.15 (17 problems lost and 7 gained, one-sided McNemar $p = 0.03$) and 0.16 at 1.20 (29 lost, 3 gained, $p = 10^{-6}$). Under-relaxation to 0.8 costs nothing at 32 steps (0.41). The break lies between 1.10 and 1.15, at or just above the spectral thresholds ($\lambda_c = 1.074$–1.126 for the GSM8K prompts; median upper stability limit 1.095 for the ARC development questions). Generation breaks earlier than log-likelihood scoring, consistent with answer changes that compound over tokens. The collapse at 1.2 is the one Wang et al. [82] reported.

A frozen test on fresh problems confirms this. Its design and analysis were pushed before any test item was read, after two rounds of independent review. On 200 GSM8K test problems, accuracy is 0.330 natively, 0.395 at $\lambda = 1.05$, 0.255 at 1.15 and 0.095 at 1.20. The two pre-stated predictions pass: accuracy declines at 1.20 (-23.5 points; 50 problems lost and 3 gained, exact one-sided McNemar $p = 3 \times 10^{-12}$), and 1.05 loses no more than four points (difference $+6.5$, one-sided 95 % lower bound $+2.5$). At 1.15 accuracy falls by 7.5 points ($p = 0.01$, reported descriptively). Development and test agree closely (Figure 31).

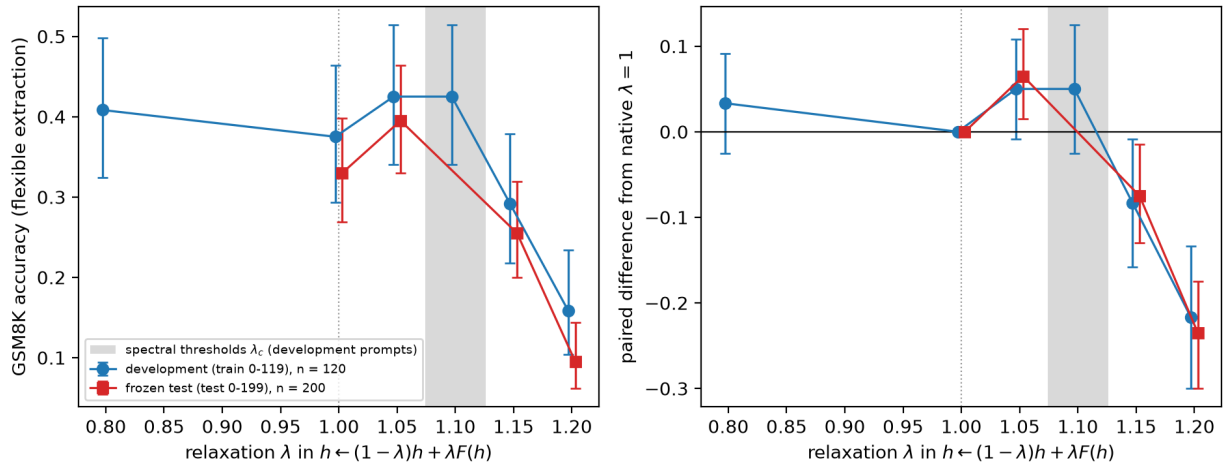


Figure 31: Huginn-0125 chain-of-thought accuracy on GSM8K against the relaxation strength λ (eight examples, greedy decoding, 32 recurrence steps per token): development problems (blue) and the frozen test on held-out problems (red; design and analysis fixed before any test problem was read). Grey: the spectral thresholds $\lambda_c = 1.074$–1.126 computed from the fixed points of five development prompts. Left: accuracy with Wilson 95 % intervals. Right: paired difference from the native loop with paired-bootstrap 95 % intervals.

The same frozen design tested the item-level prediction on 64 held-out ARC-Challenge questions. The first 16 spectra preceded the read-out generation; the remaining ones were completed afterward under a recorded amendment, with the producer attesting that read-out outcomes had not yet been inspected, and duplicate copies were resolved by a rule fixed in advance. Every question passed the admission gates (fixed-point residual at most 2.6×10^{-6}; identical gold-row token IDs in every arm); for 13 questions one of the three Arnoldi sets was partial, which the design admits for an estimated feature. The pre-stated statistic, the mean of the within-arm Spearman correlations between $\hat{\rho}_q(\lambda)$ and the gold row's residual entering the 32nd update, is 0.701 (95 % bootstrap interval $[0.575, 0.780]$; 0.68, 0.73, 0.73 and 0.67 at $\lambda = 1$, 1.05, 1.08 and 1.10); the frozen primary gate, the interval's lower end above 0.3, is met. On unseen questions, the fixed point's estimated spectrum therefore ranks each question by the size of the gold row's residual entering update 32; a residual is not in general a distance to the fixed point. Descriptively, the share of questions with a detected unstable mode rises from 8 % natively to 34, 69 and 84 % at $\lambda = 1.05$, 1.08 and 1.10. An independent implementation reproduced both parts of the frozen test from the saved data.

15 Steering toward hostility

A second steering direction tests whether the organization of the philosophy-steered map is specific to one concept. We steer along a valence direction, the contrast of hostile statements against matched kind ones, built in the manner of contrastive activation addition [72]; its cosine with the gratitude direction is -0.04. Sampled self-dialogue under the same steering is a different system, with a growing context and sampling noise; we report that study separately and follow the direction here in the deterministic map.

The hostility direction in the deterministic map. Steering the soft-token map of Section 2 along the same hostility direction gives a plane with chaos (a positive largest exponent in 10 % of a coarse 128×192 float32 scan) and with periodic text loops that are hostile: "Yes, that's crap"

(period 6), "a lie is" (3), "STUP…" (3), a loop on the Chinese word for insult and its 2-cycle, and others. They appear at α above about 0.035 (the insult loop) and above 0.065 (the rest). Around the period-6 loop the windows form a dome (Figure 43), and toward higher temperature its texts gain one "!Yes" after another (periods $6, 8, \ldots, 20$). In float64 the windows of periods 12 to 20 are born in folds on their low-temperature side and end in flips or +1 crossings on the other, like the stripes of Sections 7 and 8. The period-6 window ends in a fold at $\log_{10} T = -0.0991$, and the continuations of the period-8 and period-10 windows failed to converge.

The windows accumulate on the fold of the 2-cycle "!Yes" at $\log_{10} T_c = -0.0462465357$ ($\alpha = 0.083$; critical multiplier 0.9999999954, next modulus 0.050). From period 16 on, $1/\sqrt{\log_{10} T_c - \log_{10} T_{\text{fold}}}$ rises by 2.235 per lap. The fold's normal form ($a = -1.168$, $b = 44.53$ with ε in $\log_{10} T$) predicts $\sqrt{|ab|}/\pi = 2.296$ per lap for full passages through the bottleneck, within 3 % and without a fitted parameter. An independent float64 forward and automatic-differentiation implementation reproduces this fold (location to 6×10^{-10} relative, a simple real +1 multiplier with next modulus 0.050) and its coefficients (ab to 8 parts per million). In that implementation, controlled passages from fixed entrances at $\log_{10} T$ offsets -10^{-5}, -3×10^{-5} and -10^{-4} took 119/59, 61/31 and 27/13 iterates of F^2 (full channel/from its midpoint), against prospectively frozen quadratic forecasts of 118/61, 61/32 and 27/14. Five of the six are within one iterate; the smallest-offset midpoint missed by two, failing the frozen all-six criterion. Finite-endpoint (arctan) forecasts, also frozen, were within 0.923 iterates. These are controlled local transits. At the window distances the finite-channel correction to the asymptotic passage time is large, so the agreement of the window slope with the full-passage coefficient does not by itself show where the natural dynamics reinjects. The same type-I intermittency mechanism as in the philosophy-steered map is thus consistent with the hostile windows' organization, which the local fold normal form supports; the controlled passages do not identify the natural reinjection or the global organization.

A second region, near $(\alpha, \log_{10} T) = (0.06, 0.20)$, shows arms of period-7 and period-8 windows spiralling around the insult loop. In float64 the insult fixed point doubles into a 2-cycle at $\alpha \approx 0.041$ and 0.079, and that 2-cycle has a Neimark–Sacker bifurcation in the spiral region: at $\alpha = 0.062$ its complex pair crosses the unit circle near $\log_{10} T = 0.1656$, turning 0.12 of a revolution per step of the 2-cycle. This does not explain the arms. The two invariant circles born at a 2-cycle's Neimark–Sacker point are exchanged by the map, so orbits locked on them have even periods, near 16 for this rotation, whereas the arms include period 7; and of four float32 arm cells, only one kept its period in float64. In dissipative flows, windows wind in nested spirals around periodicity hubs [60], and in the Rössler system the hub's shape follows from a conjugacy to unimodal maps [61]; we have not tested whether the arms here wind around such a hub. Float64 zooms show more structure. In one arm a period-7 domain carries period-14 crescents with period-28 cores along its lower edge, between period-1 and period-2 regions and a chaotic band. In the cascade the windows are nested crescent arcs inside a dome bounded by the "!Yes" 2-cycle region, and near $(\alpha, \log_{10} T) = (0.087, -0.069)$ the period-12 and period-14 windows bend into hooked, nested arches. These are the map-level features that accompany shrimp (a periodic body with doubling crescents; hooked windows).

In float64 the period-7 body is bounded by folds. At $\log_{10} T = 0.1925$ the orbit exists between $\alpha = 0.065966$ and 0.071188, and at $\alpha = 0.066$ it ends in a fold at $\log_{10} T = 0.19263$. Below that, the multiplier passes through -1 twice (at 0.1919 and 0.1899), so the period-14 crescent is a doubling bubble. Further down, the branch folds twice (0.18797 and 0.18921, with three coexisting period-7 orbits in between) before a third flip. All crossings are simple (next modulus below 1.5×10^{-3}). An S-shaped pair of folds is what a cusp of the fold curves leaves on a nearby cut. When these boundaries are continued in the plane in float64 (Figure 42), each of the two fold curves reverses its parameter tangent once. There the oriented quadratic coefficient b of the fold's normal form passes

through zero (upper fold: -23.9 to $+244.6$ between adjacent points, with $a \approx 2.0$; lower fold: $+31.2$ to -58.7, with $a \approx 0.55$) while a stays away from zero. Located along the curves, the two cusps lie at $(\alpha, \log_{10} T) = (0.067213, 0.198983)$ and $(0.066611, 0.189382)$. There $b \operatorname{sign} a$ is below 1.3×10^{-5} of its bracket values, $c = -1.41 \times 10^6$ and -5.61×10^5, and $\det J = 58.2$ and 55.2, all evaluated at the located points. The flip curve of the period-14 bubble lies inside the body and closes around the map's period-14 cells. The cubic coefficient is nonzero at the points that bracket both sign changes: $c \approx -1.4 \times 10^6$ to -2.0×10^6 (step agreement 5 %; the smallest steps converge like the square of the step, and extrapolation leaves the sign) and -5.8×10^5 to -5.3×10^5. This supports nondegenerate cusps inside both brackets, and $c < 0$ agrees with the three coexisting period-7 orbits, two of them stable, found between the S-shaped folds. At a fold point near the lower cusp (where $b = 31.4$) the estimated unfolding is nonsingular too ($\det J = 53.5$ in the same coordinates, singular values 262 and 0.20). Its local predictions at six frozen offsets matched the orbits that Newton solves seeded at the predicted roots and the base state recovered: in number at all six and in stability at five. The miss, at the largest offset beyond the edge of the region with three orbits, marks the range of the cubic model. The hostile period-7 body thus has the same skeleton as the period-6 window of Section 9.

Its interior is shrimp-like too. At $\log_{10} T = 0.1830$, float64 bifurcation diagrams through the body (Figure 41; 1400 cells per line) show the period-7 orbit doubling to periods 14 and 28 and entering chaos (largest exponent up to $+0.1$) for $\alpha \approx 0.0659$–0.0673. Windows of periods 28, 35 and 42 sit inside the chaotic band, and the orbit then undoubles through 28 and 14 back to 7; the exponent touches zero at every doubling. Newton's method converges at map cells to primitive orbits of periods 14, 28 and 21 (residuals 3×10^{-15}, 2×10^{-14} and 5×10^{-13}; spectral radii 0.08, 4×10^{-4} and 0.006). Periods 21, 35 and 42 are three, five and six times the body's period: the satellite windows expected in the chaotic bands of a period-7 cascade. Continued in temperature, the period-14 orbit flips into period 28 near $\log_{10} T = 0.1841$ and back near 0.1857, the two ends of a doubling bubble that holds period 28 and chaos. Lower down, its branch is S-shaped (folds near 0.1824 and 0.18244, with three coexisting period-14 orbits between them) before another flip near 0.1819. All these crossings are simple (next modulus below 4×10^{-5}). A 160^2 float64 zoom of the lower body (Figure 40) shows this anatomy in the plane. The period-14 domain encloses a period-28 ring around a chaotic core. Period-21 windows sit in the chaotic region as thin crescents with legs, and curves of very negative exponent, where the orbits are superstable (the leading multiplier crosses zero), run through the windows as the "endoskeleton" of shrimps does [59]. In this zoom 18 % of cells are chaotic.

No one-dimensional shrimp centre. In a shrimp of a one-dimensional map, two superstable curves of the window's orbit cross at a doubly superstable centre [23, 59]. We continued one period-7 orbit over a 31×31 grid of the body's upper part and followed the sign of its leading multiplier (Figure 35). Its superstable curves form two arcs that approach within 1.3×10^{-3} in $\log_{10} T$ near $(\alpha, \log_{10} T) = (0.0700, 0.206)$ and do not cross at this resolution. A five times finer grid of that approach (1155 cells) shows why. There the leading multiplier is tiny (median modulus 9×10^{-4}) and as small as the next one, and in 221 cells the two form a complex pair, in two wedges that flank the gap. On this sampled approach the two computed leading multipliers exchange dominance and form a complex pair, so the sign of a leading real multiplier no longer defines a scalar superstable curve there. Where a one-dimensional map would place its doubly superstable centre, the orbit's local linearization has two comparable directions. Next to the body, the 7-step return maps of chaotic cells are single-humped in every phase, with doubled strands in two phases, the trace of a thin second dimension (Figure 36); cobwebs on these maps show the doubling cascade and the

chaos as on a one-humped interval map (Figure 37). On the tracked branch and in the sampled region, the body is therefore not the shrimp of a one-dimensional map. Whether it is the shrimp of an effectively two-dimensional map, as in the Hénon family, is open.

What the shrimp-like shapes are. The shapes are made of the bifurcation curves of single windows (Figure 34). A periodic window is the region where one periodic orbit attracts. The windows we continued are bounded by fold curves, where the orbit is born or dies together with a saddle (multiplier $+1$), and flip curves, where it doubles (multiplier -1); in higher dimensions other boundaries, such as a complex pair crossing the unit circle, are also possible. In a one-dimensional map the multiplier runs from $+1$ through 0 to -1 across the window, so every window holds a superstable curve: the dark "endoskeleton" lines of our maps. In a higher-dimensional map the leading multiplier can instead change sign through a complex pair, as it does in parts of our windows. Beyond some of the flip curves we followed lie the doubling sequences of Section 10; a flip need not start a complete cascade.

Steering strength and temperature move the orbit in different directions, so the fold curves bend. Where a fold curve turns back, a cusp can form (a visual turn alone is not our criterion: the located cusps satisfy the degeneracy and unfolding checks of Section 9), and between the two branches leaving the cusp the window continues as a thin leg. Bent fold curves with cusps, flip curves inside them and a superstable line through the body produce the bodies, legs, hooks and crescents of the maps. On the seven fold curves we continued, all nine tangent reversals are located cusps (Section 9); this is not a global count of cusps.

A shrimp in the sense of Gallas has one more element: a centre, where two superstable curves of the same orbit cross. There the orbit passes two critical points at once, which needs two turning points in a one-dimensional reduction. The return maps of the hostile body have one turning point per phase (Figure 36). Where the hostile body's superstable curves approach each other, the orbit becomes two-dimensional instead of doubly superstable (Figure 35). The period-6 hook's orbit never comes close to superstable: its leading multiplier stays above 0.028 in modulus.

The X-shaped crossing of a period-5 and a period-6 strip (Figure 9, bottom) is not one window's anatomy but two windows overlapping, and at the crossing both of their orbits are unstable (Section 9). The shapes in these one-block planes are therefore the two-parameter unfolding of single-humped windows. They are shrimp-like in body, legs and internal cascade, but have no shrimp centre.

Two turning points from two steering depths. Steering at two depths gives the band dynamics a second turning point. We drew 1584 chaotic cells (200 from each of the four two-block planes of Figure 12, 150 from each one-block map of Figure 11 and the 34 of the hostile line of Figure 41), counted the turning points of each band's return map and inspected every strong candidate by eye, since a projection can fold a single hump. Three cells, all in two-block planes at $\log_{10} T = -0.1$, have smooth, single-valued return maps with two or more turning points (Figure 32); the other strong candidates were point sets, split bands or folded projections, or were not reproduced. In one cell (hostility at block 4, gratitude at block 16), every one-dimensional phase of a 20-band attractor is N-shaped: a bimodal map. A 128×128 zoom around another, whose map has three maxima, shows a period-5 window with a body, legs, a period-10 and period-20 border, and patches where the fixed start settles on period 4 (Figure 33a). We continued the window's period-5 orbit over a 31×31 grid of its body: 608 cells solved, all with the same token loop, 447 of them attracting. In 61 of these 447 cells, the nearest cell of the zoom (at most 2.0×10^{-5} away in α_1 and 3.2×10^{-5} in α_2) settles on period 4, so attractors of periods 4 and 5 coexist at grid scale, which a single unimodal map

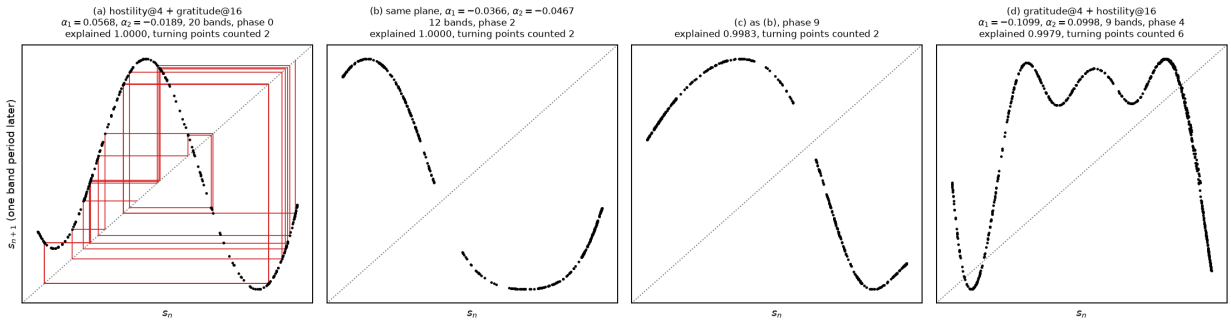


Figure 32: Band return maps with two or more turning points, from two-block planes of Qwen2.5-0.5B-Instruct (float64, $\log_{10} T = -0.1$). Each panel shows one phase of a chaotic band attractor, projected on its leading principal component s (explained variance in the title), with points (s_n, s_{n+1}) one band period apart. (a) Hostility at block 4 and gratitude at block 16, $(\alpha_1, \alpha_2) = (0.0568, -0.0189)$: phase 0 of a 20-band attractor, N-shaped, with a cobweb of consecutive visits. (b, c) The same plane at $(-0.0366, -0.0467)$: two phases of a 12-band attractor, N-shaped maps on two intervals. (d) Gratitude at block 4 and hostility at block 16, $(-0.1099, 0.0998)$: phase 4 of a 9-band attractor, with three maxima.

with negative Schwarzian derivative allows only if a basin reaches an end of its interval (Section 3).

The window still has no one-dimensional centre. The leading multiplier of the continued orbit never comes near zero: its modulus is at least 0.0139 in all 608 cells. Where a one-dimensional map would be superstable, it meets the second multiplier, and the two form a complex pair (91 cells, in narrow strips) or, where their product is negative (41 cells), pass each other as real numbers of opposite sign (Figure 33c). Per step, the second direction contracts by a median factor of 0.44 and the first by 0.76 over the attracting cells, so near these strips the orbit is two-dimensional. The curves on which the trace $\mu_1 + \mu_2$ changes sign take the place of superstable curves, and they do not cross: no 2×2 block of attracting cells has the alternating signs of a crossing (Figure 33b). One isolated cell holds a repelling orbit with the same token loop ($\mu_1 = 1.65$), and near the top of the grid one column of Newton solves failed above cells whose leading multiplier rises to 0.88; these are signs of folds inside the window that this spacing does not resolve. A second turning point in the band maps therefore does not give this window a one-dimensional shrimp centre. Like the hostile body, it is the window of a map that is two-dimensional near its superstable strips, and whether it is the shrimp of a two-dimensional map is open.

Chaos behind a repeating text. Along the temperature line of Figure 41 (left), a longer test (2000 measured steps after 1000 transient steps, twins at offsets 10^{-6} and 10^{-9}, whose exponents agree to within 0.0031 in every chaotic cell) finds 182 chaotic cells among 1400 between $\log_{10} T = 0.176$ and 0.191. In 65 of them the leading token repeats exactly for all 2000 steps while the state moves chaotically; in 59 of these the period of the text is a multiple of 7, most often 7 or 28. Figure 38 shows the strongest case ($\log_{10} T = 0.18995$, largest exponent 0.091 at both offsets). Its 2000 sampled states fall into seven separated groups, one per phase of the text, visited in a fixed order and at least 0.098 of the state norm apart (a property of the samples, not a proof of separated invariant sets), and each group lies inside one cell of the partition by leading token, so the text `I!Yes heels`↩↩`pee`↩↩ repeats at every one of the 2000 steps we followed. Inside a group the sampled states lie close to a curve (a strongly anisotropic cloud). Read once per repetition through the probability of the last token, the sampled return relation is well approximated by a smooth map with a single quadratic maximum, the shape of the logistic map: a polynomial of degree 8 fits it to 4×10^{-4} of its range, and the fitted map's Lyapunov exponent, 0.647 per repetition, is within 2 %

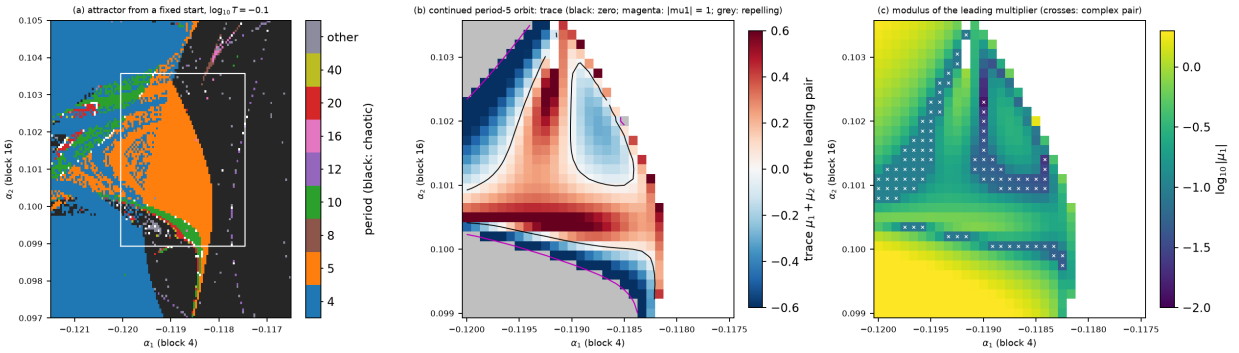


Figure 33: A period-5 window in the plane of gratitude at block 4 and hostility at block 16 (Qwen2.5-0.5B-Instruct, float64, $\log_{10} T = -0.1$). (a) Attractor reached from the fixed start on a 128×128 grid (black: chaotic), with the box of the continuation grid. (b) The window's period-5 orbit continued over the box (31×31, 608 cells solved, one token loop): trace $\mu_1 + \mu_2$ of its two leading multipliers on attracting cells, its zero curves (black) and the curves $|\mu_1| = 1$ (magenta); grey cells hold a repelling orbit. (c) Modulus of the leading multiplier; crosses mark a complex leading pair. The modulus never falls below 0.0139; the dark strips, where a one-dimensional map would be superstable, are where the two leading multipliers meet.

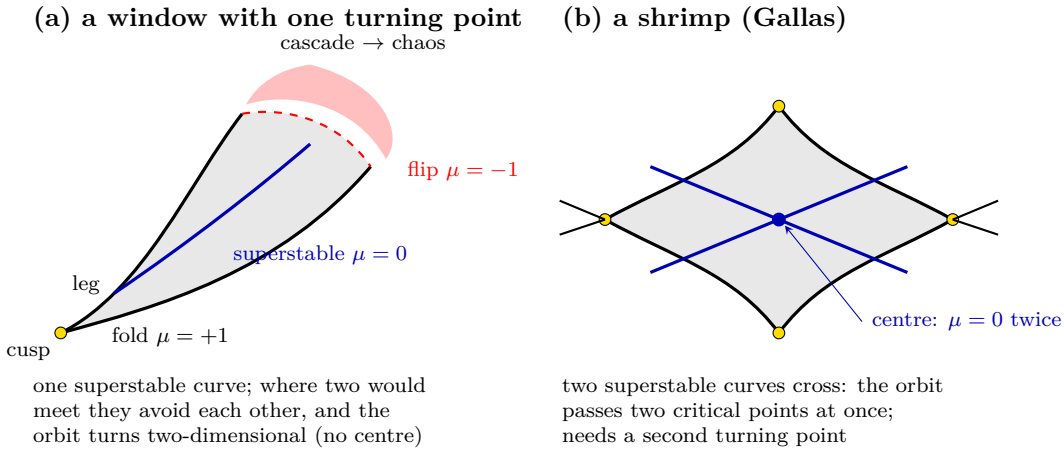


Figure 34: What the shrimp-like shapes are. (a) A periodic window of a two-parameter family whose reduced dynamics has one turning point: fold curves (black; multiplier $+1$) bound the body and meet in a cusp (gold), from which the window runs on as a thin leg; a flip curve (red, dashed; multiplier -1) opens a doubling cascade into chaos; one superstable curve (blue; multiplier 0) runs through the body. This is the anatomy we find, down to the located cusps and the cascades. (b) A shrimp in the sense of Gallas has in addition a centre where two superstable curves of the same orbit cross, which requires a second turning point. In the hostile period-7 body the superstable curves approach and turn away instead, and at their closest approach the orbit becomes two-dimensional (Figure 35).

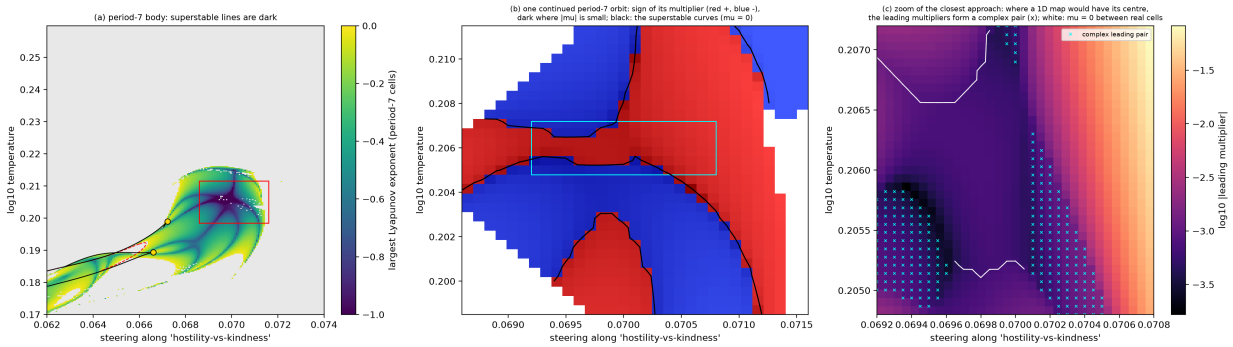


Figure 35: Superstable structure of the hostile period-7 body (Qwen2.5-0.5B-Instruct, hostility direction at layer 12, float64). (a) Largest Lyapunov exponent of the period-7 cells, with fold (black) and flip (red, dashed) curves and the two located cusps (gold); dark curves are superstable lines. (b) One period-7 orbit continued over the red box of (a): sign of its leading multiplier (red positive, blue negative), dark where it is small, and its zero curves (black). The two arcs approach near $(0.0700, 0.206)$ without crossing. (c) Five times finer zoom of the cyan box of (b): logarithm of the leading multiplier's modulus. Crosses mark cells where the two leading multipliers form a complex pair; white segments mark sign changes between cells with a real leading multiplier. Where a one-dimensional map would have its centre, the orbit is two-dimensional.

of the full map's 0.636 per seven steps, measured with twin trajectories in all 896 dimensions. The scalar approximation holds across most of the cascade. Rerun with the leading-token sequences saved, 62 of the 65 cells are again chaotic behind a repeating text, and in every one of them several phases of the text (a median of ten) give single-valued return maps; in 59 cells the best-fitting phase has a single interior stationary point (a maximum in 36, a minimum in 23). If the dynamics were exactly one-dimensional and each phase's probability a smooth invertible coordinate on it, all single-valued phases would give the same asymptotic exponent, since exponents do not change under such a change of coordinates. We therefore fit each phase's map by a polynomial whose degree is chosen by cross-validation over random folds of the return-map pairs, and take the median over the phases. It matches the full map's exponent within 10 % in 51 of the 62 cells (median discrepancy 3.7 %; within 10 % in 23 of the 24 cells with seven-token text, which give 285 repetitions in 2000 steps; Figure 39). This analysis was adopted after we inspected an outlier of our first, pre-specified analysis (a fixed degree of 8 at the single best-fitting phase), which gave 49 of 62 and underfitted one cell whose maps are steep and have two turning points; we report both. An exploratory temporal split (fit on the first 60 % of each record, test on the last 30 %) gives small value-prediction errors at the best scalar phase, chosen on the test segment itself, but in 30 of 61 cells two- or three-delay predictors improve the held-out error by 20 % or more, which the pre-stated rule counts against scalar sufficiency. These results support an empirical scalar description, not a proved one-dimensional reduction: a low-residual fit does not show that the observable is invertible, and even exact agreement of exponents would not establish a one-dimensional invariant curve (two uncoupled logistic maps observed through one coordinate agree in the same way). An independent implementation reproduced the strongest cell (the same leading token at all 2000 steps; exponent 0.089 against our 0.091; scalar discrepancy +2.3 % under the first rule and +1.5 % under the amended one) and both cross-cell counts (49 and 51 of 62). At two further cells it chose, it found good agreement after the amendment in one (text period 10, +0.2 %, although its fitted maps have both a maximum and a minimum) and a persistent discrepancy of about 18 % in the other (period 28), despite a small value error. These are the chaotic bands that follow a period-doubling cascade, and a smooth unimodal map is the setting of Feigenbaum's universality [22]; a fitted maximum on

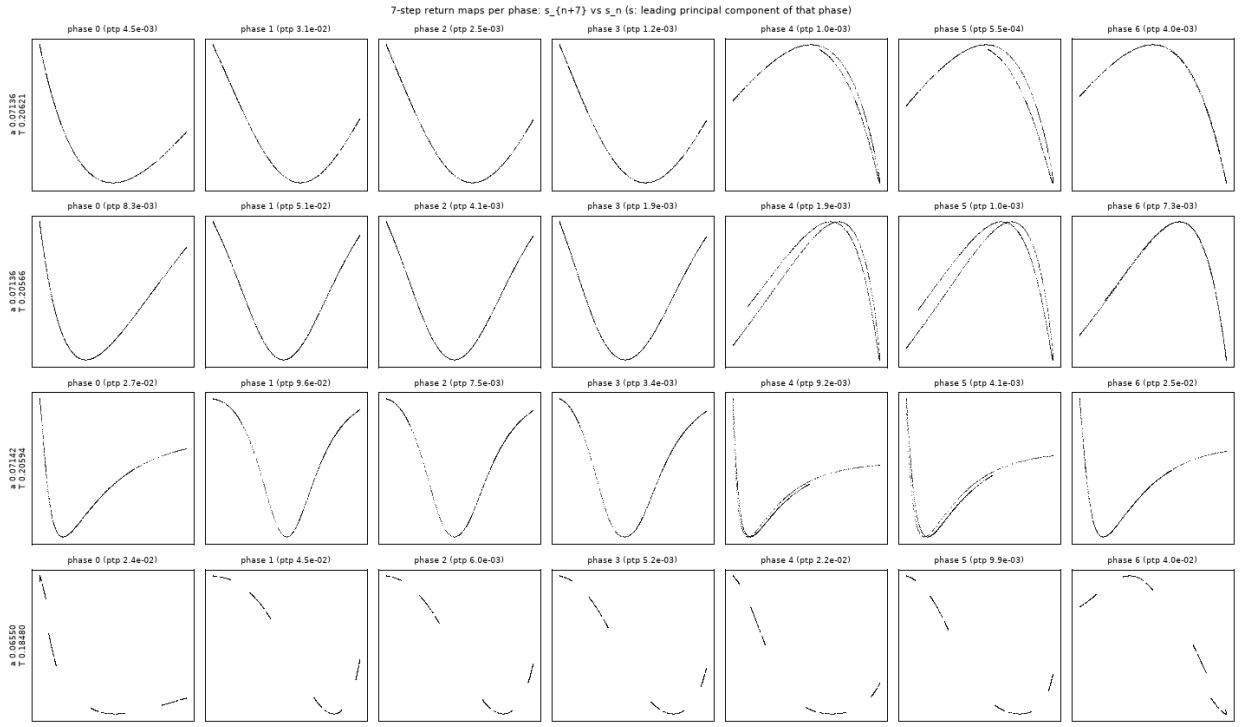


Figure 36: Seven-step return maps s_{n+7} against s_n, for each phase of the 7-periodic text, of chaotic cells next to the hostile period-7 body (top three rows: $\alpha = 0.07136$–0.07142, $\log_{10} T = 0.2057$–0.2062) and on the hostile line of Figure 41 (bottom row: $\alpha = 0.0655$, $\log_{10} T = 0.1848$); s is the projection on the leading principal component of that phase's states, which carries 89–100 % of their variance. Every phase shows a single hump. In phases 4 and 5 the curve splits into two thin strands.

the sampled range does not by itself establish that theorem's hypotheses, and we have shown the scalar approximation for this cascade only. A reader of the text sees a loop. Only the probabilities, or the state, reveal the chaos.

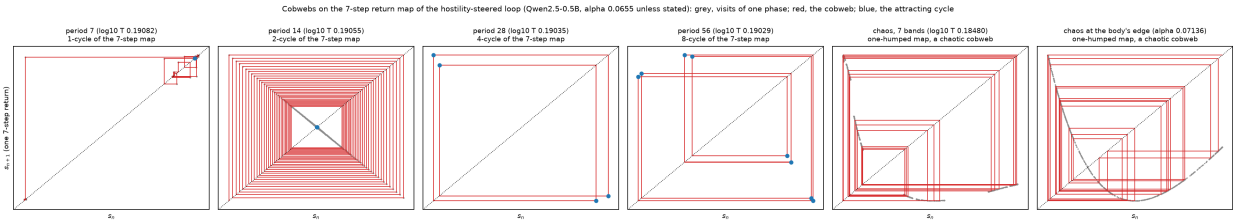


Figure 37: Cobweb diagrams of the 7-step return map of the hostility-steered loop (Qwen2.5-0.5B-Instruct, layer 12, float64). s is the projection of one phase's states on their leading principal component. Left four panels: the doubling cascade at the right edge of the hidden-chaos window along $\alpha = 0.0655$. As $\log_{10} T$ falls from 0.19082 to 0.19029, the attracting set of the 7-step map (blue) is a fixed point (period 7 of the full loop), then a 2-, 4- and 8-cycle (periods 14, 28 and 56); the red cobweb shows the approach from the fixed start (for period 7, its first visits). Right two panels: chaotic cells of Figure 36, where the visits trace a one-humped map (grey) and the cobweb wanders over it.

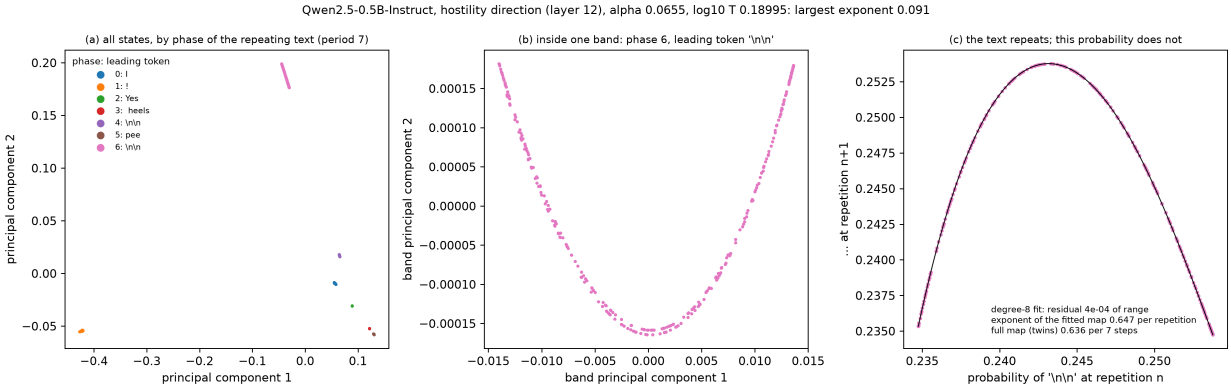


Figure 38: Chaos behind a repeating text: the soft-token map of Qwen2.5-0.5B-Instruct steered along hostility at layer 12 ($\alpha = 0.0655$, $\log_{10} T = 0.18995$; float64, 2000 steps after 1000 transient steps). (a) All states on their first two principal components, coloured by the phase of the seven-token text that the leading token repeats: seven separated sampled clouds. (b) The widest cloud (phase 6, leading token a double line break) on its own principal components: the sampled states lie close to a curve. (c) The probability of that token at successive repetitions of the text, with a fitted return curve with one quadratic maximum (black: polynomial of degree 8) whose Lyapunov exponent is within 2 % of the full map's.

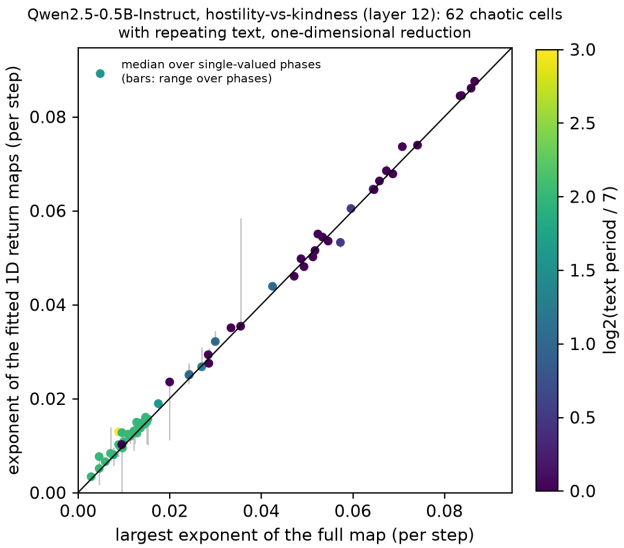


Figure 39: Scalar return-map approximation across the hostile cascade (Qwen2.5-0.5B-Instruct, hostility at layer 12, $\alpha = 0.0655$, $\log_{10} T \in [0.176, 0.191]$): for each of the 62 chaotic cells whose leading-token text repeats exactly, the Lyapunov exponent of the fitted return maps of the leading-token probability (per step; median over the single-valued phases of the text, bars: their range; polynomial degree chosen by cross-validation) against the largest exponent of the full map from twin trajectories. Colour: text period relative to 7. The line is equality. With seven-token texts the 2000 steps give 285 repetitions; with 28-token texts only 71 or 72 observations per phase (70 or 71 return pairs), and the fitted maps then overestimate small exponents.

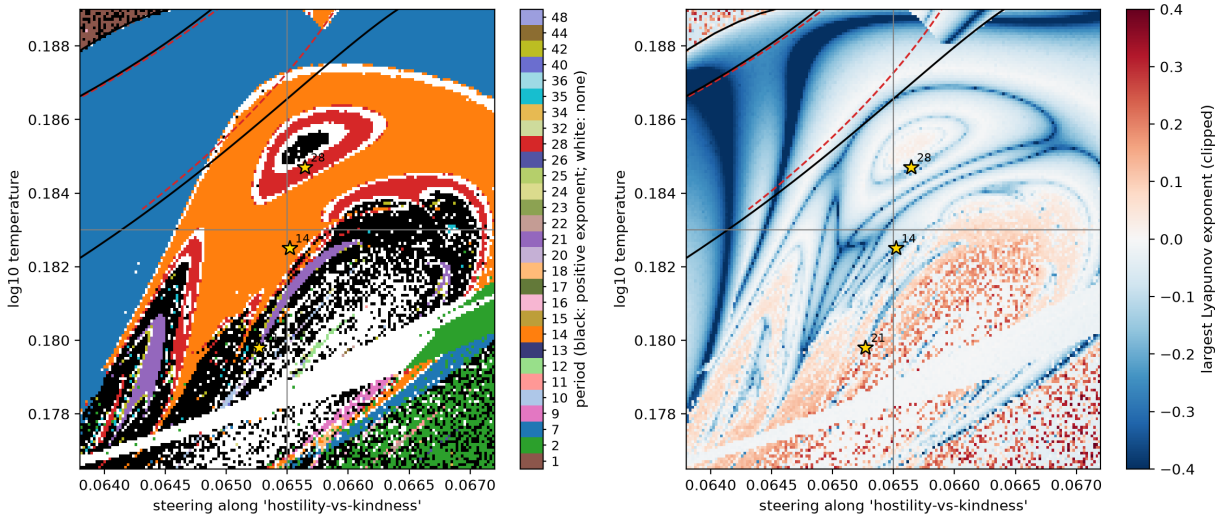


Figure 40: The lower part of the period-7 body of the hostility-steered Qwen2.5-0.5B-Instruct map (layer 12) at higher resolution (160^2 float64 cells). Left: period of the attractor (black: positive largest exponent; white: no period up to 48). Right: largest Lyapunov exponent (clipped); the darkest curves are superstable lines. Both panels: float64 fold (black) and flip (red, dashed) curves of the period-7 orbit, the two lines of Figure 41 (grey) and the Newton-verified orbits of periods 14, 28 and 21 (stars).

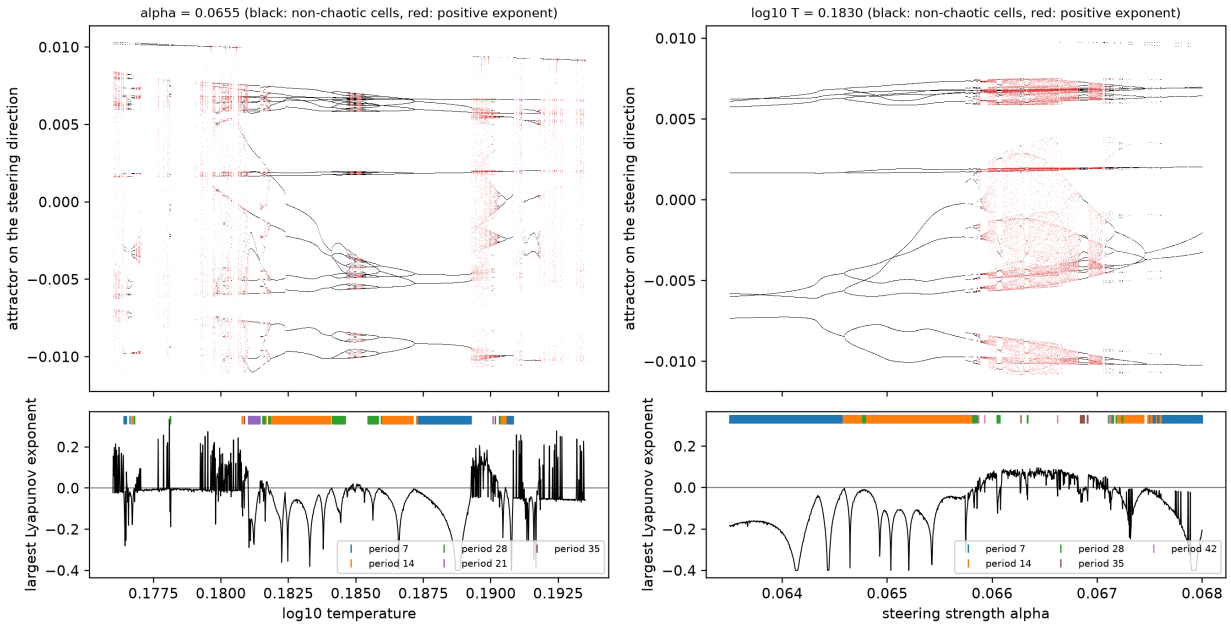


Figure 41: Inside the hostile period-7 body (Qwen2.5-0.5B-Instruct, hostility direction at layer 12): float64 bifurcation diagrams along temperature at $\alpha = 0.0655$ (left) and along the steering strength at $\log_{10} T = 0.1830$ (right). Top: the last 160 states of the attractor reached from the fixed start, projected on the steering direction (red: cells with a positive largest exponent). Bottom: largest Lyapunov exponent, with the cells of periods 7, 14, 21, 28, 35 and 42 marked (cells of other periods are not marked). On the right, period 7 doubles to 14 and 28, enters chaos with windows of periods 28, 35 and 42, and undoubles back to 7. Labels come from 400 measured steps after 1000 transient steps; with 2000 measured steps at two twin offsets, 135 of the 235 red cells on the left stay chaotic and 91 are periodic (Section 17, Limitations).

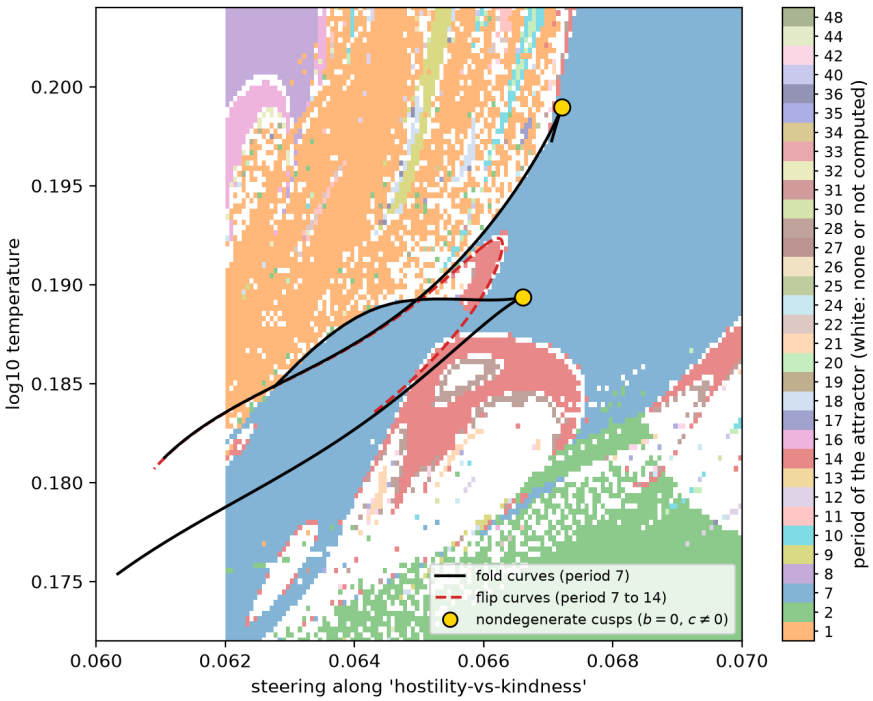


Figure 42: The period-7 body of the spiral region (Qwen2.5-0.5B-Instruct, hostility direction at layer 12): attractor periods of a float64 zoom (white at $\alpha < 0.062$: outside the zoom), the float64 fold curves of the period-7 orbit (black) and the flip curves into period 14 (red, dashed), with the two located cusps (gold).

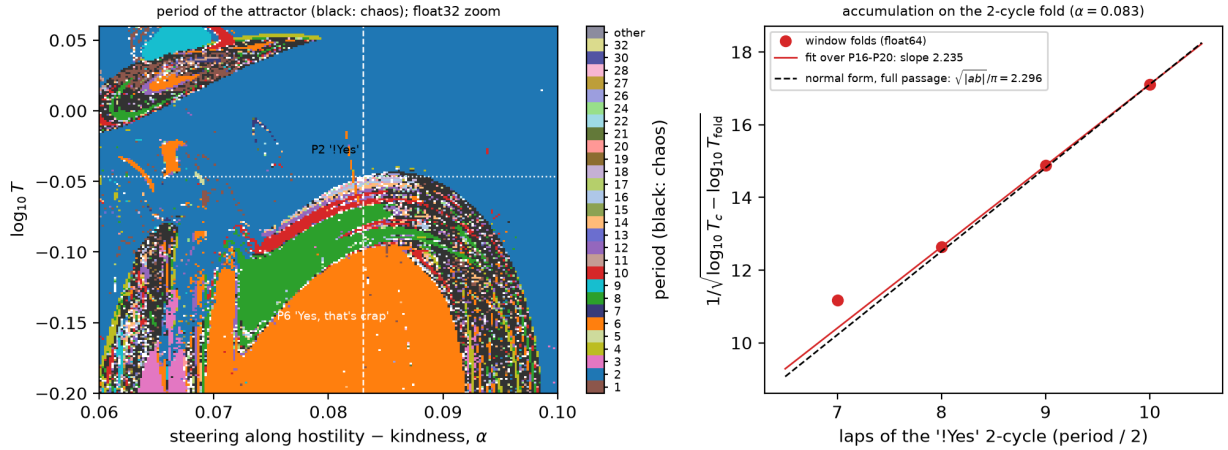


Figure 43: The hostility-steered soft-token map of Qwen2.5-0.5B-Instruct (layer 12). Left: period of the attractor in a float32 zoom (black: chaos), with the period-6 "Yes, that's crap" dome and its period-adding bands; dashed, the cut at $\alpha = 0.083$; dotted, the float64 fold of the "!Yes" 2-cycle. Right: float64 fold edges of the windows (periods 14–20) against the number of "!Yes" laps; the dashed line is the normal-form prediction for full passages, anchored at the last window.

16 Two loops in conversation

Two instances of a language model left to talk with each other drift into recurring states [12]. The best-known case is the "spiritual bliss" attractor of two Claude Opus 4 instances, whose open-ended conversations end in repeated exchanges of gratitude and silence [84]. The soft-token map poses the deterministic version of the question: can two loops that are each chaotic on their own lock each other into a periodic orbit? Agents A and B are Qwen2.5-0.5B-Instruct with the prompt of the self-dialogue pilots, each with its own steering strength along the philosophy direction at layer 12. They write one reply together, alternating soft tokens, $x_{n+1} = F_A(x_n)$ and $x_{n+2} = F_B(x_{n+1})$, so one exchange is the map $G = F_B \circ F_A$. Equal agents reproduce the single loop bitwise in float64, so on a map over (α_A, α_B) the diagonal holds each agent writing alone (Figure 44).

At $\log_{10} T = -0.10$, over $\alpha_A, \alpha_B \in [0.060, 0.082]$ (48×48 cells), 27 of the 48 agents are chaotic alone and 21 periodic. Of the 702 pairs whose two agents are both chaotic alone, 422 are periodic together; of the 420 pairs of two periodic agents, 91 are chaotic together. Writing together favours order (70 % of pairs are periodic, against 44 % of single agents), though some chaos is inherited: pairs of chaotic agents are periodic less often than pairs in general. The exponent map is plaid, as if each agent contributed its own stability, and where one agent is weakly steered ($\alpha \lesssim 0.064$) and the other is not, the pair settles into a fixed two-token exchange. Twelve entrained pairs spread over the map all converge under Newton's method to attracting primitive orbits of G (residuals at most 2×10^{-13}, leading multipliers 0.0008 to 0.79 in modulus, exchange periods 1, 3, 5 and 6). Their texts show how two chaotic writers share a loop. At period 1, A writes "is" and B "it", the single loop's 2-cycle split between two writers. At period 5, B answers "is" every time while A cycles through "This", "there", "it" and "a". That alternating two chaotic maps can produce order is known for quadratic maps [85]; here it happens in a language model's own dynamics.

Two different models behave alike. Qwen2.5-0.5B and 1.5B share a tokenizer, so a message can be passed as a distribution over the shared vocabulary and embedded by each model with its own matrix. Unsteered, the 1.5B writing with itself is chaotic at 41 of 131 temperatures ($\log_{10} T$ from 0.29 to 0.80, exponents 0.21 to 1.76 per exchange), the 0.5B at one. Writing together they are periodic at 39 of those 41 temperatures, mostly in a fixed two-token exchange: the stable partner sets the rhythm. A dialogue framing behaves differently. With each one-token message placed in a user turn and the first token of the reply as the answer, every map we computed is periodic, for one agent alone and for every pair. The reply is decided by the replier's own prompt and steering, and the partner's message barely moves it; mutual locking of the kind seen in long conversations would need longer messages than this frame provides.

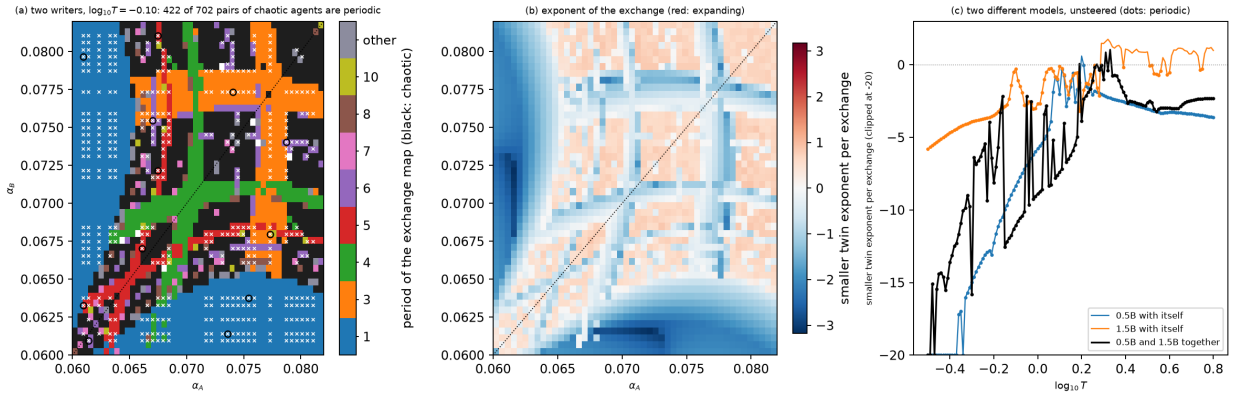


Figure 44: Two loops writing one text. (a) Agents A and B (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12, strengths α_A and α_B) alternate soft tokens at $\log_{10} T = -0.10$: period of the exchange map $G = F_B \circ F_A$ per cell (black: chaotic). The dotted diagonal holds each agent alone; white crosses mark pairs that are periodic although both agents alone are chaotic, and circles the twelve such pairs verified by Newton's method. (b) The smaller twin exponent of G per exchange on the same cells. (c) Two different models, unsteered: the smaller twin exponent per exchange of Qwen2.5-0.5B and of Qwen2.5-1.5B each writing with itself, and of the two writing together; dots mark periodic cells.

17 Discussion

What theory allows, and what is new. Chaos and periodic windows occur in many nonlinear feedback systems. In a specified ensemble of random recurrent networks with asymmetric couplings, chaos sets in above a critical gain, sharply in the large-network limit, and finite networks pass through limit cycles of increasing complexity on the way [20]; how the transition looks depends on the architecture and the transfer function [86]. Discrete-time random networks, $x(t+1) = \tanh(gJx(t))$, iterate a map as our loop does: their fixed point loses stability when the gain times the spectral radius of the couplings exceeds one, the first bifurcation of large networks is a Neimark–Sacker bifurcation, and chaos follows by a quasi-periodic route, through a torus and frequency locking, over a range of gain that narrows as the network grows [21]. Our largest model narrows differently: Qwen2.5-72B passes from a confident echo to a nearly uniform fixed point within 0.03 decades of temperature, and its chaos is confined to narrow bands, whereas in random networks chaos is generic above the transition. External input can suppress that chaos [87], and deep networks at initialization show an order-to-chaos transition in signal propagation [88, 89]. In our map, temperature plays the role of an inverse gain and the steering vector that of an input. Section 3 makes the correspondence concrete through an exact chain-rule factorization, with the softmax derivative $(\operatorname{diag} p - pp^\top)/T$ in the place of the slope of the transfer function; the identification with the random-network transition remains an analogy, not a transfer of that ensemble's theorem. Feigenbaum scaling holds for a class of one-dimensional maps with a quadratic maximum [22], and periodic windows, shrimp-shaped domains with their cusps and spirals, and doubling cascades are found in many dissipative two-parameter systems [23, 59, 60, 90] and in experiments [91]; intermittency near folds is classical [58]. These results motivate looking for chaos, windows and shrimp-like structure in a language model's feedback map, but they do not imply it. They do not say whether a trained language model realizes it, where in its parameter plane, with which texts, or whether the theory's quantitative predictions hold for that model. This paper shows that the models we studied do realize it, with concrete texts as attractors, so that a bifurcation changes what the model writes, and that several prospective quantitative predictions succeeded; the failed all-row criteria and the unresolved cases are reported

with them.

As far as we know, no earlier work has computed periodic orbits and their multipliers, continued fold and flip bifurcation curves and located cusps in the feedback map of a released pretrained language model. Nor do we know of earlier quantitative predictions from such a model's derivatives, for these objects, that were confirmed prospectively, as ours were for passage times at new folds (23 of 24 forecasts within one iterate, with the all-row criterion failing by one row) and for the orbits recovered and their stabilities at frozen offsets near numerically located cusps; derivative-based normal-form predictions in a task-trained looped transformer precede ours [19]. A frozen spectral prediction of the response to relaxation is confirmed on held-out data in Huginn, and one periodic orbit of the full transformer map of Qwen2.5-0.5B-Instruct is proved by interval arithmetic; we know of no earlier instance of either for a language model.

Is steering the origin of the chaos? Steering adds a fixed vector to the residual stream, so chaos along a steering axis could belong to the push rather than to the model. At $\alpha = 0$, however, the map is the unsteered model's own feedback map, with temperature as its only parameter. The $\alpha = 0$ columns of the hostility maps contain cells with positive finite-horizon exponents in Llama-3.1 8B (26 of 64 temperatures), Qwen2.5-7B (15 of 64) and OLMo-2 7B (11 of 64), while the near-zero columns of the 0.5B and 1.5B maps contain one such cell each (of 128 and 64). A longer float64 sweep of the unsteered 0.5B map (512 temperatures over $\log_{10} T \in [-1, 0.3]$, 1000 transient and 2000 measured steps) finds a narrow chaotic band at $\log_{10} T \approx 0.15$–0.21 (7 temperatures, exponents up to 0.28), with a period-5 loop at 437 of the 512 temperatures. The same long sweep finds 96 chaotic temperatures in Llama-3.1 8B, between $\log_{10} T = -0.20$ and 0.24 (exponents up to 0.66), interleaved with periodic windows of periods 2 to 52; within $\log_{10} T \in [-0.3, 0.3]$ its chaotic share, 41 %, equals that of the coarse column. Qwen2.5-7B has 65 chaotic temperatures between $\log_{10} T = -0.07$ and 0.28 (exponents up to 0.74), and OLMo-2 7B has 32 between 0.04 and 0.14 (up to 0.30). Steering is therefore not needed to reach chaotic regimes in any of these four models; where it does create chaos, it does so mainly through the readout, by making next tokens compete (Section 3). Temperature alone produces chaos with embedded windows, broadly in the 7B and 8B models and in a narrow band in the 0.5B model (Figure 45). Across the fifteen models of Section 13, the regular-grid sweeps find unsteered chaos in eleven of the fourteen dense models (in Llama-3.1-70B at two temperatures that pass a three-offset test), but not in Qwen2.5-72B, Llama-3.3-70B, OLMo-2-13B or the mixture of experts; finer lines found candidates in the first two, and the Qwen2.5-72B window, recomputed in float64 throughout, is periodic.

Steering does widen the chaotic regions, but not as a property of the concept: along norm-matched random directions, the 0.5B maps are as chaotic as along the concept directions (Figure 46). On the same 128^2 grid, 7.6, 8.3 and 12.5 % of cells are chaotic for three Gaussian directions with the philosophy vector's norm, against 6.8 % for philosophy and 7.5 % for gratitude. In all five maps the chaotic share rises from about 1 % at $|\alpha| < 0.01$ to between 5 and 22 % at larger $|\alpha|$, and a chaotic region at high temperature, with windows reaching down, recurs. In this model, chaos under steering is therefore a generic response to perturbing the residual stream at this norm. Rotating the steering direction makes the same point continuously: along the great circle from one Gaussian direction to another, 90° away, at constant norm, the chaotic regions move and change shape, but the chaotic share stays between 13 and 24 % at every one of 103 angles (Figure 47; a video of the rotation is in the supplementary material). The direction does select the texts. The largest attractor family in all five maps is the unsteered period-5 loop, but beyond it only 26 to 56 % of one map's cells carry an attractor text that appears anywhere in another map. Llama-3.1 8B agrees. On its hostility plane, two norm-matched random directions at the same layer give 26 and 17 %

chaotic cells against 28 % along the concept direction; all three carry 34 to 44 % near $\alpha = 0$, and the random directions thin out at large $|\alpha|$. At layer 19, where a sparse autoencoder trained on this model reads the residual stream [78], the concept vector rebuilt there gives 28 %. The decoder direction of the autoencoder feature most specific to hostility on held-out texts (chosen by a rule fixed before any map; cosine 0.10 with the concept vector) gives 29 %, a random live feature 27 % and a Gaussian direction 39 %, all at the same norm. In this model, chaos therefore occurs along autoencoder-feature, contrastive and random directions alike, with direction-dependent prevalence; at 70B the directions differ more (Section 13).

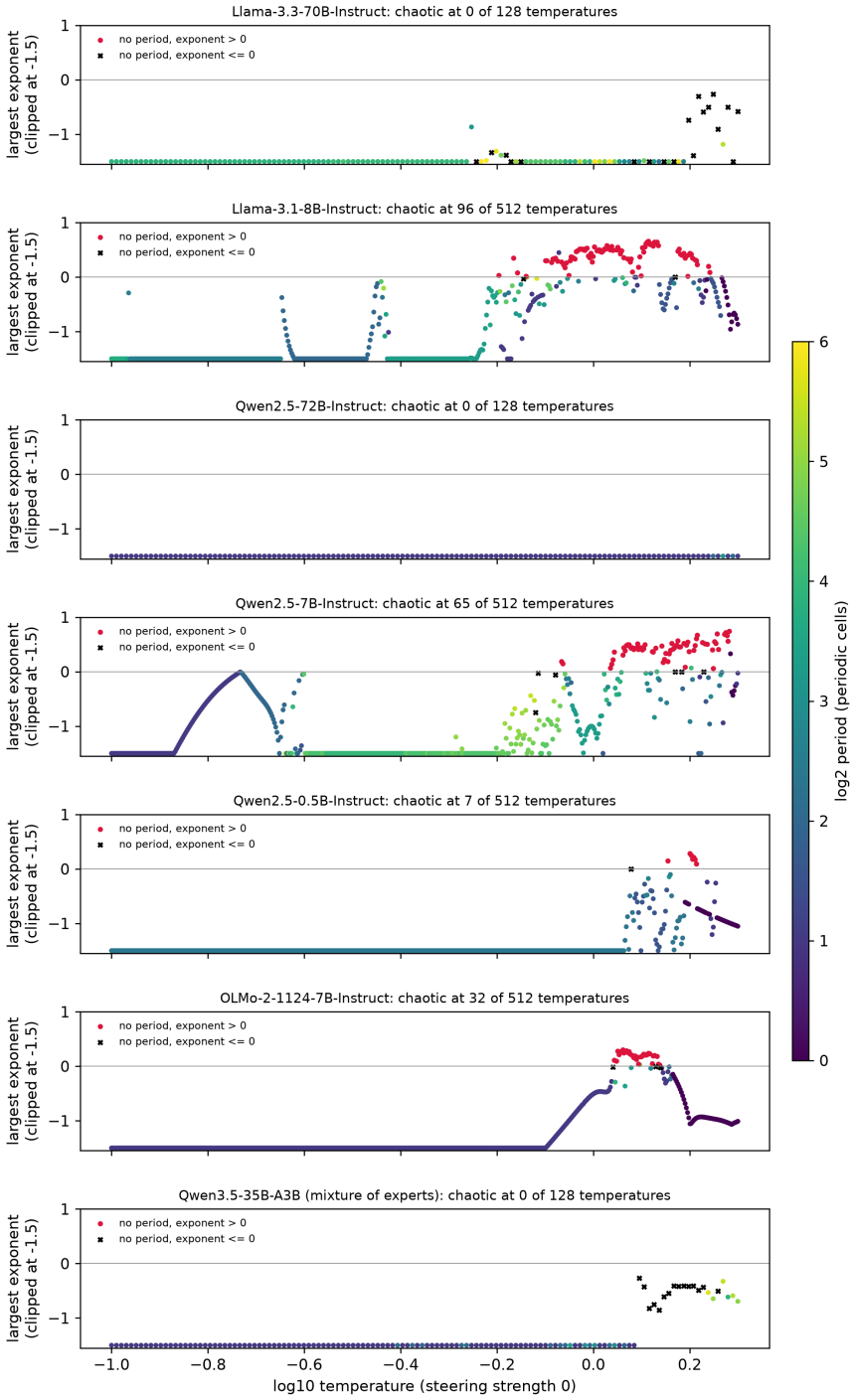


Figure 45: The unsteered soft-token map ($\alpha = 0$) along temperature, top to bottom: Llama-3.3-70B-Instruct, Llama-3.1-8B-Instruct, Qwen2.5-72B-Instruct, Qwen2.5-7B-Instruct, Qwen2.5-0.5B-Instruct, OLMo-2-1124-7B-Instruct and the mixture of experts Qwen3.5-35B-A3B. Largest Lyapunov exponent (clipped at -1.5) from float64 sweeps of 512 temperatures (128 for the 70B-class and mixture-of-experts models; 1000 transient and 2000 measured steps). Red: no period up to 64 and an exponent above 10^{-3}; black crosses: neither a detected period nor a positive exponent (unresolved); other points have a detected period, coloured by $\log_2$ of the period. Table 16 summarizes these and the wider sweeps of the other models.

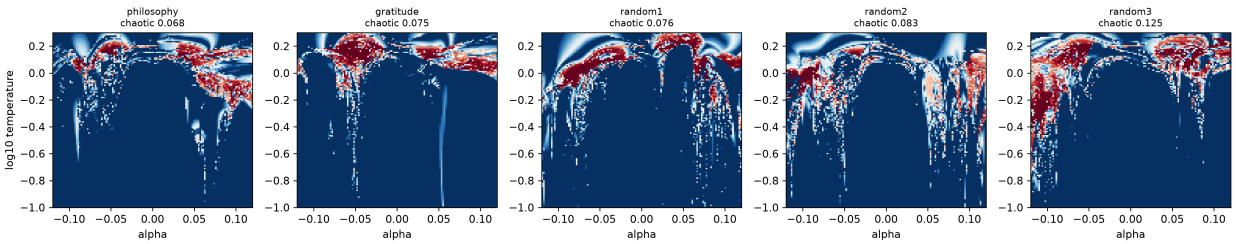


Figure 46: Steering controls in Qwen2.5-0.5B-Instruct (128^2 float64 cells, steering at layer 12, one grid and one set of settings): largest Lyapunov exponent along the philosophy and gratitude directions and along three Gaussian directions rescaled to the philosophy vector's norm. Titles give the chaotic share (finite-horizon exponent above 10^{-3}, no period up to 48).

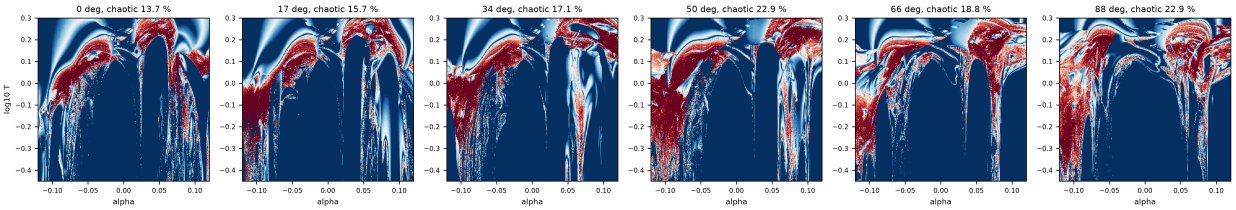


Figure 47: The soft-token map of Qwen2.5-0.5B-Instruct (steering at layer 12) while the steering direction rotates at constant norm from one Gaussian direction (0°) to another (90°): largest Lyapunov exponent (red: positive) at six of 103 angles, with the chaotic share. Float32 maps of 296×512 cells over $\alpha \in [-0.12, 0.12]$, $\log_{10} T \in [-0.45, 0.3]$; exponents from 80 measured steps at twin offset 10^{-4}; each map continues from the final states of the previous angle after 40 steps, and every tenth starts from the fixed start after 300 steps. Where attractors coexist, continued and fresh starts can reach different ones, so the maps, and the rotation video, can jump at every tenth angle and across the 17 of 120 planned angles that did not finish within the run limit. These short-horizon labels are coarser than those of the float64 maps.

What is and is not claimed. The objects studied here are invariant sets and bifurcations of an exactly defined smooth map built from a frozen model's own computation. The answer-level reproducibility tests use two Qwen2.5 models (0.5B and 7B), one task family and a soft-thinking pipeline without the top-k filtering of the published method, and their bfloat16 effect is not the float64 map's chaos (Section 6). The mechanism of Section 3 is measured along three directions in a 0.5B model and one in a 7B model, and at high temperature the network factor can carry the instability. The loop's regime does not mark the accuracy of ordinary sampled answers beyond the model's first-token uncertainty, and moving a prompt's loop into chaos does not change that accuracy (Section 6). We do not claim that ordinary sampled generation is chaotic in this sense (Section 11), nor that any window is a certified shrimp. The period-6 window and the hostile period-7 body have, numerically, shrimp-like organization: fold curves with numerically located cusps (simple critical multiplier; quadratic coefficient vanishing at a point where the cubic coefficient is nonzero and the unfolding is regular) and flip curves inside, and in the hostile body curves of very negative exponent, where the orbits are superstable, run through the windows. That is less than the full anatomy of a shrimp. For the hostile body we computed the superstable curves of one continued orbit. They do not cross, and at their closest approach the orbit is two-dimensional, so it has no one-dimensional shrimp centre (Figure 35). We have not determined the windows' symbolic sequences or tested for a periodicity hub [23, 61]. In a map that is effectively one-dimensional with a single critical point, a primitive orbit cannot be doubly superstable, so true shrimp centres

would require a second critical point in the reduced dynamics. Steering at two depths supplies one in band return maps (Figure 32), but the period-5 window we continued in such a plane is again two-dimensional where its leading multiplier would vanish (Figure 33). A generic codimension-two cusp also requires the two unfolding coefficients to depend on the two parameters with rank two. We estimated this at a fold point near one cusp of each window (determinants 2737 and 53.5), and there the local model's predictions at twelve frozen offsets matched the orbits recovered by Newton solves seeded at the predicted roots and the base state, in number at all twelve and in stability at eleven; the search was not exhaustive. At the nine located cusps themselves, $|\det J|$ lies between 14 and 2416. These are float64 computations validated by step-size checks (rung 1), not computer-assisted proofs. At the two continuation points that bracket the lower cusp of the hostile body, an independent implementation of the model reproduced the normal-form coefficients, including the sign change of b: $|b|$ within 1.2 % and c to two figures, with the opposite orientation of the null vector, which flips the signs of a and b together.

Limitations. The detailed orbit studies cover one model family (Qwen2.5, 0.5B and 1.5B), with coarse maps and three float64 orbits at 7B; the two other families (OLMo-2 7B and Llama-3.1 8B) have maps, including 128^2 refinements for Llama, but no continued orbits. The size comparison of Section 13 uses regular temperature grids that can miss narrow windows, a window that missed most of the larger models' chaos, and mostly one twin offset per label. The two-GPU runs of the 70B-class models computed their final normalization in float32 until we fixed it; their labels rest on offsets of 10^{-6} or more, and the cells we rechecked, the Qwen2.5-72B jump window and steered candidates in float64 throughout and samples of the Llama maps at three offsets, behave as stated, but the rest of those maps were not recomputed. Neither were the four coarse lines of the frozen 72B test; after the fix we computed only fine lines through their entropy jumps, one of which passes through their single chaotic temperature. The maps use one prompt (Section 13 adds ten for Qwen2.5-7B and -14B, and four for Qwen2.5-72B, its base model, and OLMo-2-7B and -32B), and steering uses contrastive vectors, checked against norm-matched random and sparse-autoencoder directions in two models. The system is a constructed loop: Section 11 extends it to memories of up to sixteen soft tokens and to the growing context of Soft Thinking, but sampled decoding of discrete tokens is a different system. The evidence consists of numerical validation, cross-checks and prospectively frozen passage tests (including failed all-row criteria), with a computer-assisted proof for only one orbit at one parameter point. Chaos labels are finite-horizon estimates (300 measured steps after 300 to 500 transient steps), and no chaotic cell is proved chaotic. A computer-assisted proof of chaos, such as a horseshoe verified by covering relations, would need enclosures of the map over boxes many orders of magnitude larger than the period-8 proof's (radius 2^{-640}). With the ball arithmetic used there, a box of radius 2^{-80} already loses the positivity of an RMS normalization at layer 3 (the enclosures overestimate through the layers), so such a proof needs other verified methods. A horizon check retested random cells of two float64 maps (Figures 1 and 7) with 1000 transient and 2000 measured steps at two twin offsets: 86 and 92 of 96 cells labelled chaotic stayed chaotic (all but one of the rest settled on periodic orbits), all 96 periodic cells in each map stayed periodic, and 18 and 20 of 32 cells with neither label turned out to be slow periodic transients. Near the accumulation of a cascade the labels are less reliable, because orbits there converge slowly or wander chaotically before they settle (transient chaos [92]): on the hostile line of Figure 41 (left), 135 of the 235 cells labelled chaotic from 400 measured steps stayed chaotic and 91 were periodic. The Feigenbaum ratios are finite-order values that rise toward δ (the last is within 1 %); the limit itself is not computed. All orbit, bifurcation and chaos results concern the constructed map, not sampled generation with a growing context. With sampling, the bridge of Section 11 tests only

whether the bottleneck's laminar phases survive the noise; its other tests are deterministic (greedy decoding, longer fed-back memories and Soft Thinking's growing context).

Steering strength is a bifurcation parameter. In the soft-token map, small changes of the steering strength switch the model between distinct repeating texts, chaos and coexisting attractors. Near a fold curve, behaviour becomes intermittent, with laminar phases whose duration the fold's normal form predicts in advance. This argues for treating an activation-steering coefficient like any control parameter of a nonlinear system. Its bifurcation set should be characterized before an operating point is chosen, history dependence should be expected near boundaries, and a monotone, smooth dose response should not be assumed. The block at which a vector enters matters as much as its strength. In a 0.5B and a 70B model, earlier injection gives more chaos and brings it to lower temperature, and two vectors at two blocks create chaos where neither does alone (Section 6). Outputs that fall into verbatim repetition at high steering doses are naturally treated as quality failures. In the present framework such loops are candidate attracting periodic orbits, and the dose at which they appear is a boundary that can be mapped like the others. The deterministic map is where such sets can be defined precisely and computed numerically. Whether its boundaries predict boundaries of sampled behaviour is open. The bridge of Section 11 shows that sampling noise shortens the laminar phases of the bottleneck, by an amount that depends on how many samples are averaged.

What this suggests for interpretability and alignment. Our results come from open models of 0.5B to 72B parameters and a constructed loop, but they suggest three things about models whose outputs feed back into their inputs, as in self-dialogue, agent loops and pipelines of model calls. First, where the loop is chaotic, a perturbation of 10^{-6} grows to order one within roughly 15 to 50 steps (the largest unsteered exponents of the 7B and 8B models are 0.30 to 0.88 per step). Even a complete account of single forward passes then predicts a long trajectory only if the state is known far more precisely than any interpretation supplies. Stepwise summaries can be inconclusive: every step of the period-8 orbit expands some direction while the cycle contracts, so the ordered product of the step Jacobians, not a summary of each step, captures the loop's stability (Section 3). What remains predictable are the invariant sets, their statistics and the boundaries between them. This extends the analysis of trained recurrent networks through their fixed points [29, 30] to the periodic orbits, chaos and bifurcations of a pretrained transformer. Second, feature dictionaries say which features an attractor occupies, but the boundaries between attractors, which decide what a small change does, are properties of the map's derivatives. In our 8B controls, a sparse-autoencoder feature direction, contrastive and random directions all drove the map into chaos (27 to 39 % of cells); at 70B the hostility-related directions gave more chaotic cells than the random ones (4.3 and 3.5 % against 1.1 and 0.2 %; Section 13). Steering acts by making next tokens compete (Section 3), which is not specific to concept-labelled directions, but a direction's geometry and its meaning can be correlated. Text can also hide the dynamics: in the hostile cascade, a seven-token text repeats exactly while the state underneath is chaotic (Figure 38), so a monitor that reads only the emitted text would call a chaotic regime periodic. Third, conclusions drawn at one operating point need not transfer: which loop a model enters depends on how much context it keeps (Table 14), on the path by which a boundary is approached, and on the sampling noise (Section 11). Treating a looped model as a dynamical system, with regimes and boundaries to be mapped, rather than as a function to be evaluated point by point, seems to us the appropriate stance for both interpretability and safety evaluation. The reproducibility tests of Section 6 make this concrete: in a chaotic regime the batch a request happens to share decides, after a predictable number of steps, what the loop writes.

In low precision, continuous feedback makes even short-budget answers depend on the batch, which greedy decoding does not.

Is the chaos useful? The chaos we map is not a source of new content. The chaotic cells of Table 4 recombine the phrases of the neighbouring loops in an order that did not settle over the recorded horizon: the trajectory wanders among the ghosts of nearby periodic orbits, which gives variety in sequence but not in substance. The periodic loops are the familiar degeneration of decoding into repetition [25, 26]. Neither regime is good text, and the diversity of ordinary generation comes from sampling and a growing context, which wash out much of this structure (Section 11). What the chaos does matter for is predictability and control. Positive finite-horizon exponents limit open-loop prediction of a trajectory at fixed state precision (our experiments do not establish mixing or limits on feedback stabilization); a stable attractor is predictable, but it can also hold a model in a state it should leave, such as the hostile loops of Section 15. On the paper's prompt, the largest Qwen2.5 and Llama models sit near the stable end, but across prompts a 14B Qwen model is more chaotic than the 7B, at higher temperatures, and OLMo-2-32B is among the most chaotic models we tested (Section 13). Whether regimes near the boundary between order and chaos are especially useful, as has been argued for computation in dynamical systems and recurrent networks [93, 94] and shown for signal propagation in deep networks at initialization [88, 89], is open for language models. We tested three versions, and none shows an advantage at the edge. Prompts whose own loop sits at the edge ($|\lambda| \leq 0.1$) are answered no better by ordinary sampling than prompts whose loop is deeply periodic or deeply chaotic (Section 6). Used as a reservoir, the loop remembers a weak injected input best in short stable cycles. Its linear memory capacity falls at every period doubling toward the onset of chaos, does not rise as the exponent approaches zero within a period, and is nearly erased by chaos (Figure 48). In Huginn, whose reasoning is itself a latent loop, a pilot on 40 development problems recorded the decay rate of every generated token's recurrence. Accuracy holds while that recurrence is comfortably stable and falls as it approaches its edge. The gain at $\lambda = 1.05$ (Section 14) comes where the token recurrence is nearly as stable as natively (median decay rate 0.921 against 0.912), and the collapse at 1.15 where it is near critical (0.981, with half of the tokens no longer converging).

Which findings might reach frontier models. Today's frontier systems include closed models and open-weight mixture-of-experts models. Some of our findings rest only on structure that every smooth softmax-readout language model with a fixed prefix shares and should carry over: the factorization of the loop's derivative (a covariance for tied embeddings, a cross-covariance otherwise), contraction at sufficiently high temperature, the reduction of zero-temperature attractors to greedy cycles of whatever memory the loop keeps, and the role of a steering coefficient as a bifurcation parameter. Others belong to particular weights and will not carry over as such: the attractor texts, the positions of windows, folds and cusps, and every number we report; a second checkpoint of the same architecture already has different attractors (Section 12). Size gives no guarantee of stability. In a fixed temperature window the unsteered chaos nearly vanished in the largest Qwen2.5 and Llama models, but across prompts a 14B Qwen model was more chaotic than the 7B, at higher temperatures, OLMo-2-32B was the most chaotic OLMo model, and post-training changed the chaos of two Llama checkpoints of the same size (Section 13). For an open frontier model we would search for chaos where its own next-token distribution spreads, across prompts. Our one mixture-of-experts model, Qwen3.5-35B-A3B, gave no positive exponent without steering and irregular cells only under hostility steering, and hard routing falls outside the smooth theory of Section 3 (Section 12). Reasoning models open a thinking block that changes the prefix, post-training may remove some

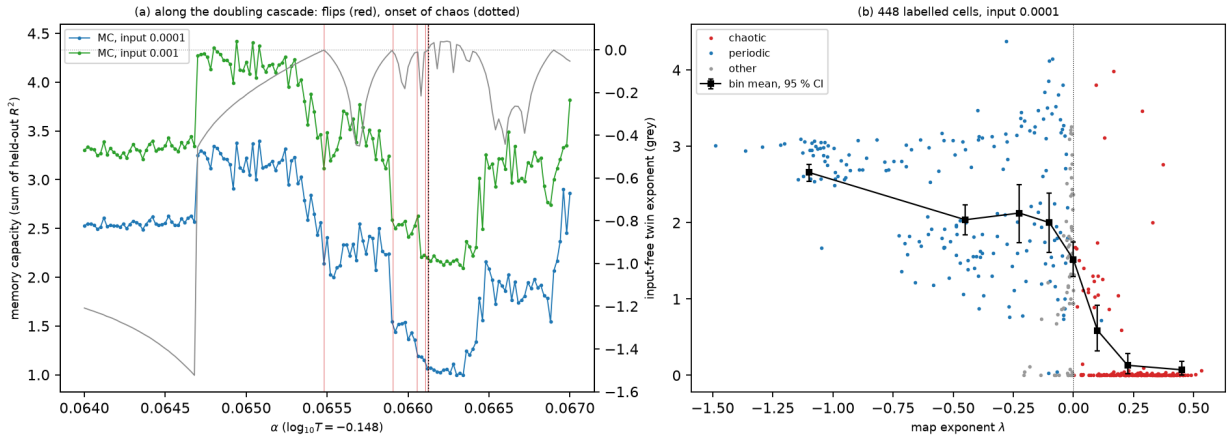


Figure 48: Memory capacity of the soft-token loop used as a reservoir (Qwen2.5-0.5B-Instruct, philosophy direction at layer 12, float64). A weak random input enters the map, $x_{n+1} = F(x_n + \epsilon|x|u_n r)$, and linear readouts of the state reconstruct the past inputs; the capacity is the sum of their held-out R^2 over 41 delays. (a) Along the doubling cascade at $\log_{10} T = -0.148$: capacity for two input sizes, the input-free twin exponent (grey), the flips (red) and the onset of chaos (dotted). (b) The 448 labelled cells of the horizon check against the map exponent, with bin means and bootstrap 95 % intervals.

attractors and create others, and deployed inference samples tokens, in low precision, over long contexts; our bridge results show that sampling shortens laminar phases and that a longer context changes which loops are entered. Closed models permit none of these computations: the map needs the input embedding matrix, full next-token distributions and a way to feed a vector in place of a token, and its analysis needs derivatives. For closed models the method and the questions transfer, not the numbers.

18 Evidence and reproducibility

We grade claims on an evidence ladder. Rung 1 means numerically validated (tolerances, residuals, convergence, precision varied); rung 2, cross-checked by an independent estimate, implementation or hardware; rung 3, a computer-assisted proof.

Search scale. The soft-token results come from an exploratory search. An initial census of the raw data at 00:19 UTC on 14 October 2026 counted 34 distinct maps (987 308 cells, 27 maps in float64) in six models, each with one prompt and one of three steering directions or none, plus one partial map. Of these, 29 maps and the partial one appear here; the five omitted are early discovery or side maps. Later additions are not in that count: the steering-control maps of Section 17, the mixture-of-experts maps, and the sweeps, lines and maps of the models of Section 13. The labelled maps contain about 2600 periodic families of at least six cells, counted per map. We singled out about 26 windows or families for float64 work: 20 are reported in detail, 2 only as map features and 4 not at all (an orbit whose continuation was abandoned, a cell that proved aperiodic, a band whose float32 orbit did not converge and a pilot that found no cusp within a bounded search). Of 54 Newton orbit attempts, 7 failed. All nine tangent reversals on continued fold curves (seven found by 14 October, and two on traces of a period-4 fold curve continued on 15 October) are numerically located cusps; the two reversals of a continued flip curve are smooth turns. Two of the five regions we flagged as shrimp candidates are reported as shrimp-like. On 15 October, the turning-point

search of the two-block planes (not counted in Table 17) led to two more windows: a period-5 window, reported with its continued orbit in Section 15, and a period-6 window that was mapped but not continued. One further language model (Ouro-1.4B) was explored and is not reported; its apparent chaos was a long transient. Every window was chosen after the maps were inspected. Only the passage forecasts, the cusp-unfolding offsets and Huginn's relaxation test were fixed before the computations that test them (Table 17).

	explored	reported	note
Soft-token maps (distinct)	34	29	plus one partial map, reported
cells	987 308	893 868	omitted: 5 discovery or side maps
float32 maps	7	4	
Models with maps	6	6	one prompt; 3 directions and $\alpha = 0$
Periodic families (≥ 6 cells)	≈ 2600	–	counted per labelled map
Windows given float64 work	≈ 26	20 (+2)	+2: map features only
Newton orbit attempts	54	–	7 failed
Tangent reversals on fold curves	9	9	all located cusps (2 found on 15 Oct.)
Unfolding offsets, frozen	12	12	counts 12/12, stability 11/12
Shrimp-candidate regions	5	2	reported as shrimp-like
Tests fixed before outcomes	6	4	2 all-row criteria failed by one row
Language models loaded	8	7	Ouro-1.4B not reported

Table 17: Search scale on 14 October 2026, before the steering-control maps; the reversal row includes the two cusps located on 15 October (Huginn's frozen test counted once; of the two tests not reported here, one belongs to the separately reported sampled-dialogue study and one, a cusp pilot in a second checkpoint, is the unreported bounded search).

- Rung 3: the period-8 stripe orbit of Table 3. At $\alpha = 0.071674$ and $\log_{10} T = -0.115$, read as exact binary64 rationals (the temperature is the real number $\exp(\ln 10 \cdot \log_{10} T)$ for that exact dyadic $\log_{10} T$), the expected-embedding map of the stored Qwen2.5-0.5B-Instruct checkpoint (all 24 layers, the full 151 936-token readout, the fixed 37-token prefix, steering after zero-based block 12) has a primitive period-8 orbit, which is the unique multiple-shooting root in a product of eight boxes of coordinate radius 2^{-640} around refined centres, and the orbit is locally attracting. Ball arithmetic (Arb [95]) encloses the map and its Jacobian over each box (8×896 directions); a Krawczyk-type inclusion [96–98] for the preconditioned shooting system gives existence and uniqueness, separation of the first point from the points one, two and four steps later gives the primitive period, and a norm bound on powers of the enclosed monodromy gives attraction. The map is the real-number map of the stored constants, read as exact dyadic rationals: rotary angles, exponentials and square roots are real functions enclosed by the balls, not floating-point tables. An independent audit in exact rational arithmetic rechecked the saved witnesses (the scalar inequalities, the primitive-period separations and the row norms of the saved monodromy powers); it did not recompute the enclosures of the model or the dense matrix products, which rest on one execution. The proof trusts the Python, FLINT and Arb stack and the stored weights. It does not exclude other period-8 points in the first box, and it says nothing about the window's boundaries or global organization. The record (`certificates/llm-p8-certificate-final-001.json`) pins the SHA-256 hashes of the checkpoint, its configuration and the steering vector, and lists the 37 prefix token IDs.
- Rung 2: the 2-cycle's fold and its normal form. An independent reimplementation of the forward map and solver recovers α_c, a and b (to 6×10^{-15}, 3×10^{-12} and 2×10^{-6} relatively),

and ab also follows from the stable-branch multipliers. The same reimplementation reproduces all 18 fixed-entrance counts of the controlled passage runs exactly (Table 5 lists 12 of them).

- Prospectively frozen: the passage forecasts at two new folds (Table 7); 23 of 24 within one iterate, and the all-row criterion failed by one row.
- Rung 2 as well: the period-2 and period-8 orbits of Table 3 (independent implementation), and the cascade's first four flip parameters (CPU and H100, agreement 4×10^{-15}).
- Rung 2 as well: the hostile 2-cycle fold and its normal form, reproduced by an independent float64 forward map with nested automatic differentiation (location to 6×10^{-10}, ab to 8 parts per million); its controlled passages were also checked independently, with the frozen all-six criterion failing by one row.
- Rung 1: the other orbits, the curves, the nine located cusps, the superstable parameters, the 1.5B flips, the hostile windows' edges, the spiral region's period-7 body and the cascade inside it, the three 7B orbits, and Huginn's float32 spectral estimates.
- Prospectively frozen, not independently reproduced: the scaling predictions P1–P3, the prompt-blocked size tests in Qwen2.5 (7B and 14B; 72B; 72B base against Instruct) and OLMo-2, and the loop-free predictor test of Section 13, all scored by code pushed before the data, together with the three-offset plateau tests of the 70B-class maps and the float64 recheck of the Qwen2.5-72B cells with exact tangent exponents. These are float64 computations at rung 1; no independent implementation has rechecked them.
- Prospectively frozen on 15 October: the answer-level reproducibility test in Qwen2.5-0.5B (positive), whose saved rows and statistics an independent implementation reproduced (the generation itself was not rerun); its 7B extension (positive) and the behaviour test (null on its primary question; no edge optimum), not independently reproduced. All are in Section 6 and each was scored by code pushed before the data (`docs/repro-task-test.md`, `docs/bpc-test.md`).
- Independently reproduced from saved data (15 October): every summary field of the 0.5B batch-layout comparison (448 cells, four comparisons) and the gain-split, logit-bias and crossover summaries of Section 3 (5353, 6837 and 24 cells); the coefficients and offset rows of the flip normal-form test (Table 13); the 1.5B and 7B fold-law scores under both period selectors (all 2700 cells); and the scalar conditions of the nine located cusps, matched one-to-one to the tangent reversals of the seven continued fold curves. This reproduces the statistics and saved scalars, not the model runs, derivatives or eigenvectors. The within-prompt steered test (null) used an analysis fixed and pushed before its data, without a separate frozen design document.
- Prospectively frozen and independently replayed: Huginn's relaxation test on held-out data, confirmed in both parts; an independent implementation reproduced both from the saved rows and spectra (GSM8K part exactly, item-level part to 10^{-12}) (Section 14).

Hardware. Float64 GPU computations ran on rented NVIDIA H100 (NVL, PCIe and 80 GB HBM3), H200, H200 NVL and A100 80 GB (SXM4 and PCIe) GPUs, one GPU per run except for the 70B-class models, whose layers were spread over two; the float32 frames of the rotation video ran on RTX 4090 GPUs; cross-checks and the interval proof ran on CPUs. The manifest

`docs/manifests/runpod-runs.json` records for each run the GPU type, the code commit, the job and its arguments, the wall time and the exit status, and the available hashes of the uploaded code and input archives and of the retrieved results, with absent fields left explicit; per-run receipts and accepted provenance limitations are kept with it. Regenerated at 12:33 UTC on 15 October 2026, after the last run, it lists 398 runs with retrieved results (code and input-archive hashes for 290, result-archive hashes for 373) and 20 pods that ended without results. GPU types round float64 reductions differently, so quantities used across runs were checked across hardware where it mattered (for example, flip parameters agree between CPU and H100 to 4×10^{-15}). Code: `tools/explore_llm_soft_loop.py`, `tools/bifurcation_curves.py`, `tools/llm_feigenbaum.py`, `tools/explore_latent_loop.py`, `tools/cusp_check.py`, `tools/huginn_relax_eval.py`, `tools/huginn_arc_analysis.py`, `tools/steer_gain.py`, `tools/crossover.py`, `tools/repro_noise.py`, `tools/repro_task.py`, `tools/repro_summary.py`, `tools/bpc_analyze.py`, `tools/turning_scan.py`. Raw-data hash receipts: `evidence/llm-dynamics-data-receipts.json` (to 12 October), the per-run receipts in the research log, and the verified snapshot manifests in `docs/backups/`; independent cross-check and audit receipts are under `codex/` (`crosscheck`, `confirm-fold`, `hostile-fold-crosscheck`, `huginn-design-review`, `huginn-independent-replay`, `paper-review`).

[**Planned:** public code and data release at submission.]

Use of AI assistance

This work was carried out with AI coding and research agents (Anthropic's Claude Code and OpenAI's Codex) working under the author's direction. The agents wrote and ran the code, executed the experiments, checked results against the raw data and drafted text; other AI systems made independent read-only checks of the results and the manuscript. Frozen designs, run cards, failed attempts, audits and hash-bound records of every run are kept in the project repository. The author directed the study, reviewed the work and takes full responsibility for its content.

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